REVIEW Artificial Intelligence

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AI for science: Progress, challenges, and perspectives

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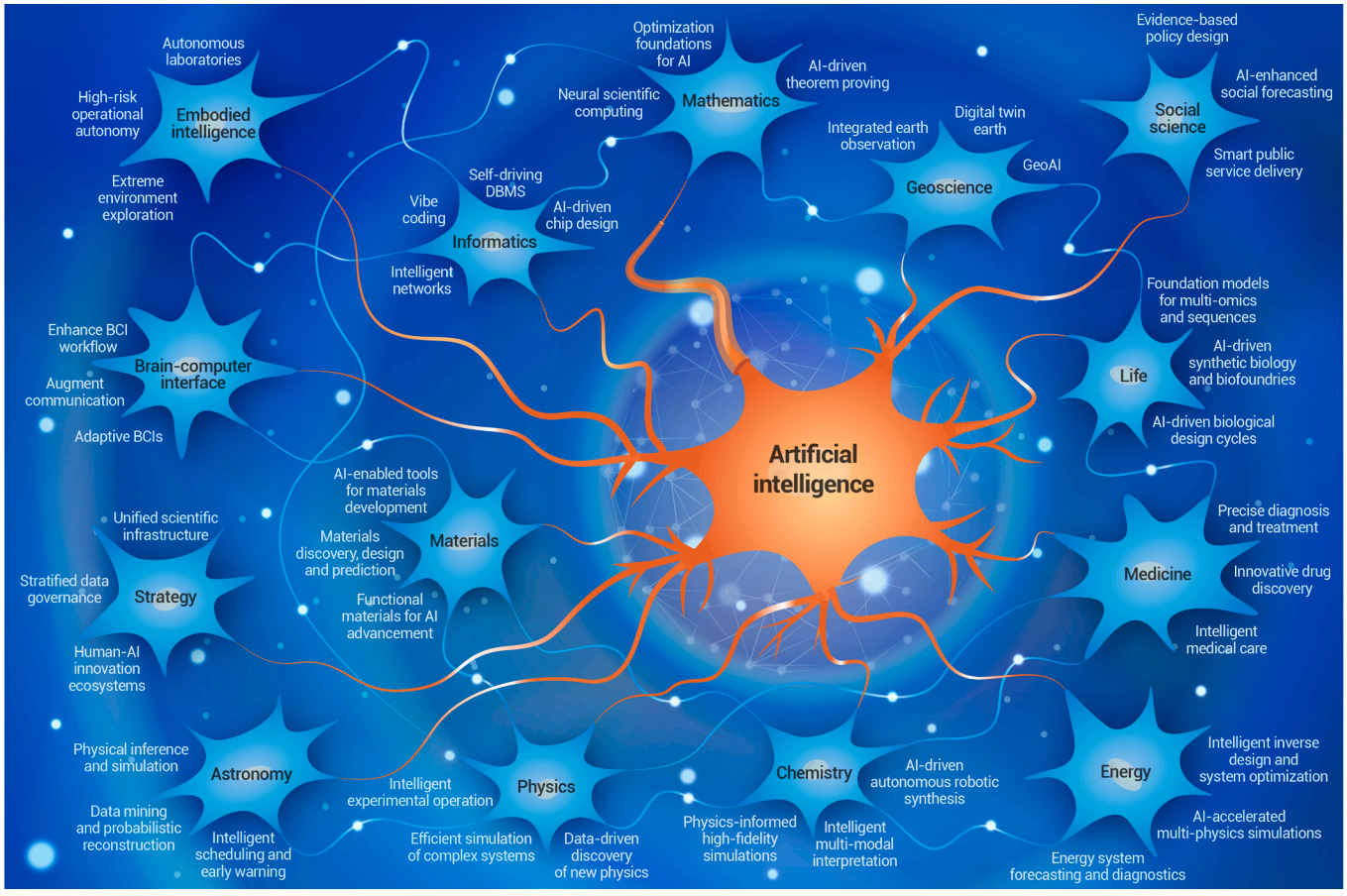
GRAPHICAL ABSTRACT



PUBLIC SUMMARY

- Artificial intelligence (AI) is shifting from task-specific tools to a foundation for scientific discovery.
- AI now builds closed-loop workflows linking literature, data, hypotheses, experiments, reports, and evolution.
- Major barriers remain in data quality, interpretability, physical consistency, and trusted human-AI research.

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Artificial Intelligence REVIEW

The Innovation

AI for science: Progress, challenges, and perspectives

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By 2026, artificial intelligence (AI) has completed a qualitative transition from specialized analytical tools to a foundational engine for scientific inquiry, fundamentally altering the paradigms of knowledge generation. In this review, we systematically evaluate the progress of AI for Science (AI4Science), characterized by the shift from text-centric generation to reasoning-oriented intelligence and the deep embedding of foundation models into the entire research life cycle. We propose a unified, closed-loop research paradigm comprising six core modules: literature research, data analysis, hypothesis formulation, experimental execution, report generation, and autonomous evolution. Furthermore, we synthesize breakthroughs across a broad spectrum of frontier domains—including information science, mathematics, energy, physics, chemistry, materials science, medicine, life sciences, astronomy, geosciences, and social science—highlighting AI's capacity for overcoming dimensionality constraints and enabling complex inverse design. We then identify critical challenges that persist, such as the scarcity of standardized high-quality data, the interpretability gap, and the physical consistency of generative outputs. Finally, we offer forward-looking perspectives on constructing cross-disciplinary scientific foundation models and fostering a future of human-AI collaborative autonomous research.

INTRODUCTION

At the conclusion of his celebrated paper "Computing Machinery and Intelligence," Turing wrote, "We can only see a short distance ahead, but we can see plenty there that needs to be done."

In 2021, we published a review entitled "Artificial intelligence: A powerful paradigm for scientific research,"[1] which surveyed how artificial intelligence (AI) technologies were then advancing scientific inquiry. Over the subsequent 5 years, AI has undergone rapid and profound progress, and its impact on

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scientific research has shifted in a qualitative manner. In particular, the emergence of large language models (LLMs) scaling from billions to over a trillion parameters (e.g., LLaMA[2], GPT-4,[3] and Gemini[4]), alongside high-capacity diffusion models, has ushered AI into a generative era.[5] This unprecedented parameter scale has unlocked emergent cognitive capabilities, such as zero-shot generalization and complex in-context learning, that align inherently with the high-dimensional, exploratory nature of scientific hypothesis generation. Consequently, these foundational models have catalyzed tangible breakthroughs across diverse scientific domains. For instance, AlphaFold[6] leverages highly parameterized attention mechanisms to capture the complex, nonlinear mapping from amino acid sequences to 3D protein structures, fundamentally transforming structural biology. Similarly, in materials science, LLM-driven autonomous laboratories[7,8] integrate multimodal reasoning with robotic execution for multiproperty-driven molecular discovery. Moreover, the introduction of techniques such as long chain-of-thought reasoning, mixture of experts (MoE), and group relative policy optimization (GRPO) has pushed large models beyond humanities-oriented text generation toward science-oriented logical reasoning. As a result, AI for Science (AI4Science) has undergone fundamental transformations in both methodology and research scope. Correspondingly, the AI4Science community is entering a new phase of paradigm transition, in which AI is evolving from a collection of fragmented, task-specific tools into a central and increasingly dominant component of scientific research. This shift enables AI methodologies to be deeply embedded into scientific workflows, systematically empowering the entire research life cycle. Motivated by these developments, we therefore undertake an update to our previous review, with the aim of presenting the latest advances in AI4Science in the era of large models.[9]

Over the past 5 years, a series of high-profile reviews has chronicled AI's growing impact on scientific discovery across disciplines. Wang et al.[10] delivered a panoramic review of AI-powered advances over the past decade, from self-supervised learning and geometric deep learning (DL) to generative models for drug and protein design. They illustrated how AI accelerates every stage of research—hypothesis generation, experimentation, and data analysis—while flagging persistent challenges in data quality, model transparency, and the need for methods that truly augment human insight.[11,12] Other authors emphasized emerging concerns: Messeri and Crockett warned that AI's allure of objectivity and productivity can foster "illusions of understanding," as scientists risk overestimating comprehension and creating monocultures of methods, potentially yielding more output but less insight.[11,13] Tang et al.[14] examined safety gaps in "AI scientists"—autonomous lab agents powered by LLMs—and proposed a triadic oversight framework to align these systems with human values and prevent misuse. Meanwhile, quantitative analysis by Gao and Wang revealed that AI adoption in science, though widespread, remains uneven: many fields face a skills gap and demographic disparities in AI benefits.[15] Musslick et al.[16] highlighted how robotic automation and AI (from robot chemists to literature-mining generative pre-trained transformers [GPTs]) are reshaping experimental science, boosting discovery rates and reproducibility, but cautioned that greater automation paradoxically heightens the need for human oversight and new ethical norms. Together, these reviews underscore a transformative trend of AI permeating all branches of science, from physics and chemistry to biology and social science, and accelerating knowledge generation. Yet they also reveal fragmentation: each study often isolates a particular aspect—theoretical understanding, technical breakthroughs, socio-technical risks, metrics of uptake, or automation—without a unifying outlook. Many earlier reviews are already dated by the rapid progress of AI (for instance, pre-ChatGPT perspectives) or limit themselves to siloed domains or concerns. This indicates an unmet need for an up-to-date, integrative synthesis.

This review is not intended as a discipline-by-discipline catalog of AI applications. Rather, it is organized around a central question: how recent AI advances are reshaping scientific workflows and practices across domains. The fields included here therefore span formal and computational sciences, natural sciences, life and medical sciences, and several cross-cutting frontiers. Taken together, these domains cover settings ranging from data-intensive analysis and surrogate modeling to multimodal reasoning, inverse design, and more integrated human-AI research workflows. Building on our 2021 review,[1] this review presents three key advances. First, it shifts focus from AI/machine learning (ML) methods used mainly for domain-specific analytical tasks to foundation models, multimodal models, and early agentic systems. Second, it transitions from surveying isolated applications to outlining a unified closed-loop perspective, covering six modules from literature research to autonomous evolution. Third, it extends the discussion from domain-level applications and enabling infrastructure to broader workflow integration, research ecosystems, and institutional conditions. Ultimately, our goal is not just to add newer examples but to clarify how AI is reshaping science at the levels of method, workflow, and research organization.

THE NEW ERA OF AI4Science

As discussed above, AI is transitioning from discriminative learning to generative modeling and from text-centric generation to reasoning-oriented intelligence. This transformation has elevated AI from a set of fragmented analytical tools to a core, and increasingly dominant, driver of scientific research. Over the past 5 years, rapid progress in foundation models, generative learning, and reasoning mechanisms has not only expanded the expressive and inferential power of AI systems but has also fundamentally altered the paradigms through which scientific discovery is conducted. This section presents a brief overview of recent advances in AI and analyzes their implications for the evolving paradigm of AI4Science, highlighting how AI is becoming deeply embedded into scientific workflows and systematically empowering the entire research life cycle.

Recent progress in AI

In recent years, AI has shifted from specialized, narrow tasks to general-purpose reasoning, driven by the rise of foundation models. This era is defined by the pursuit of artificial general intelligence (AGI), where systems demonstrate human-level versatility across distinct domains.

The catalyst for this revolution was the LLM. Built upon the transformer architecture,[17] models such as OpenAI's GPT-4[3] and Meta's Llama[2] series utilized massive scale to unlock "emergent capabilities." Researchers observed that simply predicting the next token in a sequence allowed these models to master coding, logic, and nuanced translation, adhering to "scaling laws"[18] where performance correlates predictably with compute and data size.

Simultaneously, the field of generative media exploded. Shifting away from generative adversarial networks (GANs),[19] diffusion models became the standard for text-to-image generation such as stable diffusion,[20] allowing for high-fidelity visual synthesis. This rapidly evolved into text-to-video models such as Sora,[21] which function as "world simulators." These models do not just generate pixels; they seem to learn the physics and causality of the 3D world, a crucial component for AGI.

These streams have converged into multimodal LLMs (MLLMs). Modern frontier models (such as GPT-4o or Gemini[4]) are natively multimodal, processing text, audio, and video simultaneously. This mimics human sensory integration, allowing AI to reason about the world more holistically than text-only systems ever could.

The final frontier is embodied AI. To be truly general, intelligence must interact with the physical world. Recent breakthroughs involve vision-language-action (VLA) models.[22] Instead of hard-coding robot movements, researchers are using MLLMs to serve as the "brain" of the robot. Systems such as Google's RT-2[23] demonstrate that robots can transfer semantic knowledge from the web to robotic control, enabling them to understand vague commands (e.g., "throw away the trash") and manipulate unseen objects.

In summary, the trajectory of AI has moved from processing language (LLMs) to synthesizing reality (generative media), integrating senses (multimodal AGI), and finally, acting physically (embodied AI) (Figure 1). This profound understanding of and interactive capability with the physical world is reshaping the paradigm of scientific discovery in unprecedented ways. To overcome the exponential complexity of traditional grid-based solvers, deep neural networks (DNNs) act as mesh-free function approximators, utilizing gradient optimization to make high-dimensional mappings computationally feasible. Advancing this, operator learning learns mappings between infinite-dimensional function spaces, enabling zero-shot generalization to dramatically accelerate physical simulations.[22] For complex inverse design, diffusion models[24,25] transform simple noise into highly structured configurations via stochastic differential equations. They employ conditional score matching to deterministically satisfy specific physical constraints. Because such diffusion-based workflows are often computationally intensive, recent acceleration methods such as residual mixed-precision quantization and temporal distillation have become important for enabling low-bit and resource-efficient diffusion models.[26] Finally, to navigate astronomical discrete search spaces, reinforcement learning (RL)[27] formulates optimization challenges as Markov decision processes. By leveraging value and policy networks to evaluate long-term returns, RL can help explore large sequential decision spaces more effectively.

New paradigm of AI4Science

Mainstream research paradigms in AI4Science have long been grounded in the theoretical foundation established by George Cybenko's universal approximation theorem,[28] which demonstrates that multi-layer feedforward neural networks possess the capacity to approximate continuous functions. Building upon Hinton's works on gradient descent[29] and DNN theory,[30,31] together with recent breakthroughs in GPU-accelerated computing, researchers have increasingly leveraged massive volumes of experimental and observational data to train deep neural models capable of capturing high-dimensional, complex, and highly nonlinear input-output mappings underlying scientific phenomena, such as AlphaFold[6] and Pangu.[32] This DL-driven AI4Science paradigm bypasses the search and discovery for concise and elegant formulas within scientific paradigms, instead training a black-box yet powerful model to approximate the unknown laws governing natural systems.

Despite their empirical success, such approaches typically remain task specific, with AI primarily serving as a tool for data-assisted analysis or function fitting.[33,34] The broader process of scientific discovery—spanning hypothesis formulation, experimental design, validation, and interpretation—continues to be largely human driven, with AI methods playing a supportive rather than a central, autonomous role.

In recent years, however, the AI4Science community has entered a new phase of paradigm transition. Emerging research efforts seek to embed AI methods directly into the closed-loop design-make-test-analyze scientific workflow, aiming to systematically empower the entire research life cycle. For example, advanced molecular discovery[8] targets desired molecular properties by automatically generating candidate molecular structures using AI models, predicting their properties, and further planning synthesis routes that are executed via AI-driven automated laboratory hardware. This enables an efficient closed loop from molecular design to experimental validation. Similarly, end-to-end materials discovery systems[7] employ transformer-based models to learn synthesis knowledge from scientific literature and integrate robotic laboratories to autonomously synthesize and validate candidate materials, thereby realizing a largely unmanned pathway for discovering theoretically viable materials.[35]

With the rapid advancement of LLMs, agents, and embodied intelligence, these emerging systems exhibit not only substantial improvements in model capability but also unprecedented potential for system-level integration and generalization.[36] They are progressively dissolving the long-standing boundaries of AI as isolated, tool-centric components within scientific workflows.[37,38] By positioning LLMs as knowledge-centric reasoning cores, agents as process orchestrators, and embodied intelligence models as action executors, new AI4Science systems increasingly enable end-to-end coverage of the scientific discovery pipeline.[39]

In general, the new AI4Science paradigm integrates 6 key modules—including literature research, data analysis, hypothesis formulation, experimental execution, report generation, and autonomous evolution—forming adaptive, self-optimizing workflow systems that substantially enhance both the efficiency and reproducibility of scientific discovery[40–42] (Figure 2). Analogous to human researchers, the literature review module retrieves and synthesizes relevant scientific publications for a given research topic, allowing AI models to internalize domain knowledge that underpins subsequent discovery. Acting as a data analyst, the data analysis module mines large-scale, multimodal scientific and experimental datasets to uncover patterns or validate scientific conjectures. By jointly leveraging insights from literature and data, the hypothesis generation module produces novel, testable hypotheses, serving as the primary engine of scientific innovation.[43–45] Crucially, to mitigate the inherent black-box limitations of traditional data-driven models, this closed-loop workflow increasingly embeds recent advancements in causal AI and mechanistic interpretability. By explicitly modeling cause-and-effect relationships and reverse-engineering neural representations, these techniques ensure that the AI-generated hypotheses and analytical insights are physically transparent and scientifically rigorous rather than mere statistical correlations.[46] The experimental execution module robustly validates hypotheses through experiments, encompassing experimental design, *in silico* simulations, and real-world physical experimentation, thereby significantly accelerating discovery cycles. The report generation module consolidates experimental findings and scientific insights, facilitating dissemination and discussion through compelling data visualization, structured scientific writing, and peer-review-ready outputs. Finally, scientific evolution is achieved by enabling continuous iterative discovery loops, fostering sustained long-term scientific progress.[47,48] Especially, the new AI4Science paradigm increasingly incorporates reflective verification mechanisms to identify and correct their own errors with minimal human intervention. Rather than relying solely on raw model outputs, these systems often incorporate externalized self-correction mechanisms, such as execution-and-evaluation pipelines,[49] predict-measure-update loops,[8] and discrepancy- and failure-aware experimental diagnosis,[7] to identify inconsistencies, suppress hallucinations, and iteratively refine intermediate results before they propagate through the scientific workflow. More recently, some studies have even begun to explore whether AI can participate in the peer-review process itself,[50] raising broader questions about the role of AI not only in conducting science but also in evaluating it.

Collectively, these pioneering efforts in the new AI4Science paradigm not only demonstrate the tangible value of AI in accelerating scientific discovery and reducing research costs but also exhibit distinctive advantages in overcoming critical bottlenecks related to data dimensionality, experimental complexity, and resource constraints in complex scientific domains.

Cross-domain overview and workflow mapping

To facilitate cross-domain comparison and clarify the organizing principle of this review, Tables 1 and 2 provide two complementary summaries of

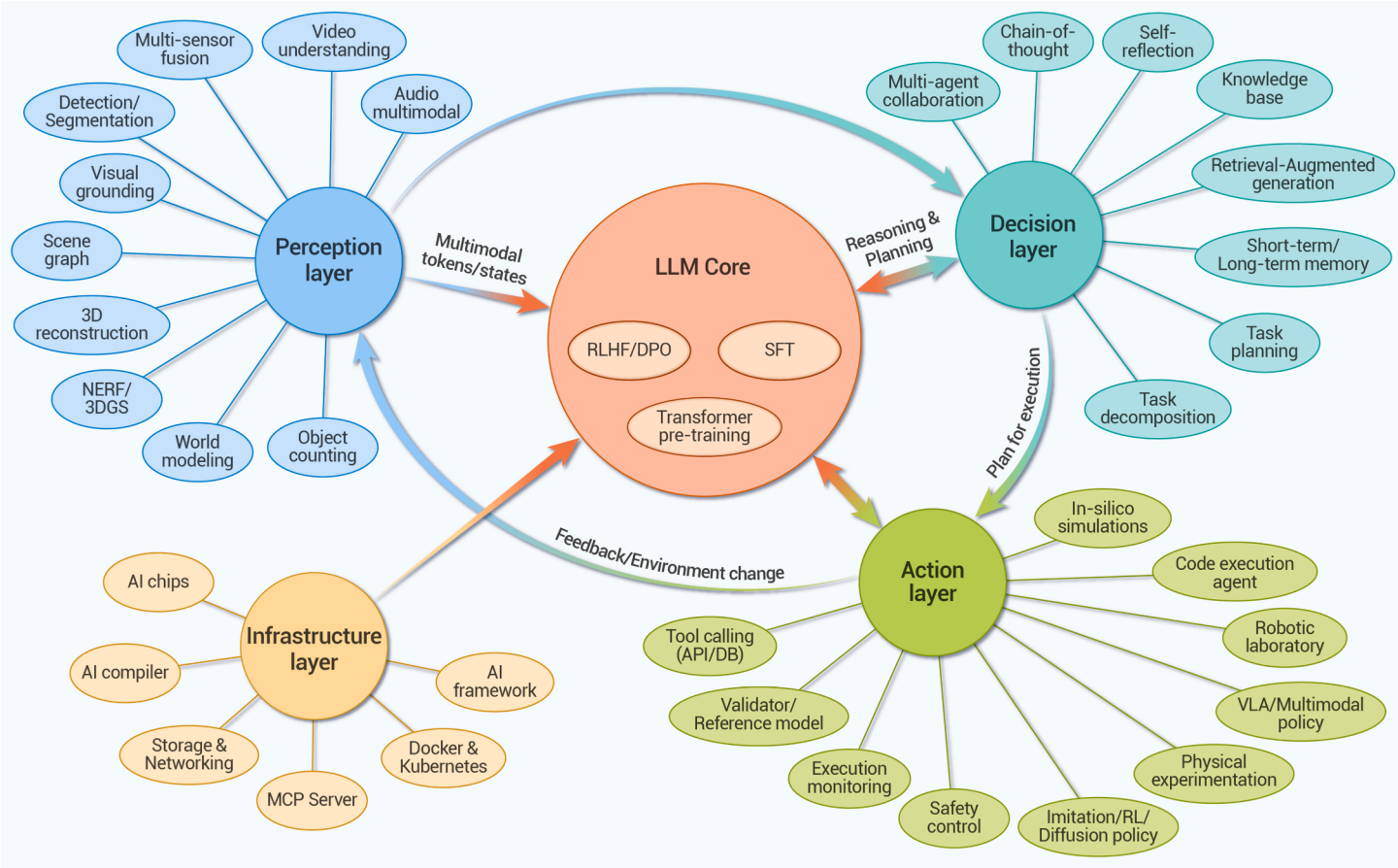

Figure 1. The knowledge graph of the AI framework

the literature discussed below. Table 1 presents representative AI4Science applications across domains in a uniform comparative framework, including typical tasks, major model families, data and benchmark resources, validation maturity, and current limitations. Table 2 further maps these domain-specific applications to the six core modules of the AI4Science workflow introduced above, namely literature research, data analysis, hypothesis formulation, experimental execution, report generation, and iterative autonomous evolution. Together, these two tables are intended to make explicit both the common analytical dimensions shared across disciplines and the uneven maturity of closed-loop AI4Science across fields, thereby providing a clearer bridge between the conceptual framework in "section AI in information science" and the domain-specific reviews that follow.

AI IN INFORMATION SCIENCE

AI is increasingly reshaping computer science (CS) itself, transforming the way core computing systems are designed, optimized, and maintained. Beyond serving as an application domain, CS has become a primary beneficiary of AI-driven methodologies, giving rise to the emerging paradigm of AI for CS.[51–53] The central objective of this paradigm is to leverage data-driven and learning-based techniques to augment or replace traditional human-designed heuristics in complex computing systems, thereby overcoming long-standing limitations imposed by system complexity, design space explosion, and reliance on expert knowledge.[54,55] This section provides a systematic overview of several mature and influential directions in AI for CS. Specifically, it covers AI-enabled automated chip design (AI for electronic design automation [AI for EDA]), where learning-based search and optimization methods are employed to accelerate and improve architecture design and physical implementation; ML-driven compiler optimization (ML for compilers), which replaces handcrafted optimization rules with models that adapt across programs, platforms, and workloads; self-driving database management systems (self-driving DBMSs), which aim to achieve autonomous configuration, optimization, and fault management with minimal human intervention; and AI-assisted software engineering, including code generation, program understanding, defect detection, and automated testing. Collectively, these directions demonstrate how AI functions as a unifying methodology that fundamentally redefines the design and optimization principles of computer systems, positioning AI for CS as a critical pillar of the broader AI4Science landscape.[27,56]

AI for EDA

As the physical scaling of transistors approaches the Angstrom era, traditional EDA algorithms face intractable computational complexity. Over the past 3 years, the research community has pivoted toward AI-driven methodologies to augment or replace conventional heuristics. This review categorizes recent breakthroughs into three domains: linguistic hardware synthesis, differentiable physical design, and autonomous verification agents.

LLM-based hardware description and synthesis. The most significant trend since 2023 has been the integration of LLMs into the register-transfer level (RTL) generation and verification phases. Researchers have developed domain-specific models, such as ChipLlama[57] and VeriGen,[58] which are fine-tuned on Verilog datasets. These models can generate hardware description language (HDL) code from natural-language specifications and assist in debugging complex logic. By 2025, the focus had shifted toward "agentic EDA," where LLMs such as ChatEDA[59] act as controllers that orchestrate traditional EDA scripts (Tcl/Python), effectively lowering the barrier to entry in IC design and reducing human-induced errors in the front-end flow.

Graph learning and physical design optimization. In the back-end flow, where spatial relationships are critical, graph neural networks (GNNs) and RL have achieved breakthroughs in placement and routing. Traditional algorithms often struggle with the massive search space of modern 2 or 3 nm nodes. Recent advances utilize GNNs to predict congestion and timing violations before the actual routing takes place, significantly speeding up design space exploration (DSE).[60] Furthermore, Google's latest iterations of RL-based floor planning have demonstrated that AI can achieve human-level or even superior power-performance-area (PPA) results in a fraction of the time required by manual iterations.

Autonomous verification and formal methods. Verification remains the most resource-intensive phase, often consuming 70% of the design cycle.

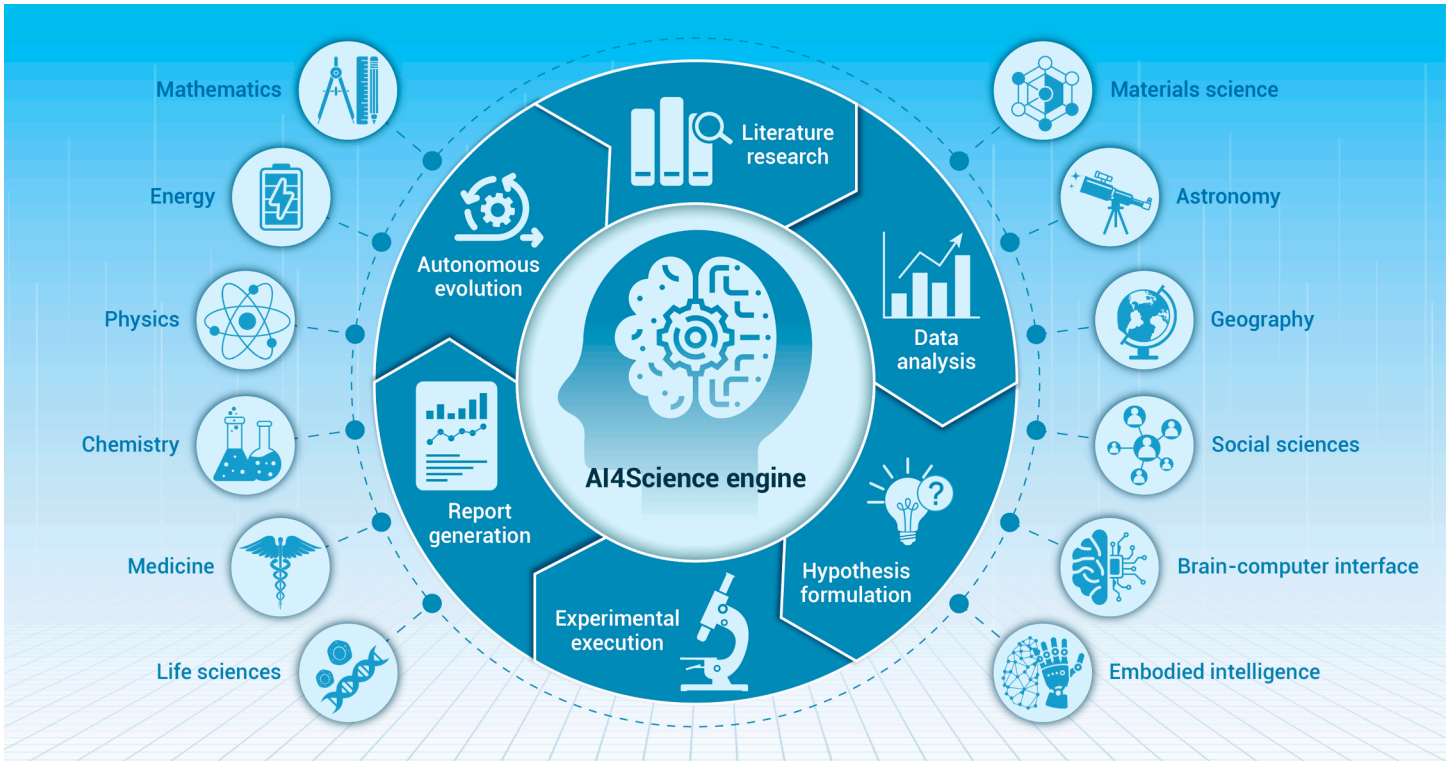

Figure 2. AI as a new engine of scientific discovery

Between 2024 and 2025, the emergence of AssertLLM[61] and similar multi-agent frameworks has revolutionized formal verification. These systems autonomously extract structural information from design specifications to generate comprehensive verification assertions. By closing the loop between RTL generation and automated testbench creation, these AI-driven systems can identify corner-case bugs that traditional simulation might miss. This shift toward "simulation-light" verification through AI surrogate modeling is becoming a standard in analog and mixed-signal (AMS) design, where SPICE simulations are prohibitively expensive.

AI-driven compiler optimization

Compiler optimization is a foundational problem in CS and a critical enabler of scientific computing. It determines how efficiently algorithms are executed and how effectively hardware resources are utilized. Conventional compilers rely on manually designed heuristics to guide optimization decisions such as phase ordering, inlining, and resource allocation. These heuristics encode expert knowledge accumulated over decades and have proven effective for stable architectures and predictable workloads.[27] However, they are inherently static. They do not adapt easily to rapidly evolving processor designs, deep memory hierarchies, or heterogeneous computing platforms. As a result, manual tuning remains necessary in many scientific applications. This growing gap between system complexity and heuristic design has become a structural limitation rather than an implementation detail. In this context, AI introduces a fundamentally different approach. It treats compiler optimization as a data-driven process that can be continuously refined through empirical evidence and performance feedback.[62]

ML reframes compiler optimization decisions as prediction and control problems.[63,64] Instead of encoding fixed rules, learning-based compilers infer optimization strategies from program features, runtime behavior, and hardware characteristics.[65] A seminal synthesis by Wang and O'Boyle established this perspective and demonstrated that learned models can outperform hand-crafted heuristics across architectures and workloads.[66] This work marked a clear conceptual shift. Compiler optimization was no longer viewed solely as an algorithmic design problem but as an empirical process grounded in observation and measurement. Subsequent research consolidated this direction. A comprehensive survey in *ACM Computing Surveys* showed that compiler autotuning using supervised learning and RL provides a principled framework for addressing long-standing challenges such as phase ordering, optimization selection, and performance portability.[67] These studies reposition compiler optimization as an experimental science. Optimization strategies are evaluated, compared, and refined based on evidence rather than fixed assumptions.

Recent systems illustrate how this perspective translates into practical compiler infrastructures. MLGO integrates RL policies into production compilers and replaces selected heuristic-driven passes with learned decision models.[68] As more programs are compiled and evaluated, optimization strategies improve without manual intervention. This capability represents a shift from static compiler expertise to accumulated system knowledge. At the same time, standardized environments such as CompilerGym enable reproducible experimentation by exposing compiler behavior as a learning problem.[69] These environments support systematic comparison of optimization strategies and accelerate progress by lowering the barrier between AI research and compiler development. For scientific computing, this shift is particularly significant. Performance portability across CPUs, GPUs, and emerging accelerators remains a central challenge. Learning-based compilers reduce the need for manual tuning and allow computational systems to adapt to evolving scientific workloads. This role aligns with broader perspectives that view AI as an enabling infrastructure for scientific research rather than a domain-specific tool.[70] In this sense, AI-driven compiler optimization exemplifies how AI reshapes core computing foundations to support scalable and adaptive scientific inquiry.

AI for self-driving DBMS

Self-driving DBMSs aim to build a closed-loop system capable of autonomous optimization, management, and operation, thereby minimizing or even eliminating reliance on human database administrators (DBAs).[71,72] Among the foundational frameworks in this field, NoisePage proposes a three-tier self-driving architecture, including workload forecasting, behavior modeling, and action planning.[73] NeurDB further embeds AI workflows (training, inference, and fine-tuning) into the database architecture, enabling the system to dynamically adjust to changing workloads.[74]

Learned query optimization serves as a critical component of self-driving DBMSs. Deep RL (DRL), which combines the perception capabilities of DL with the decision-making power of RL, is particularly suitable for sequential decision problems such as parameter tuning, index recommendation, and query plan selection.[46,75] Along this direction, Bao pioneers a practical paradigm for learning-based query optimization by selecting optimal combinations of query

Table 1. Cross-domain comparison of representative AI4Science applications discussed in this review

Domain	Representative tasks	Model families	Data/benchmark	Validation level	Main limitations
Information science	EDA; compiler optimization; database management; software engineering; networking	LLMs; RAG; agents; RL	code repositories; logs; code benchmarks; system traces	(A−) to (B−)	verifiability[a]; data leakage; cross-project robustness; need for human review
Mathematics	theorem proving; conjecture and method discovery; auto-formalization; neural PDE/ eigenvalue solving	LLM-proof-assistant agents; RL/search/evolutionary methods; PINNs; neural operators	Lean/mathlib; miniF2F and Olympiad sets; proof corpora; PDE/operator benchmarks	(B−) to (C+)	library and evaluator dependence; formalization bottlenecks[a]; limited long-horizon reasoning; high compute cost
Physics	event reconstruction; triggering/ control; detector surrogates; inverse modeling; simulation acceleration; anomaly detection; semi-automated experiments	GNNs and geometric DL; CNNs/transformers; GANs, diffusion models, and flows; self-supervised models; RL/BO; ML potentials; agents	detector streams; scattering, spectroscopy, microscopy, and electronic-structure data; many-body/atomistic simulations; selected experiments	(A+) to C	high-throughput rare events; multi-scale inverse problems; non-ideal experiments; strict demands on physical consistency,[a] uncertainty, and real-time reliability
Chemistry	molecule generation; interatomic interaction modeling; reaction prediction; synthesis planning; catalyst/drug design	GNNs; equivariant models; transformers; diffusion/ foundation models; agentic workflows; LLMs	molecular and reaction databases; *in silico* screening benchmarks	(B−) to C	synthesizability[a]; inconsistent benchmarks; limited wet-lab confirmation; data bias
Materials science	property prediction; structure/ material generation; performance optimization; processing control; autonomous materials discovery	GNNs; transformers; generative models; Bayesian optimization; lab agents	materials databases; multimodal materials datasets; crystal/property benchmarks	B to C	data quality; stability and synthesizability[a]; lack of unified metrics; limited multi-objective optimization
Energy	battery/fuel-cell optimization; materials screening; system control and forecasting	physics-informed DL; GNNs; surrogate models; RL; generative design models	simulation data; operational data; materials and energy benchmarks	(B−) to (C+)	safety constraints[a]; long-horizon validation; multi-objective trade-offs; fragmented benchmarks
Geoscience	hydrological forecasting; subsurface characterization; geohazard assessment; mineral prospectivity mapping	foundation models; vision-language models; PINNs; neural operators; GANs/diffusion; GNNs; LSTMs; physics-ML hybrids	satellite imagery; station observations; SoilGrids; geophysical/geochemical surveys; mineral datasets	(A−) to (C+)	non-stationarity[a]; sparse subsurface/cryosphere data; physical consistency; uncertainty under regime shifts; cross-scale transfer; high emulation cost
Life science	protein/genome/cell modeling; synthetic biology; pathway design; smart breeding; ecology monitoring	sequence/protein LMs; structural diffusion; GNNs and causal models; multimodal foundation models; closed-loop agents	omics and single-cell atlases; sequence/structure repositories; screening and biofoundry logs; field, remote-sensing, and eDNA data	(B−) to (C−)	limited wet-lab throughput; noisy heterogeneous data; weak standards; causal/mechanistic gaps[a]; generalization, biosafety, privacy, and auditability
Medical science	clinical decision support; imaging; intelligent surgery; drug-response simulation; digital twins; target/ molecule design; TCM/acupuncture robotics; risk and wearable monitoring	multimodal LLMs and VLMs; embodied agents; GNNs and causal models; diffusion models; protein/ chemical LMs	EHRs; medical images; multi-omics; protein structures; trials; wearables; TCM corpora; aging cohorts	(B−) to (C+)	opacity; bias and fairness; privacy/ regulatory gaps; hallucination[a]; limited prospective validation; weak causal interpretability
Astronomy	source detection; object classification; transient discovery; inverse modeling; survey analysis; observation planning	self-supervised models; transformers; diffusion models; multimodal learning; agents	raw images; catalogs; telescope streams; simulated and observational benchmarks	(A−) to C	large data volume; low signal-to-noise ratio; domain shift[a]; interpretability; limited follow-up validation
Brain-computer interface	signal enhancement; motor-intent decoding; linguistic decoding	CNNs; GANs; feedforward neural networks	neural signals	A to C	data scarcity[a] and heterogeneity; computational efficiency; energy consumption
Embodied intelligence	planning/control; robotic experimentation; low-altitude and near-space missions; extraterrestrial/deep-sea exploration	VLM/VLA models; Sim2Real; RL; LLMs	synthetic data	(B−) to C	sim-to-real gap[a]; weak integrated world models; long-horizon planning limits; governance and collaboration hurdles
Social science	policy evaluation; causal inference; behavior modeling; inequality/ mobility analysis; public opinion/ media analysis; institutional decision-making; system optimization	causal models; statistical learning; agent-based and network models; LLMs; multimodal learning; RL policy simulation	surveys; administrative records; social media traces; panel data; RCTs; simulation environments	(B−) to C	limited ground truth and causal identification[a]; bias and representativeness; ethics/privacy; weak external validity; difficult real-world experimentation; context dependence

Validation level is coded as A–D according to the degree of experimental or real-world validation. Validation levels: A, experimentally validated or already deployed in real-world workflows; B, prospective or closed-loop prototype with controlled validation; C, retrospective benchmarking, simulation, or *in silico* validation; D, conceptual, policy-level, or vision-oriented discussion; +, validated across many domains; −, validated in only a few domains. EDA, electronic design automation; LLMs, large language models; RAG, retrieval-augmented generation; RL, reinforcement learning; PINNs, physics-informed neural networks; GNNs, graph neural networks; CNNs, convolutional neural networks; GANs, generative adversarial networks; BO, Bayesian optimization; ML, machine learning; VLMs, vision-language models; LSTMs, long short-term memory networks; VLA, vision-language-action; EHRs, electronic health records; TCM, traditional Chinese medicine; DL, deep learning; PDE, partial differential equation; RCT, randomized controlled trial.

[a]To enhance readability and address the reality gap between benchmark success and real-world deployment, these core limitations represent the primary structural bottlenecks preventing the transition from *in silico* modeling to closed-loop, real-world execution in each respective domain.

Table 2. Domain-wise synthetic assessment of how AI4Science has been realized across the proposed six-module scientific discovery framework

Domain	Literature research	Data analysis	Hypothesis formulation	Experimental execution	Report generation	Autonomous evaluation
Information science	∘	·	∘	·	∘	∘
Mathematics	∘	∘	·	∘	–	–
Medical science	∘	·	∘	∘	∘	∘
Materials science	∘	·	∘	·	–	∘
Geoscience	∘	·	∘	–	–	∘
Life science	∘	·	∘	–	∘	·
Physics	–	·	∘	∘	–	∘
Chemistry	–	·	∘	·	–	·
Energy	∘	·	∘	∘	∘	–
Astronomy	–	·	∘	·	∘	∘
Brain-computer interface	∘	·	∘	·	∘	∘
Embodied intelligence	·	–	–	–	·	∘
Social science	·	·	∘	∘	·	–

Filled circles (·) indicate modules that are represented by relatively mature and representative AI4Science applications in the corresponding domain. Open circles (∘) indicate modules that are present in an emerging, partial, or early-stage form. Dashes (–) indicate modules that are not yet a major or established focus in that domain or are not supported by sufficient representative evidence in the present review.

hints on top of existing optimizers.[76] Similarly, openGauss incorporates a learning-based optimizer within its database kernel that can rewrite slow SQL queries into equivalent high-performance queries.[77] To address the performance regression issue in learning-based optimizers, Lero introduces a lightweight learning-to-rank approach: since query optimization only requires the relative order of plans rather than precise cost estimation, this method achieves more robust performance.[78]

Automatic configuration tuning has evolved from Bayesian optimization to LLM-driven approaches. GPTuner systematically integrates LLM knowledge into database tuning.[79] Its knowledge preparation pipeline automates the collection, cleaning, and integration of tuning knowledge, while the prompt ensemble algorithm unifies structured knowledge representations. Furthermore, the coarse-to-fine Bayesian optimization framework enables efficient and rapid convergence. Beyond tuning, D-Bot extends LLM capabilities to database diagnosis, utilizing tree-of-thought reasoning for root cause analysis and supporting multi-LLM collaborative diagnosis.[80] Taking a more holistic approach, the DB-GPT framework integrates knowledge base retrieval, tool invocation, and diagnostic report generation, providing a comprehensive solution for LLM-driven database management.[81]

Future research in self-driving DBMSs exhibits trends toward deeper integration with AI across three dimensions. First, global collaborative optimization will become a key focus. Through unified objective functions and coordination mechanisms, various components contribute to making collaborative decisions, thereby minimizing strategic conflicts. Second, explainability and controllability urgently need enhancement. As technologies such as LLMs become integrated, systems may not only provide optimization decisions but also offer understandable justifications. Third, generalizability and transferability are essential to reducing deployment costs. By leveraging meta-learning and transfer learning techniques, systems can quickly adapt to new database environments or workload patterns with minimal samples, reducing the burden of retraining models for different scenarios. In summary, future self-driving DBMSs will evolve toward being more collaborative, transparent, and versatile. The ideal form is not a complete "black-box automation" but a system that achieves a high degree of autonomy. It should maintain explainable decision-making processes and preserve the possibility of human supervision, ultimately enabling efficient, reliable, and trustworthy intelligent data management.

AI for software engineering

Across the software engineering life cycle, LLMs are moving from one-shot text assistants to workflow components embedded in grounded and verifiable pipelines. Their strongest gains come when generation is bound to project context (artifacts, repositories, specs, and logs), constrained by explicit gates (traceability checks, compilation, tests, and static analysis), and improved through execution feedback loops. Under this paradigm, LLMs support (1) requirements structuring, (2) code development and generation, (3) testing and quality assurance with correctness-first metrics, and (4) maintenance and operations via repository-grounded repair and incident response. The subsections below summarize this "ground-constrain-verify" pattern across the four stages.

LLMs are emerging as "requirements copilots" that help analysts transform messy stakeholder input into structured artifacts such as candidate user stories, acceptance criteria, and trace links. A recent systematic literature review of LLMs in software engineering synthesizes early evidence that LLMs can support requirements elicitation and analysis tasks—summarization, classification, conflict detection, and traceability—while also highlighting reliability, evaluation methodology, and data-leakage risks that are particularly acute in requirements engineering (RE) settings.[82] To move from promising demos to dependable practice, Ferrari and Spoletini propose a two-way roadmap: RE can provide formalisms and verification hooks that constrain LLM outputs, and LLMs can automate labor-intensive RE activities when embedded in checkable workflows.[83] Beyond direct text processing, language model representations can be used to mine operational logs and user interactions for latent intent and constraints; for instance, LLM-based modeling of complex real-world route optimization demonstrates how sequential decision data can be cast into language-like structures for downstream analysis that is analogous to extracting requirements from behavior.[84] Finally, governance is not optional: evidence on LLM-assisted scientific writing shows that fluent outputs can still be incorrect or misleading, motivating provenance, review gates, and accountability mechanisms throughout the RE life cycle.[85]

For code development and generation, LLMs are most effective when integrated into closed-loop engineering pipelines that couple synthesis with execution feedback, testing, and automated review. AlphaCode demonstrated competition-level program synthesis by generating many candidate solutions and selecting among them via filtering and clustering, illustrating that "generate and rank" can overcome the brittleness of single-pass generation.[86] Extending this idea, FunSearch shows that pairing LLM sampling with automated evaluation can yield new mathematical constructions, reinforcing a general software engineering pattern: LLMs become more reliable when they operate inside program search or repair loops with objective scoring signals.[87] In applied settings, LLM-orchestrated tool use is increasingly prominent; for example, LLM-driven AutoML workflows can call external tools to structure tasks end to end, offering a transferable blueprint for integrating code generation with build systems, test harnesses, and CI/CD gates rather than relying on raw text outputs.[88] Domain

reviews also emphasize that LLM gains often come from retrieval and structured knowledge integration, suggesting that production-grade coding assistants should bind generation to project-specific context (APIs, style guides, and invariants) and enforce verification as a first-class step rather than an afterthought.[89]

For testing and quality assurance, earlier work often prioritized generating more tests (or higher coverage) but struggled to consistently satisfy engineering-grade correctness gates such as compilability and valid assertions. More recently, LLM-based research has made this gap explicit: an FSE study reports that LLM-generated unit tests can resemble developer-written artifacts in readability/usability yet frequently fail compilation or contain faulty assertions, motivating validate-and-fix loops that iterate with execution feedback to reach practical quality.[90] To align evaluation with acceptance criteria, an ASE paper argues that compilation success and assertion validity should be treated as first-class metrics rather than secondary considerations.[91] In parallel, methods are expanding to harder tasks and more realistic settings: differential prompting improves the chance of finding failure-inducing tests by steering the model toward subtle differences between buggy and correct versions[92]; dependency-aware API testing from OpenAPI specifications addresses operation sequencing and parameter constraints[93]; and empirical work on web-form test generation clarifies real-world capabilities and limitations and suggests reusable strategies.[94] Industry efforts also increasingly emphasize improving existing tests under filtering and verification constraints for measurable gains,[95] while flaky-test repair benefits from structured guidance such as fix-category prediction to reduce irrelevant edits and improve targeted repair.[96] Overall, the direction is shifting from quantity-driven generation to dependable testing under verifiable gates and feedback loops.

For software maintenance and operations, earlier automation tended to rely on localized patches and brittle heuristics, making it difficult to handle multi-file changes and CI-style verification. More recently, LLM-era systems have become increasingly repository grounded and pipeline friendly. InferFix exemplifies this transition by combining static analysis (localization/categorization) with retrieval-augmented prompting so that patch generation is guided by concrete evidence and fits CI workflows.[97] Meanwhile, D4C questions whether infilling-style repair aligns with decoder-only LLM objectives and reframes APR as completion-style function refinement guided by artifacts such as failing tests and documentation.[98] Evaluation has also moved beyond snippets: SWE-bench anchors assessment in real GitHub issues and repositories, forcing methods to handle multi-file edits and executable verification loops,[99] and SWE-agent shows that agent-computer interfaces enabling repository navigation, file editing, and running tests are crucial for reliable end-to-end maintenance.[100] Upstream, LLMs can support test-free fault localization when coverage is unavailable, strengthening "locate-then-fix" pipelines.[101] In operations, LLMs increasingly generate actionable artifacts: recommending incident-specific KQL diagnostic queries from historical incident data,[102] improving log parsing via adaptive caching for consistency and cost control,[103] and organizing telemetry into automated RCA narratives and reports to accelerate incident response.[104] In short, the recent trend is to embed LLMs into constrained, executable workflows with feedback rather than rely on one-shot textual assistance.

AI for networking

The rapid advancements in AI are driving a fundamental transformation in networking, heralding an era of AI-integrated networks and a future internet. Traditional network optimization, regulated by mathematical models, often falls short in addressing complex scenarios with many imperfect and highly dynamic network environments, such as unmanned aerial vehicles (UAVs) and satellite networks. In contrast, data-driven and AI-enabled approaches have demonstrated superior performance by efficiently approximating computationally intractable algorithms, capturing nonlinear and unknown network patterns, and significantly accelerating traditional model-based approaches.

Conventional AI technologies, including DL and DRL, have become widespread in various research areas of networking, facilitating advancements in network feature extraction and intelligent decision-making. This ubiquitous impact of AI is observed across all layers of the network stack. More specifically, in the physical layer, DL can be used for interference alignment. It can also be used to classify modulation modes, design efficient network coding[105] and error correction codes, etc. In the data link layer, DL and DRL can be used for resource (such as channels) allocation,[106] medium access control,[107,108] link quality evaluation, and so on. In the network (routing) layer, routing establishment and routing optimization[109] can help to obtain an optimal routing path. In higher layers (such as the application layer), enhanced data compression and task allocation are used. Besides the above protocol stack, one critical area of using DL is network security. Nevertheless, conventional AI technologies rely on data generated by physical devices (e.g., networking devices and network nodes) or specific applications (e.g., mobile gaming) that are only trained on a certain dataset to create mapping abstractions to make highly accurate decisions.

Recently, generative AI (GAI), exemplified by LLMs such as GPT, is a prominent branch of AI capable of creating AI-generated content (AIGC). These models can engage in conversational interactions with humans and generate novel data based on input and contextual information. With billions of parameters, GAI models are adept at processing vast quantities of network data to extract complex behavioral patterns. The integration of GAI into networking is anticipated to significantly improve existing AI-enabled algorithms, especially in areas such as context-aware generation, adaptive reasoning, and dynamic decision-making.[110,111] GAI applications extend to critical domains such as network management, network routing, and network security.[112] For instance, in physical-layer communications, GAI enhances or complements AI models for tasks including signal detection and physical-layer security.[113] Furthermore, GAI can boost the efficiency of network routing by simulating, creating, and analyzing synthetic network scenarios, thereby optimizing routing algorithm selection.

While GAI offers substantial opportunities in network design and optimization, it also encounters several significant challenges, notably regarding latency and real-time performance.[114] For instance, GAI models must drastically reduce inference times to satisfy the low-latency demands of critical network applications such as autonomous vehicles and industrial control. Moreover, in resource-constrained edge and mobile environments, there is a pressing need for the development of more lightweight and resource-efficient GAI models.

AI IN MATHEMATICS

The relationship between mathematics and AI is fundamentally bidirectional. Mathematics provides the formal language and core methodologies underlying modern AI, including logic, probability, statistics, linear algebra, numerical analysis, and optimization. In turn, AI is increasingly contributing back to mathematics and mathematically grounded sciences through theorem proving, conjecture generation, symbolic reasoning, and scientific computing. Accordingly, this section interprets "AI and mathematics" in this broad reciprocal sense: we review not only how AI assists mathematics and scientific discovery but also how mathematically grounded advances make modern AI systems trainable, scalable, and reliable. From this perspective, optimization is included here not as an isolated engineering topic but as a central mathematical tool for variable AI tasks, including formal reasoning, partial differential equation (PDE) solving, operator learning, and quantum simulation, as illustrated in Figure 3.

In the early stages of AI, mathematical theories and methodologies played a decisive role that extended far beyond basic optimization or approximation theory. While stochastic gradient descent and Taylor expansions provide the practical means for ML, more abstract branches of mathematics—such as linear algebra, information theory, and Bayesian statistics—define the very conceptual boundaries of what a machine could "learn." The formalization of neural networks, for instance, relied heavily on the multi-dimensional calculus and statistical mechanics of the time. During this foundational period, mathematics provided the essential rigor that transitioned AI from a collection of heuristic experiments into a structured scientific discipline.

In recent years, the rapid advancement of LLMs and RL has accelerated the shift toward "AI for math."[115] This evolution has moved from simple numerical computation to complex symbolic reasoning and the generation of novel conjectures. The impact is particularly evident in fields such as number theory, combinatorics, and graph theory. For example, AI systems such as DeepMind's FunSearch[87] have successfully discovered new solutions to the

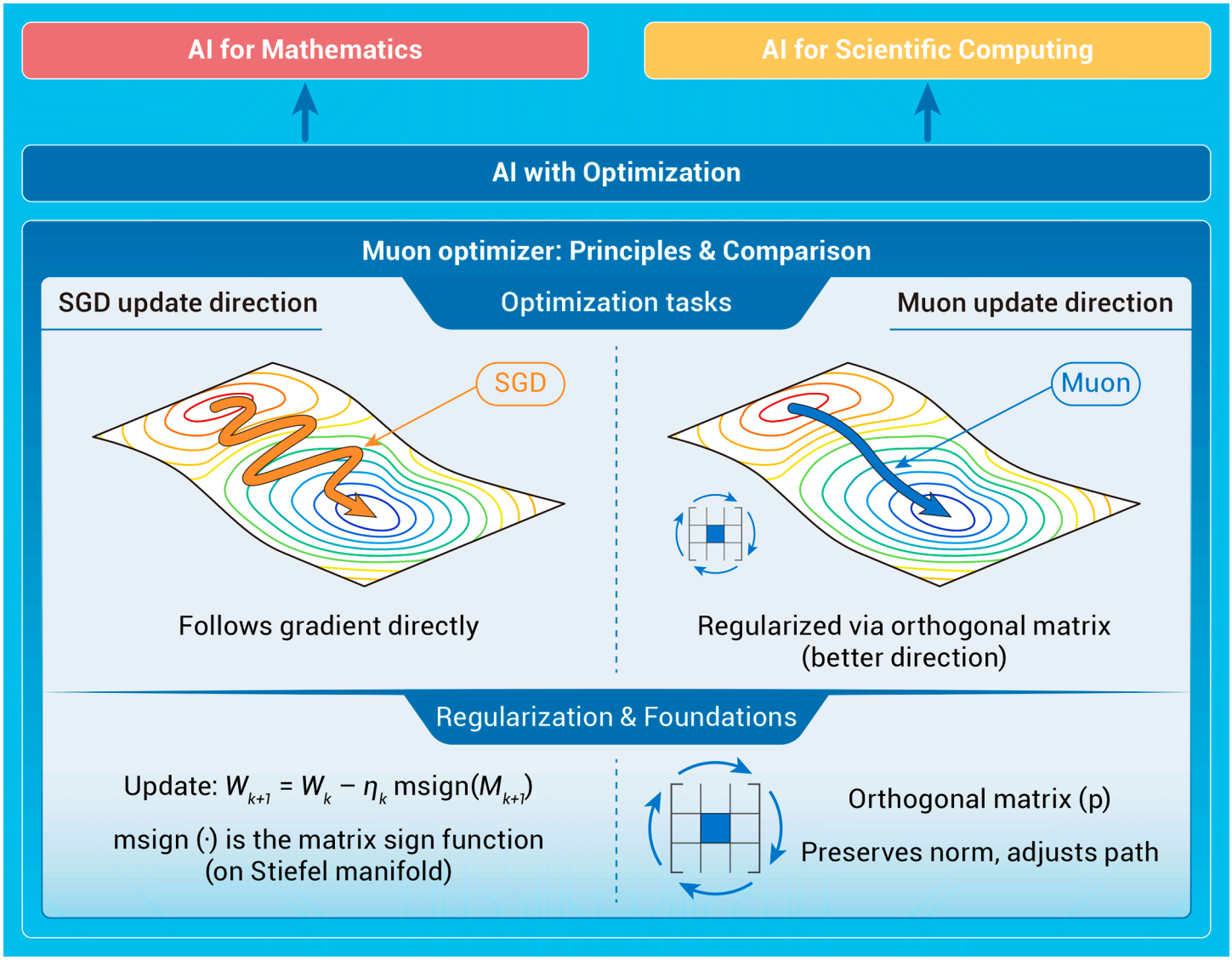

Figure 3. AI in mathematics

cap set problem, surpassing previously known human-established bounds in extremal combinatorics. In the realm of scientific computing and optimization, neural operators and AI-driven solvers are now tackling high-dimensional PDEs that were once computationally intractable.

Furthermore, the integration of formal verification tools with AI "provers" is bridging the gap between intuitive discovery and rigorous proof. By leveraging massive knowledge bases and automated formalization, AI is now capable of identifying deep patterns across disparate mathematical domains, leading to "surprising breakthroughs" that challenge our traditional understanding of the discipline. This paper explores the mechanisms behind this reciprocal empowerment, arguing that the future of mathematical inquiry lies in the seamless integration of human intuition and AI.

AI for mathematical reasoning

AI, particularly LLMs, has begun to fundamentally reshape mathematical research. AI systems can now formally prove nontrivial theorems, generate novel conjectures and algorithms, and assist in translating informal mathematics into machine-verifiable form. It can be emphasized that the role of AI as an accelerator of mathematical discovery will become much more important in future mathematical research directions, even though there are still many limitations.

Recent breakthroughs in AI, especially LLMs and RL, have enabled machines to engage with mathematics at unprecedented levels. Systems such as AlphaProof[116] have achieved medal-level performance on International Mathematical Olympiad (IMO) problems, while LLM-driven search frameworks like FunSearch[87] have produced new results in open combinatorial problems. These developments signal a shift: AI is no longer limited to numerical computation or symbolic manipulation but is becoming an active participant in mathematical reasoning and discovery. Four core components of AI for mathematics should be considered, i.e., (1) automated theorem provers, (2) conjecture and algorithm generation, (3) auto-formalization of mathematical text, and (4) mathematical knowledge bases. We conclude with open challenges and future research directions.

AI-based automated math theorem proving. Automated theorem proving has been a central goal of AI since its inception. Modern systems combine neural models with formal proof assistants such as Lean, Rocq, and Isabelle. AlphaProof[116] represents a major milestone: it integrates an LLM with RL in Lean, enabling automated searches over proof states. AlphaProof solved three out of five non-geometry problems at the IMO 2024, achieving a silver medal performance. Parallel efforts include the DeepSeek-Prover family,[117–119] which trains LLMs on millions of synthetic Lean proofs and further improves performance via RL and subgoal decomposition. The Seed-prover first focuses on deep and broad reasoning for automated theorem proving[120] and further enhances undergraduate-level theorem proving through learning from experience.[121] The Kimina-prover preview version[122] attempts to train large formal reasoning models with RL. More recently, the Intern-S1 model[123] achieved a high score of 102, not only winning the Chinese Mathematical Olympiad (CMO) 2025 gold medal but also far exceeding the gold medal score line of

78 points. Recently, some human-AI interactive theorem proving[124,125] frameworks or agents have been introduced for advancing math research. Despite these advances, current provers remain limited. They require large computational resources, rely heavily on existing formal libraries, and struggle with long or conceptually novel proofs. Fully autonomous theorem proving at the frontier of mathematical research remains an open challenge.

Math conjecture and method discovery. Beyond verification, AI has demonstrated the ability to generate new conjectures and methods. FunSearch[87] combines LLMs with evolutionary search, enabling programmatic exploration of solution spaces. While applied to the classical cap set problem, FunSearch discovered constructions that surpass the previously best-known bounds, marking a rare instance of AI contributing to an open mathematical problem. Considered an extension of FunSearch, AlphaEvolve[49] extends the paradigm to method design. AlphaEvolve rediscovered and improved matrix multiplication algorithms, including a 4×4 multiplication scheme using only 48 scalar multiplications, improving upon Strassen's result. Across dozens of open problems, the AlphaEvolve systems frequently rediscovered state-of-the-art solutions and occasionally produced genuine improvements.[126] These approaches highlight AI's strength in large-scale exploration. However, they depend on well-defined evaluation functions and do not replace rigorous proof. Human interpretation and formal verification remain essential to transform AI-generated hypotheses into accepted mathematical results.

Auto-formalization and proof assistants. A critical bottleneck in AI for mathematics is the translation of informal mathematical language into formal representations. Auto-formalization seeks to bridge this gap. LLMs were first introduced for auto-formalization,[127] and furthermore, the combination of auto-formalization and LLM-driven provers can be considered as the mainstream technique. The Lean Workbook project[128] constructed a large dataset of natural-language problems paired with Lean formalizations, significantly expanding the training data for formal reasoning models. ProofBridge[129] advances this direction by aligning natural-language proofs with Lean code using joint embeddings and retrieval-augmented generation (RAG). By iteratively repairing generated proofs using Lean's type checker, ProofBridge substantially improves the correctness of auto-formalized proofs. Some agents have been established for auto-formalization based on Lean4.[130–134] Despite progress, auto-formalization remains difficult due to the ambiguity in mathematical language and the precision demanded by formal systems. Current tools function best as assistants to expert users rather than as fully autonomous translators.

AI has already proven capable of assisting with complex proofs, generating new methods, and accelerating exploration in mathematics. However, major challenges remain: scaling to deep conceptual reasoning, reducing dependence on massive computation, improving robustness and interpretability, and achieving reliable end-to-end auto-formalization. Future research will likely focus on tighter integration of symbolic reasoning and learning, improved human-AI collaboration, and the expansion of formal mathematical infrastructure. While fully autonomous mathematical discovery remains a long-term goal, current evidence suggests that AI will play an increasingly central role in advancing mathematical science.

AI for scientific computing

Scientific computing traditionally relies on rigorous numerical methods to solve PDEs and eigenvalue problems and to simulate physical systems. These mesh-based classical approaches face the curse of dimensionality and rely on problem-specific discretization. In addition, many scientific computing pipelines involve large-scale optimization as a computational primitive. Such limitations have motivated the exploration of AI techniques as alternatives. In particular, DNNs can serve as flexible function approximators, with a proven capability to approximate functions relevant to PDE and quantum problems,[135–144] offering the potential to bypass some bottlenecks of conventional algorithms. Recent progress suggests that AI-driven methods can be roughly organized into three themes: (1) learning solutions for specific scientific problems, (2) learning general solution mappings applicable across families of problems, and (3) learning-enhanced optimization and solvers. The following paragraphs discuss these two paradigms in turn.

Learning problem-specific neural network solutions. One line of research leverages neural networks as mesh-free solution approximators for individual scientific computing tasks. Here, we introduce physics-informed neural networks (PINNs)[145] primarily as a representative framework for neural PDE solving from the viewpoint of scientific computing and numerical analysis, while their diverse domain-specific applications across AI4Science are left to the corresponding later sections. For PDEs, PINNs incorporate strong-form residuals into the loss and train neural networks to minimize these residuals at collocation points, sharing a similar spirit with the deep Galerkin method (DGM).[146] Alternatively, the deep Ritz method (DRM)[147] and deep energy method (DEM)[148] minimize the variational energy, while the weak adversarial network (WAN)[149] adopts a min-max weak-form formulation with separate trial and test networks. In recent years, neural PDE solvers have advanced in several directions, including numerically inspired architectures,[150–154] improved training strategies such as residual adaptive sampling,[155,156] advanced optimization methods,[157,158] and deeper theoretical understanding.[159–161] For high-dimensional eigenvalue problems, especially in quantum many-body physics, neural-network-based variational Monte Carlo (NN-VMC) has attracted considerable attention.[162,163] From the pioneering use of restricted Boltzmann machines for spin models[164] to neural wavefunctions incorporating permutation-equivariant network layers,[165–168] a variety of architectures have been developed. In parallel, improved sampling techniques[169,170] and quantum natural gradient-based optimization methods[167–173] have reduced variance and accelerated convergence, achieving near-chemical accuracy on small molecules. Moreover, NN-VMC has been extended beyond ground states to excited states,[174] underscoring the flexibility of this approach.

Learning generalized neural network solution mappings. While the above neural solvers are trained separately for each problem instance, an alternative paradigm is to learn models that generalize across a family of related problems. The pioneering Fourier neural operator (FNO)[175] learns operators through spectral convolutions based on truncated Fourier representations, enabling compression and reconstruction of function-space information and achieving efficient, mesh-invariant performance on parametric PDE problems. Another influential framework is deep operator networks (DeepONets),[176] which employ a branch-trunk architecture to encode input functions and spatial coordinates and to learn the corresponding operator mapping. Building on these foundations, subsequent developments such as MgNO,[177] which incorporates multigrid ideas, and MIONet,[178] which extends operator learning to multiple input functions, have further broadened the scope of operator learning to more complex settings.[179] Beyond these neural operators, recent research has begun to integrate physical principles more directly into operator learning. A representative example is the variational physics-informed neural operator (VINO),[180] which trains neural operators by minimizing the energy formulation of PDEs. In simulating physical systems, especially atomistic problems in materials science, transferable neural models serve as data-driven surrogates that generalize across different systems. In Kohn-Sham density functional theory (KS-DFT), neural networks have been used to approximate unknown functional components, including learning the kinetic energy functional via convolutions or graph transformers,[181,182] learning the exchange-correlation functional via operator learning methods,[183] and learning electron densities or Hamiltonians via rotationally equivariant architectures.[184–187] In molecular dynamics (MD), beginning with the seminal deep potential MD (DeePMD) method,[188] pre-trained neural interatomic potentials have become increasingly established across systems[189,190]; with downstream tuning, they can further improve MD accuracy.

Learning-enhanced optimization and solvers

As an important sub-area of scientific computing, optimization has significant benefits from AI. RL has been successfully deployed to solve large-scale linear programming (LP) and mixed-integer programming (MIP) problems. Seminal work by Tang et al.,[191] Huang et al.,[192] Wu et al.,[193] and Wang et al.[194] demonstrated that neural networks can learn to replace computationally expensive heuristics (e.g., branching rules), resulting in solvers that are significantly faster than commercial solvers on specific data distributions. Furthermore, the concept of "learning to optimize" (L2O)[195,196] has gained attention for optimization tasks. The L2O frameworks utilize meta-learning to automatically discover novel update rules parameterized by neural networks. These learned optimizers can capture complex dependencies in the loss landscape that analytical algorithms might miss, offering superior empirical performance for specific classes of optimization problems.

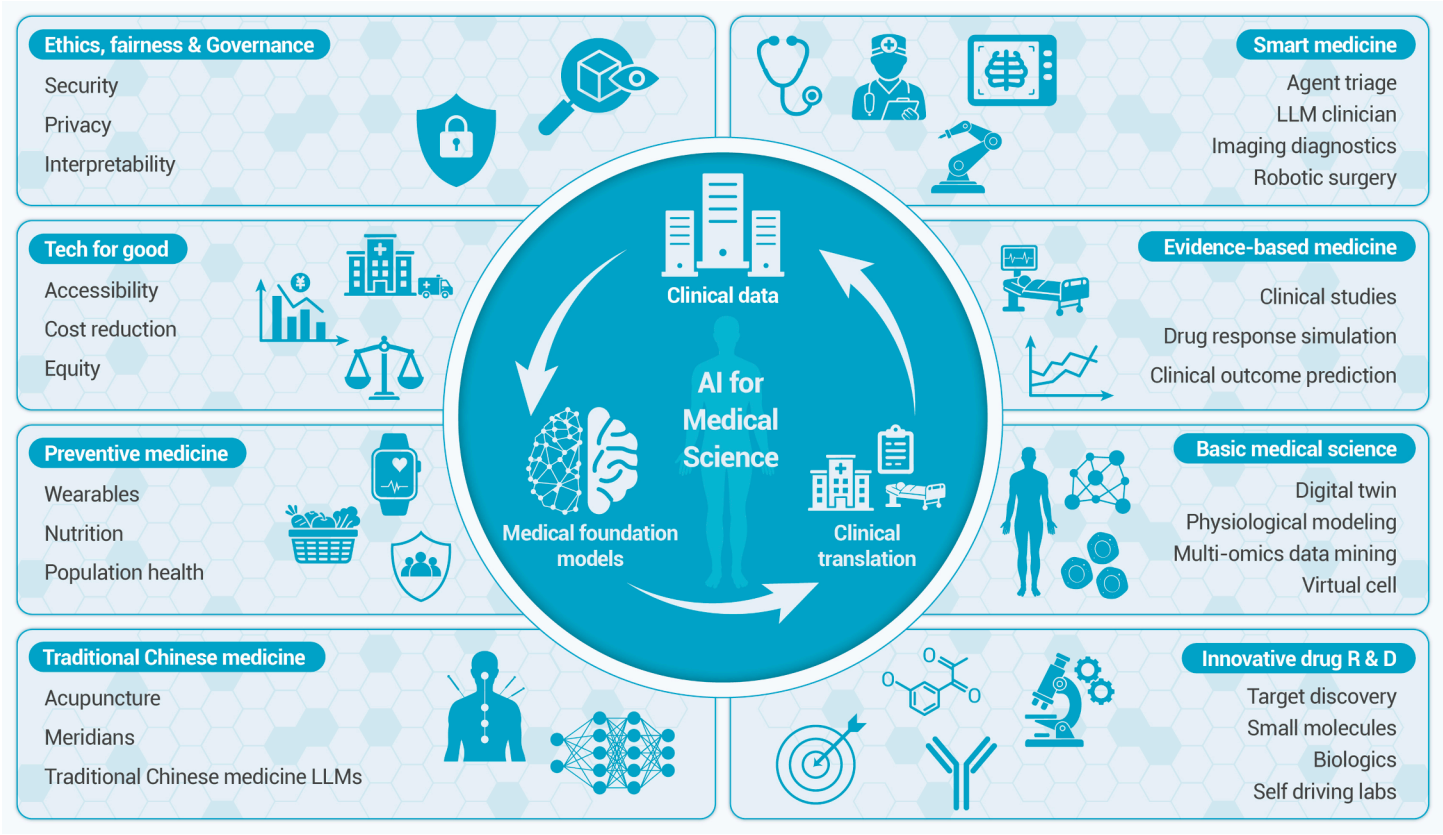

Figure 4. AI in medical science

Despite these advances, AI for scientific computing still faces several fundamental challenges. Model reliability under distribution-shift and out-of-distribution inputs remains a key concern. Generalization is not yet fully characterized from a mathematical perspective, especially when one seeks guarantees compatible with physical constraints, conservation laws, and symmetry structures. Moreover, the practical trade-offs between accuracy and computational cost are still being clarified; for example, scaling laws are important for allocating computing resources effectively. Addressing these issues will likely require tighter connections between numerical analysis, statistical learning theory, and physics-informed modeling principles.

Mathematical optimization enabling AI

Optimization techniques are a cornerstone of modern AI tasks. They often determine whether a model can be trained at all, how quickly training can be carried out, and how reliably it proceeds under stringent computational and statistical constraints. Various important AI tasks, including large models for theorem proving and auto-formalization and PINNs and neural operators for PDEs, require the models to be trained robustly, efficiently, and at scale. In all of these settings, training leads to high-dimensional, stochastic, non-convex, and often anisotropic optimization problems. Therefore, optimization is included here not as a detached survey of existing training algorithms but as a mathematical foundation that directly affects whether AI can succeed in various important tasks, including mathematical reasoning and scientific computing. In this sense, this section presents the mathematics for the AI side of the interaction emphasized throughout this review. We organize the discussion into three representative directions: adaptive moment methods, geometry-aware matrix updates, and convergence theory for automatic-differentiation-based training in nonsmooth settings.

Adaptive moment estimation methods. The efficacy of modern AI is intrinsically tied to the evolution of optimization algorithms capable of navigating high-dimensional, non-convex landscapes. A key difficulty in neural network training is that different parameters exhibit significantly different gradient scales and noise levels, especially under mini-batch settings. Motivated by the need for an optimizer that is robust to ill conditioning and reduces the effort required for manual hyperparameter tuning, ADAM[197] employs element-wise adaptive learning rates through first-order moment and second-order moment estimation, making training significantly more stable across various architectures and tasks. For nearly a decade, ADAM and its variants, such as decoupled weight decay and related adaptive methods,[198–201] have served as default choices for large-scale neural network training. The robustness and efficiency of ADAM in training foundation models were recently recognized with the ICLR 2025 Test of Time Award.

Optimization methods with geometry-aware matrix updates. As mathematical reasoning systems and scientific AI models increasingly rely on transformer-like architectures, it becomes natural to exploit the matrix structure of their parameters rather than treating all coordinates independently. Motivated by the requirement to better exploit matrix geometry and improve scaling behavior, the MUON (momentum orthogonalized) optimizer applies orthogonalized matrix updates that effectively constrain the update directions within the Stiefel manifold.[202,203] By explicitly controlling the spectrum of updates by enforcing the update directions that lie within the Stiefel manifold, MUON addresses the scaling challenges of modern transformers effectively, providing a geometry-aware alternative to purely element-wise adaptive methods (e.g., ADAM and its variants) and showing highly efficient empirical performance.[204–206] A possible future direction is to develop a more unified geometric theory and scalable implementations for MUON and its variants, such as identifying the learning rate scheduling schemes,[204] the efficient computations of orthomornalization to matrices on GPUs, and the analysis for when manifold constraints to update directions could provide guaranteed advantages.

Convergence guarantees from conservative fields. Beyond algorithm design, modern AI also raises new mathematical questions about why practical training works. Classical convergence analysis is often inadequate for real neural network training because automatic differentiation is applied to highly composite objectives and sometimes nonsmooth building blocks, where standard smooth or Clarke-regular assumptions may fail. Motivated by the need for convergence tools that remain valid under nonsmoothness and non-Clarke-regular losses, researchers employ differential inclusion frameworks to analyze the convergence of optimizers in training nonsmooth neural networks.[207–212] Most notably, Bolte et al.[208] introduced the theory of conservative fields, providing a

rigorous foundation for stochastic gradient methods driven by automatic differentiation, overcoming the failure of the chain rule for the Clarke subdifferential. This perspective provides theoretical guarantees for real-world applications of automatic differentiation and DNN training, and its foundational impact was recognized with the 2024 Lagrange Prize in Continuous Optimization. Future progress may focus on employing conservative fields as theoretical foundations for real-world training tasks, such as the convergence of adaptive methods, their corresponding convergence rates, and the generalization analysis for specific optimizers.

More importantly, as most of the models discussed in this review are trained using automatic differentiation, such theory is directly relevant to the mathematical foundations of those applications. It therefore illustrates another way in which AI and mathematics interact: practical AI systems motivate new optimization theory and that theory in turn strengthens our understanding of AI training.

AI IN MEDICINE SCIENCE

AI is fundamentally reshaping the paradigm of modern medicine. This shift moves beyond the enhancement of specific tools, representing a core transition from traditional, hypothesis-driven research toward a data-centric model of "discovery science." AI reconfigures much of the entire pipeline of how medical knowledge is generated, validated, and translated into practice. Figure 4 summarizes major application areas from ethical frameworks and smart medicine to preventive care, traditional medicine, and drug discovery, all supported by foundation models.

Conventional medical research is constrained by the inherent limits of human cognition and the physical, ethical, and temporal boundaries of clinical and experimental systems. In contrast, AI leverages its capacity to analyze complex, multimodal data, including integrating genomic, imaging, and clinical records, to uncover non-intuitive patterns, generate novel mechanistic hypotheses, and enable sophisticated *in silico* modeling. In some contexts, this can enhance the scale and predictive power of medical inquiry, potentially shortening the path from biological insight to therapeutic strategy.

This transformation permeates the entire medical continuum. It changes how we identify clinical knowledge gaps through data exploration, how we investigate disease mechanisms and design treatments via computational simulation, and ultimately how we practice medicine through dynamic risk assessment and personalized care. The following discussion will examine this structural change across critical domains: from ethical frameworks and intelligent diagnostics to evidence generation, drug development, and preventive health. It will further argue that AI serves as a significant force in advancing a future of medicine that is more predictive, precise, and equitable.

Ethical, fairness, and interpretability challenges in clinical AI

Recent advances in AI-driven healthcare promise enhanced diagnostic and predictive capabilities, yet they also raise important ethical concerns regarding privacy, fairness, and explainability.

First, the privacy and ethical challenges associated with medical AI are rooted in the inherently sensitive nature of health data. Protected health information demands rigorous governance to prevent unauthorized access, re-identification attacks, or misuse, yet current legal frameworks often lag behind rapid technological innovation, leaving responsibility for harms ambiguous among developers, clinicians, and institutions (e.g., unclear liability when AI diagnostic errors cause patient harm).[213,214] Inadequate regulatory standards thus can exacerbate risks to patient privacy and trust.[215]

Second, bias and fairness issues emerge when training datasets lack representative sampling. Medical data are seldom random or equitable across demographics—web-based access biases, skewed clinical encounter distributions, and overrepresentation of certain age or socioeconomic groups lead to models that inadvertently reinforce existing health disparities. Under-sampling exacerbates systemic algorithmic bias across gender and racial lines in prediction models; for instance, studies reveal algorithmic biases disadvantaging Black patients via biased data, inappropriate proxies, or racially skewed clinical recommendations.[216–218]

Third, explainability and transparency remain notable barriers to clinical adoption.[219] High-complexity models such as DNNs often function as black boxes, obscuring the rationale underlying specific predictions.[220] In public health, such opacity complicates resource allocation and interventions for outbreaks. Similarly, clinicians consistently report that the black-box nature of many ML models impedes their willingness to adopt AI in clinical practice.[221,222]

Addressing these interrelated challenges highlights the need for inclusive dataset curation, robust privacy-preserving methods, fairness audits, and development of interpretable models, all underpinned by clear regulatory and ethical frameworks to ensure AI promotes equitable, transparent, and patient-centered care.

AI-driven intelligent medicine: Intelligent healthcare agent, medical LLM, and intelligent surgery

For smart healthcare, AI is driving medicine from discrete auxiliary tools toward an integrated ecosystem.[223] Medical agents, empowered by multimodal perception, long-term personalized memory, tool-chain orchestration, and complex logical reasoning, are evolving into cognitive hubs that bridge patients with resources. In the future, they will enable highly personalized and precise prevention and treatment tailored to individual needs. Meanwhile, they will facilitate proactive health management by constructing continuous health profiles, performing dynamic risk assessment, and providing context-aware clinical decision support. This transformation relies on the agent's precise ability to call and understand evidence-based medical evidence.

Recently, Ferber et al.[224] integrated visual foundation models such as MedSAM for medical image segmentation to quantify lesion progression while simultaneously retrieving clinical guidelines and medical knowledge bases in real time. In complex oncology decision-making tasks, this architecture substantially mitigates hallucinations and improves decision accuracy from 30.3% with general LLMs to 87.2%. Similarly, real-world deployments such as Ant Group's "A-Fu" medical agents have demonstrated the enormous potential of GAI in handling heterogeneous medical data and delivering accessible healthcare services, providing important empirical evidence for the clinical translation of digital medicine. The semantic alignment between vision foundation models and LLMs endows medical imaging diagnosis with cross-modal perceptual capabilities. At the macroscopic imaging level, Li et al.[225] proposed BrainGPT to overcome the limitations of conventional 2D slice-based analysis. Through clinical visual instruction tuning (CVIT), the model is enabled to understand 3D cranial CT volumetric data, achieving deep alignment between lesion severity and clinical semantics. At the microscopic pathological level, Tran et al.[226] developed HistoGPT to process gigapixel whole-slide images. By employing cross-attention mechanisms to map visual features into the clinical text space, the model not only generates high-quality pathology reports but also accurately predicts tumor subtypes, tumor thickness, and tumor margins under zero-shot settings.

Beyond decision-making in the digital space, an important direction for smart healthcare involves precise intervention in the physical world. Embodied AI agents are revolutionizing surgical robotics by integrating multimodal perception with precise physical manipulation. The future of intelligent surgery lies in the deep integration of "proactive cognition" and "precise execution." Traditional surgical robots are often constrained by perceptual blind spots in unstructured environments. In contrast, the new generation of surgical robots is evolving beyond the role of passive executors, transforming into embodied intelligent agents equipped with the brains of MLLMs. These embodied intelligent agents can be designed to process complex anatomical environments and dynamic surgical scenes, enabling context-aware responses.

For example, the SUFIA framework developed by Moghani et al.[227] demonstrates language-guided robotic dexterity by integrating advanced perception modules such as LangSAM to accurately localize surgical instruments and tissues, translating natural-language instructions into millimeter-level surgical actions. AI-driven 3D reconstruction systems[228] enhance preoperative planning for lung surgeries, improving anatomical variant identification by 8% and reducing errors by 41%, as demonstrated in a multicenter study.

Similarly, real-time AI-assisted recognition[229] of intra-abdominal metastases during laparoscopic gastric cancer surgery achieves accurate tumor staging and personalized treatment strategies.[230] Even non-electronic innovations, such as slipknot-gauged mechanical transmission, offer resource-efficient force control in robotic operations, improving knotting precision by 121% for novice surgeons.[231] These advances foster autonomous surgical agents

capable of adaptive decision-making in dynamic environments, minimizing human error and optimizing outcomes. Future developments will combine AI cognition with mechanical intelligence, paving the way for equitable access to advanced care.

Evidence-based medicine: AI-assisted clinical research, drug response simulation, and treatment effect prediction

Traditionally grounded in systematic reviews and randomized controlled trials, evidence-based medicine increasingly struggles with efficiency, individualization, and timely updating in the face of heterogeneous data, disease complexity, and evolving clinical decision demands.[232,233] AI has the potential to gradually reshape evidence-based medicine from a retrospective, static model toward a more computable and continuously updatable paradigm.[234,235]

AI is increasingly evolving into an end-to-end methodological framework for clinical research, enabling multimodal integration of electronic health records (EHRs), imaging, omics, and wearable data to predict high-risk subgroups, disease trajectories, and intervention windows.[236,237] Within paradigms such as target trial emulation, AI is being explored to refine eligibility criteria, outcome definitions, and follow-up strategies, improving generalizability while reducing trial time and resource demands without compromising rigor. These applications remain largely at the stage of methodological development and single-center validation rather than established practice.

Drug response simulation and treatment effect prediction represent another major frontier. GNNs, causal inference methods, and generative models aim to integrate drug-target interactions, signaling pathways, clinical phenotypes, and endpoint outcomes into unified computational frameworks. These approaches support a shift from population-level average treatment effects toward individualized treatment effect prediction. Counterfactual inference-based simulation is currently best regarded as an exploratory tool for pre-evaluating alternative dosing regimens, therapeutic strategies, and risk-benefit trade-offs prior to real-world studies or clinical trials, with prospective clinical validation still limited.[238,239]

At the same time, AI-assisted diagnostic and reasoning workflows should not be equated with evidence itself. Improvements in model performance metrics do not necessarily translate into better patient outcomes. Risks such as bias propagation, human-AI interaction effects, and broader systemic consequences require rigorous validation through prospective, multicenter studies, real-world evidence generation, and randomized controlled trials.[240,241] Consequently, evidence-based medicine must also evolve, moving from a static hierarchy of evidence toward adaptive validation and governance frameworks capable of continuous evaluation, dynamic recalibration, and clear delineation of clinical applicability boundaries for AI models.[242]

AI for digital twins and medical data mining

In basic medicine, AI is revolutionizing the simulation of biological systems through digital twins—virtual replicas of living organisms or cells that enable predictive modeling and hypothesis testing. These tools integrate multi-omics data with ML to simulate life processes, bridge gaps in experimental biology, and accelerate discoveries in disease mechanisms and drug exploitation.

Digital twins for life simulation create dynamic, personalized models of physiological systems. Bunne et al.[243] outlined the framework for building AI virtual cells (AIVCs), leveraging multimodal AI to simulate molecular, cellular, and tissue behaviors across scales. These frameworks enable virtual laboratories that support *in silico* experimentation and prediction of cellular responses to perturbations, addressing nonlinear dynamics and multi-scale interactions through neural network integration of multi-omics data. Virtual cells, a core component of digital twins, model intracellular processes and responses. Liu et al.[244] introduced UNICORN, a multi-task learning framework that predicts gene expression from DNA sequences using embeddings from genomic language models. Pre-trained on single-cell data, UNICORN captures cell-type-specific phenotypes and uncertainties, outperforming Enformer in multi-omics predictions. He et al.[245] proposed Squidiff, a diffusion-based model for forecasting transcriptomic changes during cell differentiation and perturbations. Furthermore, Johnson et al.[246] advanced virtual cells with a human-interpretable grammar for encoding multi-cellular models, using plain-language rules to simulate tumor microenvironments and neural development. This grammar integrates multi-omics data for personalization, enabling virtual experiments that link theory to empirical observations.

Beyond mechanistic modeling, AI-driven data mining enables scalable biological inference. Valanarasu et al.[247] developed GigaTIME, a multimodal AI that generates virtual multiplex immunofluorescence profiles from H&E slides, creating a virtual cohort of 299,376 slides across 24 cancers and uncovering 1,234 associations among proteins, staging, and survival. Ding et al.[248] proposed TITAN, a whole-slide foundation model pre-trained via vision-language alignment, enabling rare cancer retrieval, zero-shot classification, and report generation in low-data settings.

These advancements highlight AI's paradigm shift in medicine research, enabling scalable, interpretable simulations that inform personalized therapies and biological discovery.

AI-driven innovative drug discovery

The integration of AI into pharmaceutical R&D has contributed to a noticeable evolution away from purely serendipitous discovery toward more rational, data-driven design. This transformation permeates the entire drug development life cycle, fundamentally reshaping the efficiency and success rates of novel therapeutics.

Target discovery: Synergistic decoding of disease biology through AI and multi-omics. AI algorithms now synergize with multi-omics technologies to decode complex disease biology. Open Targets Genetics[249] and the reimagined Open Targets Platform[250] integrate genome-wide association study (GWAS) fine-mapping, functional genomics, and drug/biology evidence to prioritize causal genes and target-disease associations. Deep sequence models further sharpen mechanistic hypotheses by predicting gene regulation from noncoding DNA (e.g., Enformer[251]) and by enabling scalable variant interpretation, which helps connect patient genotypes to actionable pathways. In parallel, the 2021 breakthrough in protein structure prediction (AlphaFold2[6]) and its rapid scaling to proteome-wide coverage[6] unlocked structure-based workflows for many previously "undruggable" proteins, while interaction-focused models (AlphaFold 3[252]) extend prediction to multicomponent complexes including proteins, nucleic acids, and small-molecule ligands. Together with fast single-sequence predictors (e.g., ESMFold[253]) and improved multimer modeling (e.g., RoseTTAFold[254]), these tools have shifted early discovery from sparse experimental structures to an abundance of computable 3D hypotheses. This enables rapid pocket assessment, target tractability analysis, virtual screening, and rational design of perturbagens for PPIs, RNA-binding proteins, and allosteric sites while also supporting evidence triangulation across structural bioinformatics, pathway models, and real-world patient data.

Molecular design: From predictive screening to generative, 3D-consistent rational design. In terms of molecular design, generative and foundation models are moving from predictive screening to generative design with an explicit focus on 3D consistency and multi-objective constraints. For small molecules, 3D-aware pre-training (e.g., Uni-Mol[255]) and SE (3)-equivariant diffusion models[256] support conformation generation and property-conditioned sampling, while diffusion-based docking (DiffDock[257]) reframes pose prediction as probabilistic generation with calibrated confidence—useful for prioritizing synthesis when experimental capacity is limited. Pocket-centric generation (e.g., PocketGen[258]) further expands data augmentation by proposing realistic binding pockets to stress-test design algorithms. Beyond binding, modern pipelines couple generators to fast predictors for solubility, permeability, metabolic stability, hERG/CYP liabilities, and off-target risk, increasingly augmented with physics-based scoring (e.g., MD/FEP surrogates) to reduce late-stage attrition. Chemical language models (MolGPT[259] and ChemGPT[260]) and emerging MLLM systems are being used for scaffold hopping, reaction prediction, and retrosynthesis-aware optimization, increasingly paired with large-scale retrosynthesis planners (e.g., RSGPT[261]) to reduce synthesis feasibility risk earlier in the DMTA loop. For macromolecular therapeutics, ProteinMPNN[262] provides scalable sequence design for given backbones; RFdiffusion[263] demonstrates *de novo* backbone generation and binder/enzyme scaffolding; and diffusion-based pipelines now cover antibodies, peptides, and engineered proteins[264,265] when coupled to experimental selection. Critically, these model classes are converging with automation: autonomous robotic chemists and lab-in-the-loop active learning (including transcriptomics-driven phenotypic campaigns[266]) enable rapid DMTA cycles with fewer wet-lab iterations and better exploration-exploitation balance. The pharmaceutical industry can further enhance model generalizability and development efficiency by combining AI platforms with

cross-institutional data-sharing frameworks, thereby accelerating the entry of AI-designed antibodies, fusion proteins, and other complex biologics into preclinical pipelines. Representative progress has already emerged in early clinical translation. For example, PandaOmics integrates multi-omics data and biological network analysis to identify TRAF2-NCK interacting kinase (TNIK) as a potential antifibrotic target, facilitating the development of a selective TNIK inhibitor (INS018_055), which has progressed into phase 2 trials for idiopathic pulmonary fibrosis.[267] AI also supports downstream development by learning exposure-response relationships, enabling adaptive trial enrichment and biomarker-driven patient stratification from real-world data and digital endpoints, and by generating regulatory-ready documentation artifacts (model cards, data lineage, and audit trails). Looking forward, the bottleneck is shifting from model capacity to rigor: prospective and blinded benchmarks, uncertainty quantification, physics-informed constraints, and regulatory-grade traceability will determine which AI-designed candidates reproducibly achieve clinical success.

AI across the full R&D continuum: Bridging basic research and clinical translation. ML, DL, GNNs, and natural-language processing methods have now systematically penetrated the entire drug R&D continuum, substantially improving predictive accuracy and screening efficiency at key stages including target identification, virtual screening of lead compounds, ADMET property prediction, and clinical trial design. AI agents built around LLMs further enable deep integration of heterogeneous tools and modules across the workflow, shifting drug development from isolated local optimizations toward coordinated, process-level collaboration, thereby increasing overall success rates.

On this basis, AI is driving a more structured and continuous linkage between basic research and clinical translation. Self-driving laboratories (SDLs) integrate AI agents with robotic automation to execute closed-loop design-make-test cycles, accelerating lead optimization. At the clinical stage, AI-enhanced digital twins, patient stratification strategies, and real-world evidence analyses are improving the design and execution of clinical trials, with early successes already reported in AI-guided precision oncology studies.

Collectively, these technologies hold potential to reduce the cost and timeline of drug discovery, and some observers suggest they may contribute to what could be called an era of programmable medicines.

AI application trends in TCM and acupuncture: Digital empowerment and barrier-breaking

The integration of AI and digital technologies into traditional Chinese medicine (TCM) and acupuncture is a transformative trend centered on digital empowerment and breaking disciplinary barriers. In acupuncture, intelligent robotic systems (e.g., an 8-DOF dexterous robot with trajectory-constrained MPC models and needle-bending correction) address issues such as practitioner shortages and subjective efficacy, ensuring precise, stable treatment via engineering integration. In broader TCM, specialized LLMs such as Tianyi, trained on TCM corpora and evaluated via TCMEval, overcome limitations of general AI to democratize expert knowledge and bridge theory-practice gaps. Advanced NLP techniques such as the DHGCN model enable document-level relation extraction, efficiently mining symptom-acupoint links from unstructured data to structure a TCM knowledge base. Collectively, these innovations integrate TCM with engineering, CS, and data science, standardizing practices, enhancing accessibility, and providing evidence for global acceptance—potentially marking a phase of interdisciplinary synergy that preserves traditional wisdom while amplifying its impact through digital precision.[268–270]

Preventive medicine: AI-enabled health defense, wearables, and precision nutrition

Modern healthcare systems focus predominantly on treating established disease, while comparatively little attention is paid to identifying risk decades before clinical onset. Yet many high-burden conditions—including Alzheimer's disease, cardiometabolic disorders, and age-related functional decline—develop silently over long preclinical phases.[271–273] A central challenge in preventive medicine is therefore temporal: can medical decision-making be shifted forward by decades rather than being confined to years—or to the moment when disease has already become clinically apparent? For example, rather than diagnosing dementia after irreversible neurodegeneration has occurred, could high-risk individuals be identified in midlife and guided toward effective prevention? Similarly, female aging is increasingly recognized to begin with ovarian aging. If the timing and pace of ovarian reserve decline, including milestones such as the menopausal transition, could be predicted in advance, proactive, stage-specific health management would become possible. AI has been proposed as a potential pathway toward individualized risk prediction long before disease manifestation, although further validation is needed. In women's health, AI-based tools, such as OvaRePred, model ovarian aging trajectories and predict stages of decline, including perimenopause and menopause.[274–284] Analogously, long before clinical symptoms emerge, APOE $\varepsilon 4$ carriers already exhibit detectable metabolic and biomarker abnormalities.[285,286] Integrating APOE $\varepsilon 4$ status with age, metabolic profiles, and longitudinal lifestyle data may allow AI models to identify individuals at high risk for Alzheimer's disease decades before symptom onset, opening a genuine window for preventive intervention.[287–291]

Once high-risk individuals are identified, effective preventive medicine relies on continuous health management rather than one-time risk prediction.[292] Wearable devices play a central role by enabling long-term monitoring of lifestyle and physiological states, including physical activity, sleep quality, heart rate variability, and glucose dynamics.[293–295] These data provide objective feedback on daily behaviors and allow AI systems to support personalized lifestyle modification, adherence monitoring, and early detection of functional decline.

At the biochemical level, mass spectrometry (MS)-based profiling of steroid hormones and key nutritional biomarkers enables sensitive detection of early endocrine and metabolic deviations that precede overt disease.[296–304] AI will further integrate these measurements into precision endocrinology and nutrition frameworks. In women, endocrine indicators can be interpreted using AI-derived ovarian endocrine age reference ranges, such as those generated by OvaRePred,[284] rather than relying solely on chronological age. This strategy enables earlier identification of subtle abnormalities and supports timely, stage-specific interventions through targeted nutrition, lifestyle modification, or clinical follow-up.

Together, AI-enabled prediction, continuous monitoring, and precision nutrition reframe preventive medicine as a proactive, life-course-oriented health strategy, with the potential to become a cornerstone of sustainable healthcare systems as data governance and ethical standards continue to evolve.

AI for good: Scaling expertise and advancing equitable, precision care

AI in medicine should be evaluated not only by algorithmic performance but by its capacity to generate equitable, scalable, and clinically meaningful impact in real-world practice.[305,306] In this context, AI for good emphasizes applications grounded in real-world clinical data and designed to benefit populations that are often underserved, including women, older adults, and individuals with metabolic or subclinical endocrine disorders. By operating in heterogeneous care environments rather than narrowly selected research cohorts, AI systems can reduce unwarranted practice variation and expand access to standardized, high-quality care.

A central societal value of medical AI lies in its ability to translate individual clinical expertise into population-level benefit. Much of high-quality medical decision-making relies on tacit knowledge accumulated through years of clinical experience.[305] AI offers a mechanism to codify this expertise by learning from large-scale, high-quality clinical data generated by experienced physicians, thereby transforming expert judgment into reproducible, scalable decision-support tools.[276,284,307,308] In doing so, AI enables expert-level reasoning to extend beyond tertiary centers and senior clinicians, allowing more patients to benefit from consistent and evidence-informed care regardless of geography or provider seniority.

Beyond expertise dissemination, AI enables a paradigm shift from single-marker interpretation to system-level physiological intelligence.[292,305] Through integration of multi-dimensional data—including comprehensive biomarker panels, metabolic features, age, body composition, and longitudinal trends—AI supports refined phenotyping that better reflects biological complexity.[305,309] This approach moves clinical assessment away from binary normal-abnormal thresholds toward individualized, context-aware risk stratification and management.

Importantly, AI for good should augment rather than replace clinicians.[305,310] Transparent and interpretable clinical decision-support systems can enhance diagnostic consistency, reduce uncertainty, and improve patient safety, particularly in fields such as endocrinology, reproductive medicine, and chronic disease management.[1,219,278,280,291,311–316] When embedded into routine

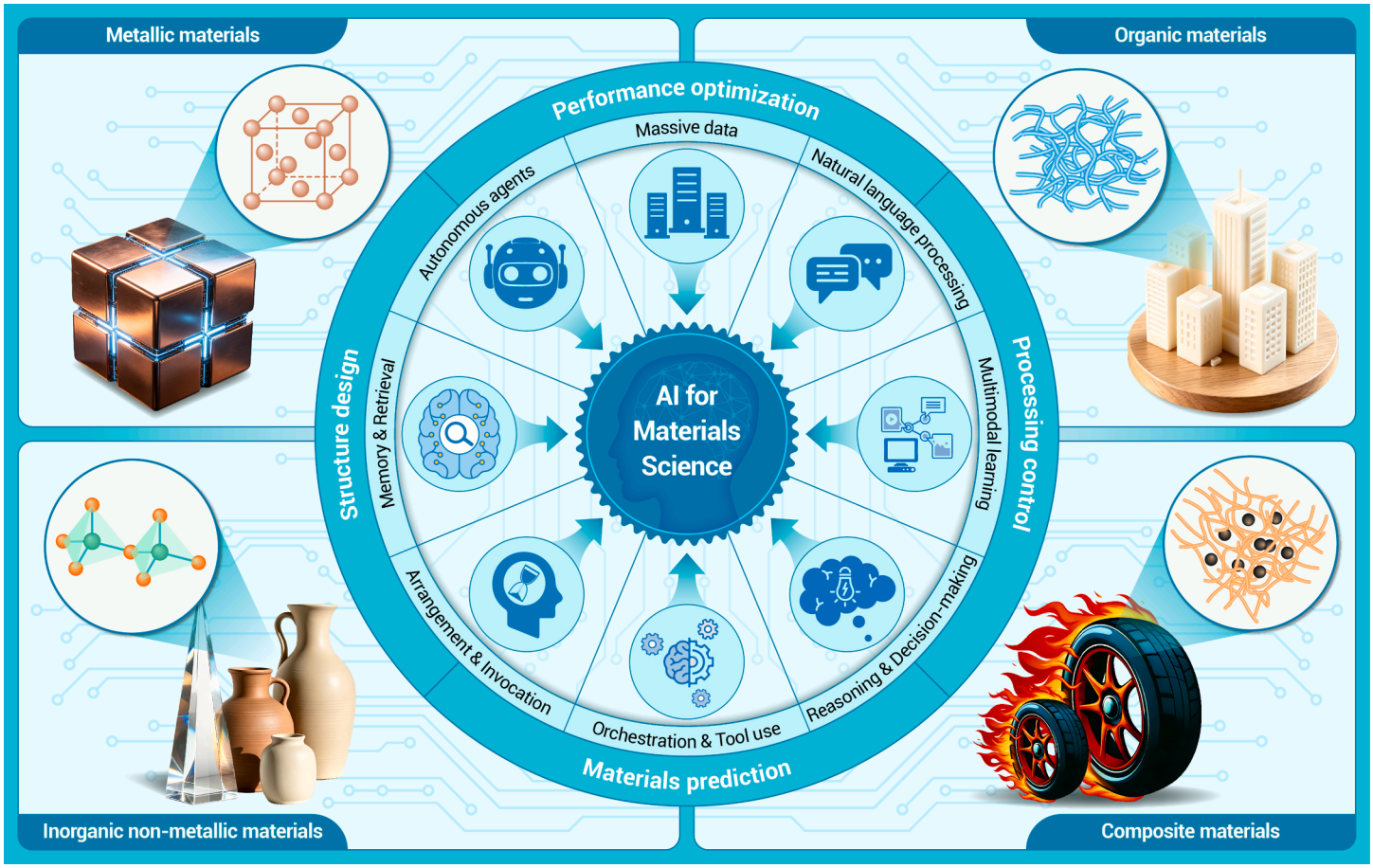

Figure 5. AI-powered materials science

workflows, AI facilitates collective clinical upskilling rather than automation-driven substitution.

Finally, integration of AI with high-throughput diagnostics, EHRs, and digital health platforms enables continuous, low-burden monitoring across the life course, prioritizing within-person change over isolated measurements.[276,317,318] At the population level, aggregated and anonymized AI outputs can inform public health surveillance and policy development. Together, these applications illustrate how AI might contribute to a gradual transition of medicine from episodic, reactive care toward proactive, data-driven, and person-centered health systems.

AI IN MATERIALS SCIENCE

Human civilization could not have evolved to the extent it has without the development and harnessing of materials. Both naturally occurring matter (e.g., stone, wood, cotton, linen, and leather) and synthetic materials (e.g., steel, glass, and plastic) have continuously shaped human society. As such, bridging material structure design with processing control to achieve performance optimization and ultimately expanding applications has become a central objective of modern materials science. The integration of materials science and information technology has fostered AI-driven material research, significantly accelerating progress in materials design and discovery, advancing the Materials Genome Initiative, and promoting practical applications.[1] In particular, the explosive growth of AI technologies has transformed materials science from the traditional trial-and-error paradigm approach into an intelligent, end-to-end exploration framework. The development of AI-powered data analysis tools, together with the emergence of the materials genome, has accelerated the evolution and iteration of materials research across diverse fields, including autonomous decision-making.[110,319] This paradigm shift facilitates the autonomous design, synthesis, performance evaluation, and prediction of materials. In this section, AI-enabled tools and approaches for materials science research have been systematically summarized by discussing relevant breakthroughs in key areas, including inorganic, organic, and composite materials (Figure 5). We aim to better illustrate the characteristics of mutual promotion and interdisciplinary integration between materials science and AI technologies to drive the rapid development of this exciting field.

AI-enabled tools for materials science

With the development of AI and big data technologies, building a data-driven rational design framework has emerged as a pivotal new paradigm shift to break the bottlenecks of traditional materials R&D. Accordingly, data and algorithms now serve as indispensable tools underpinning progress in this all-important, epoch-defining area of research.

In the early stage, databases such as the Materials Project,[320] the Open Quantum Materials Database[321] (OQMD), and Atomly[322] accumulated massive volumes of data through high-throughput computations, laying the foundation for AI model training. Subsequently, the emergence and widespread adoption of databases including CatTestHub,[323] CORE MOF,[324] and MOSAEC-DB[325] promoted the diversification of data infrastructures and led to the establishment of standardized data platforms compliant with the FAIR principles. Representative examples include Meta's OMat24 dataset platform,[326] the Materials Project API, and large-language-model-based literature data extraction tools such as MaTableGPT.[327] Together with the introduction and application of federated learning methods in materials science,[328,329] these efforts have significantly improved the usability of materials databases and enhanced the generalization capability of ML models.

At the core algorithmic level, AI models have evolved from property prediction to synthesis pathway inference and further toward inverse structure generation. Between 2020 and 2024, GNN models such as M3GNet,[330] CHGNet,[331] TiraCGCNN,[332] and GPTFF[333] enabled rapid property prediction for inorganic materials. By 2025, in addition to large-scale models focused on materials property prediction, such as Harvard University's Nequip-OAM-L,[334] AI research in materials science increasingly shifted its focus toward generative inverse design. State-of-the-art models represented by Microsoft's MatterGen[335] and MIT's SCIGEN[336] are capable of directly generating entirely new crystal

structures based on target physical properties, enabling true on-demand materials creation.

Meanwhile, LLMs have begun to play an active role in materials synthesis. These include MatChat,[337] released in 2023 for predicting synthesis routes of inorganic materials, as well as MatterChat[338] and CataLM,[339] released in 2025, which significantly improve materials property prediction and human-AI interaction. In addition, Prollama[340] and OpenFOAMGPT[341] introduce iterative correction loops and RAG, demonstrating strong capabilities in contextual understanding and reasoning.

More recently, materials science has advanced from early explorations of automated platforms such as A-Lab[7] toward systems that emphasize autonomous decision-making in multi-agent architectures. Representative examples include MIT's AtomAgents,[342] Edison Scientific's Kosmos system,[343] MatAgent,[344] and SciAgents.[345] These systems not only support literature analysis and experimental planning but also perform logical reasoning and iterative feedback optimization in a manner analogous to human scientists. Such intelligent agents are gradually taking over the entire experimental pipeline, from materials design and synthesis to characterization, thereby establishing closed-loop, largely autonomous research workflows. This transition marks the entry of materials science intelligence into a new era of development.

AI-driven discovery of inorganic materials

The integration of AI into inorganic materials discovery has recently catalyzed a paradigm shift from traditional trial-and-error approaches to data-driven, generative design. Diffusion models and generative frameworks, which have demonstrated remarkable capabilities in predicting and synthesizing novel crystalline structures, play a central role in this progress. For instance, MatterGen employs a diffusion-based process to generate diverse inorganic materials, achieving a higher success rate in producing stable, new materials.[346,347] The crystal structure prediction is enhanced in CrystalFlow by incorporating physical symmetries and thermodynamic constraints into normalizing flow models,[348] which reduces the root-mean-square deviation between generated and DFT-relaxed structures. These models excel in property-guided inverse design, enabling targeted generation of materials with specific electronic, magnetic, or mechanical properties.[346]

In parallel, autonomous experimental platforms, such as the A-Lab,[7] leverage robotics and AI to close the loop between computational prediction and synthesis. By combining literature-mined synthesis recipes with active learning, the A-Lab synthesized 41 of 58 target compounds in 17 days, highlighting the potential and efficiency of AI and robotics in accelerating the discovery of materials. However, the characterization of these autonomously synthesized compounds still has challenges. Automated Rietveld refinement of X-ray diffraction data might misidentify phases and lead to overestimation of discovery success.[349] Some structures claimed as "new" were later found to be disordered variants of known compounds,[349] indicating the need for robust experimental validation. Structurally constrained generative models advance the field by embedding geometric patterns into the generation process. This approach enables the design of quantum materials with tailored symmetries, though its efficacy depends on the quality of training data and the integration of physics-based constraints.[336]

Accelerating the discovery of inorganic materials with AI and autonomous experimental platforms is a critical direction for the future of materials science. However, several key challenges remain to be solved. First, generative models might produce structures that are thermodynamically metastable or synthetically inaccessible due to training data biases toward known compositions.[350] Second, there is a lack of standardized metrics for assessing new and stable materials, which might lead to inflated success rates in reported discoveries.[349] Third, while current models can benefit from large-scale training data, their applicability to complex, multiproperty optimization remains limited.[346] Therefore, more efforts are still needed to improve automated characterization and develop unified benchmarks to bridge the gap between *in silico* design and experimental synthesis.[351,352]

AI-driven development of organic materials

The application of AI in the field of organic polymer materials is transitioning from single-point breakthroughs to full-process integration. The design, synthesis, processing, and performance optimization of polymer materials are no longer isolated stages but are rather closely integrated through data-driven approaches and algorithm-based decision-making.

At the source of polymer discovery, design, and characterization, traditional trial-and-error methods and simple group contribution techniques are in the process of being progressively phased out. Although language models based on simplified molecular input line entry system (SMILES) perform well, they may fail to account for the distinctive characteristics of polymer materials possessing high molecular weights, molecular weight distribution, complex macromolecular chains, and aggregation morphologies. Neglecting the periodic and multi-scale features of organic polymers may lead to an incomplete understanding of their structure-property relationships in materials synthesis and characterization.[353,354] Therefore, GNNs and transformer architectures (e.g., PerioGT) that can capture topological structures are emerging as critical tools for bridging microscale organic molecule structures with macro-scale material properties.[354–356] More importantly, to solve the stubborn problem of synthetic inaccessibility in AI-designed molecules, pure generative models, such as variational autoencoders and GANs, are evolving toward virtual forward synthesis.[357,358] This strategy constrains the space of materials design to accessible monomers and established reaction pathways, which essentially prioritizes synthetic feasibility at the design stage, thereby avoiding the awkward situation where a theoretically perfect molecule is designed but cannot be practically synthesized.

In the realm of polymer processing control and performance optimization, the role of AI has advanced from static prediction to dynamic, real-time decision-making. While DL excels in handling large-scale data streams, small-sample learning strategies remain essential when confronted with costly and complex experimental setups, such as thin-film optimization of organic electrochemical transistors and microfluidic synthesis.[359,360] Bayesian optimization continues to serve as the benchmark approach for guiding SDLs in navigating multi-dimensional process parameters (e.g., temperature, solvent, flow rate, and reactant ratios), owing to its effective balance between exploration and exploitation.[360,361] Importantly, the optimization framework itself is not fixed. Recent studies have shown that dynamically switching between surrogate models, such as transitioning from Gaussian process regression (GPR) to random forest (RF) algorithms, can more effectively overcome performance bottlenecks.[359]

The competitive advantage of AI in this field will not stem from the superiority or inferiority of any single model but rather from the ability to achieve "explicitly encoding." This is achieved by embedding fundamental thermodynamics and kinetics principles of materials behavior as well as processing conditions into AI architectures through PINNs and multimodal learning (integration of imaging and spectroscopic data).[358,362] Such integration can enhance model interpretability under data-limited conditions and enables a closed-loop workflow spanning from organic molecular design to practical applications.

AI-accelerated development of composite materials

Nature has created numerous composite materials that exhibit exceptional performance unattainable using conventional human designs.[363] Composite materials, made up of two or more distinct constituents, integrate the advantageous features of each component, thereby achieving properties that no single material can provide alone, including enhanced mechanical, optical, thermal, and electrical performance.[364] Typically, a composite material consists of a continuous dominant phase, referred to as the matrix, and one or more dispersed minor phases, known as fillers. Achieving an optimal match between the matrix and fillers to maximize overall performance remains a grand challenge, as it has traditionally relied on extensive experimentation and trial-and-error processes.

With the rapid advancement and growing capabilities of AI, these techniques are emerging as powerful tools to address this challenge, with the potential to drastically shorten the development cycle of innovative high-performance composite materials.[352,365–371] AI approaches, including supervised learning, unsupervised learning, DL, and RL, are expected to play an increasingly important role in accelerating composite materials discovery from several key aspects.

Material discovery. To date, millions of materials have been discovered, creating vast materials databases for various application purposes. However, selecting suitable materials that meet specific performance requirements remains a formidable task. Increasingly, AI-driven methods have been employed to efficiently screen optimal material combinations for targeted applications.

For example, Gong's group proposed a descriptor-based strategy that integrates data mining, experimentation, and ML to rapidly screen biocompatible hydrogels with super-adhesive properties under water using protein databases.[367] With the assistance of AI techniques, scientists and engineers can therefore identify promising matrix and filler candidates tailored to end-use requirements, significantly reducing the need for exhaustive literature review and empirical trial-and-error experimentation.

Solvent selection. Achieving uniform filler dispersion is critical for the performance, reproducibility, and stability of composite materials, making solvent selection a key processing parameter. Data-driven AI models can identify suitable solvents for polymer matrices by learning solvent-material relationships based on solubility parameters and molecular descriptors.[372] By incorporating filler properties such as polarity and surface chemistry, AI can further predict solvent compatibility in multicomponent systems. As single solvents rarely satisfy all components, AI-enabled analysis of prior experimental data enables rapid optimization of cosolvent compositions and ratios, replacing labor-intensive trial-and-error approaches and accelerating the preparation of homogeneous composites.

Interfacial compatibility. Interfacial interactions between the matrix and fillers critically govern the performance of composite materials. In separation membranes, interfacial defects can create nonselective transport pathways, leading to a pronounced loss in selectivity and necessitating extensive experimental screening. AI techniques have been employed to identify optimal polymer-porous nanoparticle pairs that simultaneously enhance gas permeability and selectivity.[373] By constructing datasets that capture interfacial descriptors, such as interaction energy, functional group compatibility, and interfacial affinity, AI enables rapid prediction of optimal material pairings across diverse composite applications.

Manufacture technique. The performance of composite materials is dictated by their multiscale structures, which are strongly governed by the choice of manufacturing techniques and processing parameters. AI-driven approaches enable the digitalization of structure-process relationships and automated optimization of fabrication conditions, accelerating the development of high-performance composites.[353] By integrating databases that capture material descriptors and manufacturing characteristics, AI can intelligently match composite systems with optimal fabrication routes and parameter windows to meet targeted performance requirements.

Beyond the abovementioned roles, AI can substantially shorten the optimization cycle for key composite design variables, including filler loading, size, morphology, and matrix modification strategies. More importantly, when trained on sufficiently rich datasets, AI models can directly predict the final performance of composite materials, enabling efficient, task-oriented, and tailor-made material design.

AI for materials and materials for AI

The past years have witnessed significant progress in AI for accelerating materials discovery, design, fabrication, and deployment. Notably, AI-assisted methodologies are transforming materials development from conventional experimental research on metals, inorganic compounds, organic polymers, and composite materials toward data-driven, autonomous, and intelligent manufacturing of predictive materials. This paradigm shift is expanding the scope of AI-driven materials from traditional applications to emerging fields, such as adaptive systems operating under extreme environmental conditions.[374–376] For instance, AI-enabled quantum materials are poised to advance quantum sensing, storage, and biomimetic computing. Concurrently, intelligent sensing materials, such as multifunctional hydrogels, smart polymer dielectrics, and mechanically responsive metamaterials, are serving as essential components for signal detection and perception, enabling promising applications in embodied intelligence, brain-computer interfaces (BCIs), and beyond.[377–381] AI-enabled advanced semiconductors, such as perovskites and 2D materials, also hold promise for improving digitalized energy conversion and storage techniques. However, materials rarely exist in an ideal, defect-free state in the real world; instead, they commonly contain a high density of defects. Such defects, arising from deviations in stoichiometry during synthesis, often give rise to unexpected optical, electrical, magnetic, and catalytic behaviors. Due to their inherent complexity and limited controllability, defective materials are frequently neglected from AI-based materials discovery frameworks, potentially hindering the identification of unconventional and high-performance material properties. Thus, it remains crucial to establish high-quality materials databases at the foundational level, along with developing reliable large models and AI systems in materials science, which enable systematic integration across multiple dimensions of materials to enhance capabilities in bidirectional design and optimization. Additionally, AI-driven materials research facilitates the generation of inorganic and organic candidates but may neglect critical assessments of synthetic feasibility. Some computationally predicted structures are theoretically possible yet experimentally inaccessible due to kinetic limitations, thermodynamic instability, and practical reaction constraints. For real applications, rigorous synthetic viability evaluation should be integrated into AI-powered materials discovery.

Moreover, the foundational technologies underpinning AI, such as computers, the internet, and data centers, rely fundamentally on integrated circuit chips. Thus, developing low-power chip materials is essential for building high-performance hardware infrastructures to support AI systems. Furthermore, advancements in massive training and inference of AI models and intelligent agents require substantial energy resources, thereby largely increasing the demand for power support and management. High-performance energy materials that enable efficient power generation, conversion, and storage are therefore indispensable to meet this growing demand. Reliable computing operations in AI systems depend critically on a stable electricity supply, underscoring the importance of advanced semiconductor materials; insulating materials, including inorganic, organic, and hybrid dielectrics; and conductive materials such as non-ferrous metals. These materials are the fundamental blocks for constructing various electrical/optoelectronic devices such as power devices, transformers, ultra-high-voltage transmission lines, photovoltaic panels, and energy storage systems. They will provide essential support for scalable and sustainable power operations, accommodating the explosive growth of AI. Looking ahead, the deep integration of AI and materials science will foster mutual advancements, driving transformative technological innovations in materials informatics.

AI IN GEOSCIENCE

The development of AI techniques/models in geoscience

The integration of AI into geoscience has evolved beyond its early role as an auxiliary analytical tool and is increasingly contributing to discovery in Earth system science.[382,383] Spurred by the rapid growth in multimodal observational data and advances in high-performance computing (HPC), this convergence is expanding our ability to simulate and analyze interactions across the atmosphere, hydrosphere, lithosphere, biosphere, and cryosphere,[384] as shown in Figure 6.

Within the realm of geospatial AI (GeoAI), foundation models are reshaping how Earth systems are modeled and understood. A pivotal change is evident in model architecture, characterized by a shift from narrowly focused, task-specific models toward universal foundation models. Propelled by the evolution of the transformer architecture, the field has widened from traditional convolutional neural networks (CNNs) and recurrent neural networks (RNNs) toward attention-based mechanisms and GNNs.[385–388] In meteorology, large-scale models such as Pangu-Weather, GraphCast, and FengWu have been shown to achieve competitive forecast skill at substantially reduced computational cost compared with traditional numerical weather prediction (NWP), thereby enabling rapid global forecasting in recent studies and reports.[389,390] Similarly, in remote sensing, vision-language multimodal models such as SpectralGPT and SkySense, as explored in emerging approaches, have been developed to leverage self-supervised pre-training on petabyte-scale heterogeneous data (i.e., satellite imagery, reanalysis, and station observations).[391,392] These models demonstrate robust generalization, requiring only minimal fine-tuning to adapt efficiently to downstream tasks ranging from climate prediction to mineral exploration.

Crucially, physics-informed ML (PIML) has emerged as a vital nexus connecting data-driven approaches with physical laws.[393] Techniques represented by PINNs and neural operators are embedding physical consistency into DL.[179,394] Rather than relying solely on statistical pattern recognition, these approaches incorporate conservation principles such as mass, momentum, and energy either within network architectures or through physically informed loss functions.[395,396] PINNs have effectively mitigated physical inconsistencies

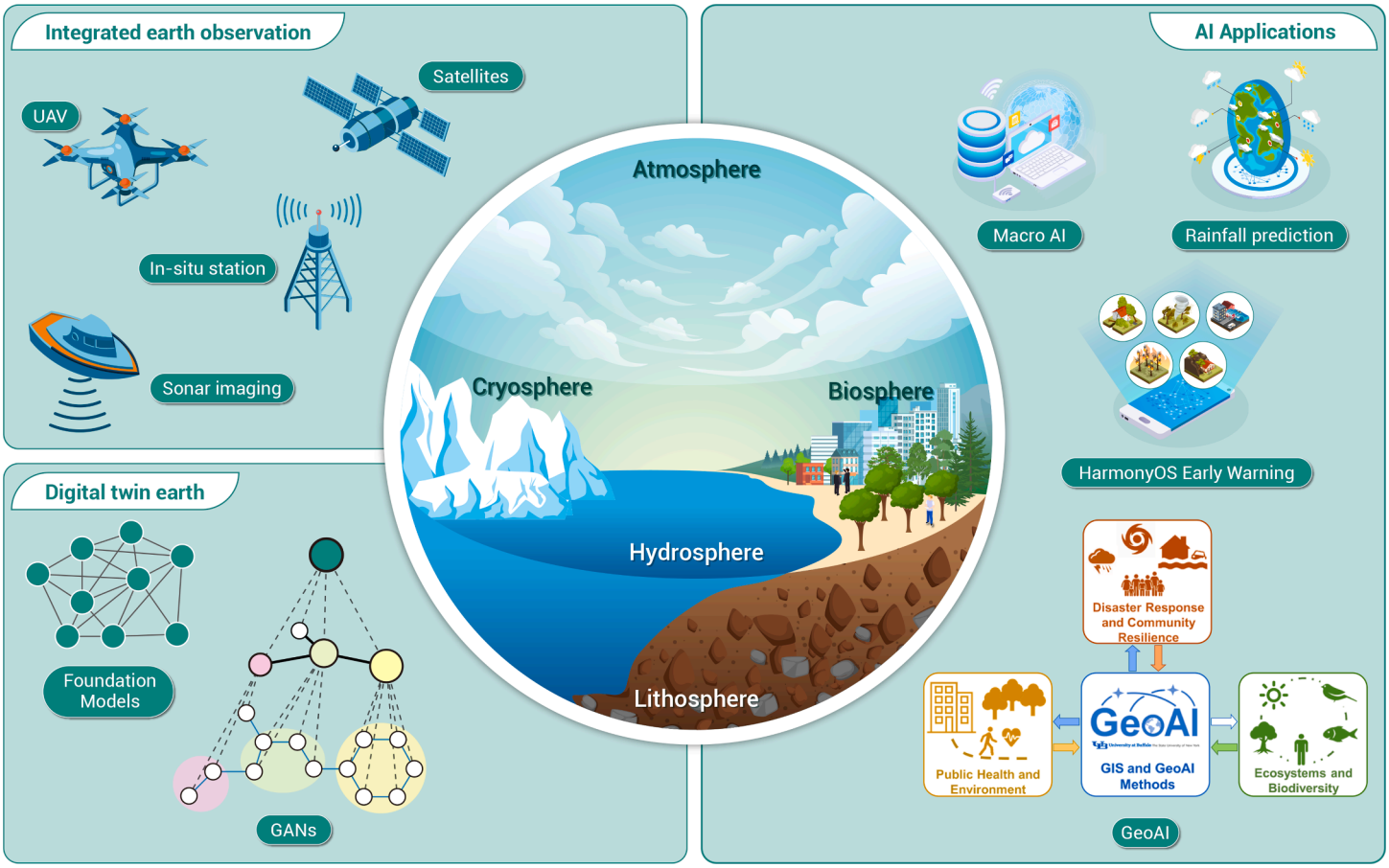

Figure 6. AI in geoscience: The key AI techniques/models and their applications

in data-sparse scenarios, such as deep Earth inversion and extreme event modeling.[397] Meanwhile, neural operators (e.g., FNOs) learn mappings between function spaces to accelerate high-dimensional, nonlinear geodynamic simulations.[398] This physics-aware modeling paradigm diminishes reliance on labeled data and enhances interpretability, physical fidelity, and applicability across diverse geographic regions.

However, a fundamental challenge in Earth system modeling lies in the inherent non-stationarity of geoscientific processes across both spatial and temporal scales.[399] Climate dynamics, hydrological processes, and geological evolution are influenced by both natural variability and anthropogenic forcing, leading to continuous distribution shifts in observational data, including both changes in input distributions and shifts in underlying process relationships.[400] As a result, AI models trained on historical datasets may experience performance degradation when applied to unseen regions, future climate regimes, or extreme events that lie outside the support of the training data.[383] This limitation is particularly critical in Earth system applications, where models are frequently required to extrapolate across spatial domains, temporal regimes, and boundary conditions rather than operate within well-sampled interpolation ranges. In addition, mismatches between the spatial and temporal scales of observations, processes, and model representations further exacerbate these challenges, as relationships learned at one scale may not hold when transferred across scales. To address these limitations, emerging research emphasizes the importance of domain adaptation and transfer learning to improve model generalization across heterogeneous environments.[401] Nevertheless, these approaches typically rely on some degree of shared structure between source and target domains and may be insufficient under strong regime shifts or when underlying process relationships change. Further, out-of-distribution generalization has become a critical requirement for ensuring the robustness and reliability of AI systems in geoscientific applications,[402,403] particularly under increasing climate variability and extremes, and manifests differently across atmospheric, hydrological, and subsurface systems.

Concurrently, the maturation of GAI offers novel solutions for data scarcity and spatiotemporal discontinuity. Technologies based on diffusion models and GANs are now effectively tackling challenges such as cloud removal in satellite imagery and the reconstruction of historical climate records.[404–406] This generate-and-verify mode is redefining our ability to reconstruct missing Earth observations, providing high-fidelity baselines for long-term analysis.

Earth science research worldwide is undergoing profound transformations, characterized by more integrated research scopes, more advanced technologies and methods, tighter coupling between basic research and applications, expanding spatiotemporal scales of study objects, coordinated multidisciplinary and multi-sectoral development, and a research paradigm where model-driven and data-driven approaches advance side by side. A key challenge for the new generation of Earth observation systems lies in how to capture the dynamic states of Earth surface hydrology, ecology, weather, climate, and the solid earth and how to quantify their associated uncertainties.

Collectively, these developments, from foundation models to physics-informed learning and generative reconstruction, indicate a structural evolution in geoscientific modeling. AI systems are increasingly used to emulate, accelerate, and integrate components of complex Earth system models while remaining embedded within expert-driven workflows. Although recent demonstrations of autonomous AI systems in other scientific domains suggest that such approaches may offer potential future directions for geoscience applications,[407] applications in geoscience continue to rely on hybrid frameworks that combine physical understanding, domain expertise, and uncertainty quantification. Within this context, AI contributes to the development of Earth digital twin initiatives by enhancing computational efficiency, data integration, and cross-sphere coupling.

The application of AI in distinct Earth spheres

AI for atmosphere. The integration of AI into atmospheric science is reshaping approaches to recognition, simulation, and prediction across multiple atmospheric layers. Increasingly, AI systems combine large-scale data-driven

learning with physically informed constraints and multi-source observations, enabling new forms of model emulation and forecasting.[408] This progress is distinctly reflected in the lower and upper atmosphere domains. In the lower atmosphere, AI is bridging the gap between weather forecast timeliness and climate simulation feasibility.[409,410] It is enabling foundational models and neural emulators that complement and, in specific benchmark settings, have been reported to match or exceed traditional computationally intensive numerical models. Notable AI weather models include Google DeepMind's GraphCast and GenCast,[390] the European Centre's operational AIFS, and China's Pangu, Fengwu, and Fuxi models.[32] These models learn complex atmospheric dynamics from vast historical data, with forecast accuracy reported to be comparable to top numerical models. However, the assumption of statistical stationarity is often violated in atmospheric systems, particularly under rapid climate change, and model performance may degrade when encountering extreme or previously unseen events that fall outside the training data distribution,[390,411] including events associated with shifts in large-scale circulation regimes and extremes that are underrepresented in historical records. This limitation is particularly relevant for atmospheric prediction, where models are required to generalize across evolving climate states and increasing variability rather than operate within stationary climatological conditions. Furthermore, geophysical foundation models such as Microsoft's Aurora demonstrate superior forecast skill at a fraction of the computational cost of traditional methods.

In the upper atmosphere, where data are sparse, AI's contribution lies in integrating sparse multi-source observations with physically informed modeling to improve representation of upper-atmospheric processes. It addresses the complexity of physical processes in the stratosphere, mesosphere, thermosphere, and ionosphere.[387] For instance, AI models have systematically revealed coupling mechanisms in the sporadic E layer, improving the prediction of ionospheric structures. In space weather forecasting, foundation models are being developed to predict solar activity and its terrestrial impacts. Examples include the solar model Solaris, NASA and IBM's Surya model, and China's Fengyu model, which uses a unique chain architecture to simulate the solar wind-magnetosphere-ionosphere coupling.[412] It is worth noting that the study of the lower and upper atmospheres is not isolated, and AI is becoming a catalyst for achieving cross-sphere coupling and full atmosphere modeling.[413] By coupling trained atmospheric and oceanic AI modules, a DL Earth system emulator reproduced key large-scale climate features, including long-term warming trends and El Niño variability, while completing a 1,000-year simulation on a single GPU within 12 h.[414] This shows that AI has great potential in the efficient integration of the interactions of various layers.

The strong performance of AI-based weather models does not reflect a replacement of physical modeling but rather a reconfiguration of how atmospheric dynamics are represented. Traditional NWP explicitly solves discretized governing equations forward in time. In contrast, AI-based models learn effective dynamical representations from large archives of physically consistent simulations and reanalysis data. These systems therefore encode statistical representations of atmospheric behavior that are grounded in physically generated training data. The shift is thus better understood as a transition from explicit equation solving to data-driven dynamical emulation rather than from physics to data.

In view of the algorithm procedure, AI fundamentally modifies the traditional data assimilation pipeline in three ways. First, AI helps create highly accurate and fast neural network emulators of these complex physical operators, significantly speeding up the traditional assimilation cycle. Second, AI, and specifically DL, helps learn a more flow-dependent and dynamically evolving representation of the fusion covariance for traditional data assimilation, replacing the static climatological approximations used in many systems. This leads to a more accurate weighting of new observations against the forecast. More radically, AI is enabling a new paradigm. Instead of a separate assimilation step followed by a forecast, models such as Google's GraphCast or the AIFS are trained on historical data that have already been assimilated. They implicitly learn the mapping from a noisy and sparse observation space to a complete and coherent atmospheric state, creating a single, unified pipeline that ingests observations and produces a forecast directly.

Overall, AI is rapidly reshaping atmospheric science by enabling more efficient, accurate, and physically informed analysis and prediction across the Earth system. The progress of AI in atmospheric research extends beyond analytical assistance and increasingly influences both methodological development and operational practice.[382]

AI for the hydrogeosphere. In recent years, advancements in AI technology have increasingly influenced research approaches in hydrosphere science.[415] By integrating ML, DL, and physical mechanisms, AI has achieved remarkable improvements in the integration and estimation of multi-source hydrological variables, the precision of hydro-meteorological forecasting, and the comprehension of extreme hydrological and climatic events.[416]

The current research in AI in the hydrosphere is transitioning from a purely data-driven approach to a more profound integration of physical mechanisms alongside AI.[417] The focus has shifted from basic loss function constraints to sophisticated differentiable modeling that embeds physical mechanisms within neural network architectures, thereby enabling closer integration of physical constraints within neural network architectures.[418] This evolution supports the transition from localized static calibrations toward more dynamically consistent representations of global water systems, with demonstrated improvements in large-scale water balance estimation and parameter field consistency.[419,420] This deep integration approach extends the applicability of AI, demonstrating robustness and physical consistency compared with purely data-driven methodologies, while offering complementary strengths to traditional numerical models.[421–423]

AI has emerged as the pivotal technology for synthesizing multi-platform and multi-scale observational data in the hydrosphere, facilitating high-precision estimations of essential hydrological variables.[424] By developing ML and DL frameworks, researchers can effectively merge diverse data sources, such as remote sensing and ground observations, thereby enhancing the estimation of critical variables such as precipitation, soil moisture, and water bodies.[425,426] For instance, the combination of Sentinel-1 SAR and CYGNSS data within ML frameworks has markedly improved the accuracy of daily river water-level estimations.[427] Likewise, integrating ML with various remote-sensing soil moisture datasets can produce fusion products with superior resolution and accuracy compared to individual data sources[428] while also significantly broadening their spatiotemporal coverage.[429]

The integration of AI into hydro-meteorological forecasting is contributing to enhanced accuracy, quicker response times, and a deeper understanding of the intricate processes within river basins. Traditional physical models often exhibit structural biases due to the oversimplification of complex hydrological dynamics. In contrast, data-driven models, particularly those utilizing DL, can learn directly from vast datasets, leading to significant enhancements in forecasting accuracy.[430,431] Nevertheless, hydrological systems are inherently non-stationary due to climate variability, land-use change, and human interventions, which introduce substantial distribution shifts and pose challenges for model transferability across basins and temporal regimes,[432] including shifts in hydrological regimes, extremes, and human-regulated flow conditions that may not be represented in the training data. This limitation is particularly critical in hydrology, where models are frequently transferred across catchments and required to generalize beyond observed conditions despite strong spatial heterogeneity and scale dependence.

AI models are adept at capturing the dynamic features of radar echo sequences, enabling the prediction of storm paths and intensities in real time.[433] Long short-term memory networks and their hybrid architectures are capable of modeling nonlinear hydrological processes and long-term temporal variations, while the incorporation of GNNs further boosts the model's capacity to represent spatial interactions within watersheds.[434,435] Despite these advances, a central requirement for operational hydrological forecasting remains robust uncertainty quantification. Recent developments in probabilistic DL, ensemble prediction frameworks, and Bayesian approaches aim to provide calibrated predictive distributions rather than deterministic point estimates. Integrating uncertainty-aware AI models into hydro-meteorological forecasting is essential for risk-based flood management and climate adaptation planning.

Furthermore, the convergence of AI with multi-source remote-sensing data is propelling extreme hydroclimate research from mere statistical descriptions to in-depth mechanistic analyses. Utilizing multi-source satellite data alongside ML techniques, researchers have successfully quantified the sensitivity of global vegetation to daily fluctuations in rainfall[436] and mapped the spatial patterns of critical soil moisture under conditions of global vegetation water stress.[437] Explainable ML analyses have further uncovered that the sensitivity

of vegetation in semi-arid regions worldwide to soil moisture has markedly increased over the past four decades,[438] and the area of vegetation affected by moisture constraints in tropical regions outside the Northern Hemisphere continues to expand.[439] At a mechanistic level, methodologies such as Granger causality tests, ridge regression, and RFs have been employed to clarify essential processes, including feedback loops between soil moisture and precipitation, as well as the influence of cloud cover variations on day-night asymmetric warming.[440]

AI for pedosphere/lithosphere. AI technology is increasingly influencing lithospheric research by strengthening hybrid frameworks that integrate data-driven methods with hypothesis- and process-based modeling approaches.[434,441] Large-scale geoscientific initiatives and coordinated observation programs have generated unprecedented multi-source datasets spanning satellite remote sensing, airborne geophysics, and surface and subsurface measurements.[442–446] The availability of harmonized, global-scale data infrastructures has enabled the development of AI geospatial models that integrate petabytes of Earth observation data for mapping the pedosphere, lithosphere, and associated ecosystems. Foundational pre-trained models, such as Prithhvi, Clay, and GEOS3 at 30 m resolution,[447] illustrate how large-scale spectral and geophysical datasets can support generalized Earth representation learning. However, also at plot and landscape scales, combinations of various ML techniques with spectroscopy greatly advanced the rapid and non-invasive sensing of soil properties down to the scale of microbiome and facilitated the use of digital smart technologies in modern agriculture using (e.g., autonomous light robots and drones for precision farming), real-time field management (e.g., fertilization and selective weeding), or the management of novel cropping systems.[448]

However, a critical limitation in pedosphere and lithosphere applications arises from the pronounced spatial heterogeneity and temporal non-stationarity of subsurface systems,[449] including strong variability in material properties, boundary conditions, and governing processes across scales. Geological formations evolve over long timescales and exhibit strong variability across regions, while observational data are often sparse and focus on events (e.g., tectonic shifts, landslides, and volcanic eruptions) with local shifts impacting local occurrence spaces of the pedo-/lithosphere, are unevenly distributed, and are biased toward accessible areas,[450] leading to incomplete and uneven sampling of subsurface states. These factors result in significant distribution shifts between training and deployment conditions, thereby constraining the extrapolation capability of current AI models,[451] particularly when applied to new geological settings or depth domains. To mitigate these challenges, recent research has increasingly explored domain adaptation and transfer learning approaches to improve model generalization across regions, scales, and data domains.[452,453] Despite these advances, robust extrapolation beyond the support of training data remains an open challenge in the pedosphere and lithosphere, particularly for rare events and poorly observed environments.

Leveraging their capacity for nonlinear modeling and multimodal data integration, AI techniques have been increasingly applied across lithospheric research, including studies of structure and evolution, resource exploration, and geohazard assessment.[454–456] For instance, the integration of global basalt geochemistry with big data AI techniques has enabled global-scale reconstructions of the evolutionary history of lithospheric thickness.[457] In applied geoscience, recent reports indicate that AI-based workflows integrating remote-sensing, geochemical, and geological data are being applied for mineral prospectivity analysis, supporting exploration targeting and predictive analysis.[458,459] Furthermore, AI demonstrates unique advantages in revealing geological patterns that are difficult to identify using conventional methods, such as lithological mapping in covered terrains[460] and the detection of weak geophysical and geochemical anomalies,[461] thereby significantly accelerating the pace of geological discovery.

Beyond these advances, a defining challenge in pedosphere and lithosphere research lies in the inherently inverse nature of subsurface problems, where sparse observations, strong heterogeneity, and scale mismatch lead to substantial uncertainty.[418] The harmonization of regional soil measurements into large global soil databases has provided labels for predictive AI modeling to produce the global SoilGrid at 250 m spatial resolution.[462] Recently, a new extended harmonized global soil database extended soil labels for AI Earth observation modeling at 90 m pixel resolution, linking soil predictions (u.a., soil organic carbon and clay content) to environmental geo-covariates and socioeconomic factors.[463] Nonetheless, one should be aware that such models may still fail to predict local system behavior due to erratic downscaling of soil and other ecosystem controls to specific site conditions.

Recent AI frameworks increasingly move beyond deterministic predictions toward probabilistic and ensemble-based inference, aiming to quantify the range of subsurface states consistent with available observations.[462] These methods integrate data-driven learning with physical constraints by employing physics-informed approaches, such as PINNs and related PIML techniques. These techniques embed governing equations of wave propagation, elasticity, and transport, thereby enhancing the robustness, interpretability, and physical consistency of subsurface characterization—particularly in data-sparse and structurally complex settings.[418,464] Meanwhile, data-driven equation discovery methods are emerging to directly derive governing equations from observational data, offering a complementary pathway to uncover the underlying physics of complex geosystems.[465]

Within the broader Earth system context, AI-enabled representations of the pedosphere and lithosphere are increasingly evolving from static mapping products toward dynamically updated components of coupled Earth system models as proposed in recent work.[466] Beyond structural characterization, similar AI principles can be extended to the reactive and heterogeneous nature of soil systems. DNNs indicate that nonlinear interactions between sorbent properties and molecular descriptors can be learned to predict sorption behavior of diverse organic contaminants across material classes without site-specific parameterization.[467]

While AI faces persistent challenges in pedosphere and lithosphere research, including data scarcity, high computational demands, and interpretability constraints, the evolution of next-generation explainable AI (XAI) and general-purpose AI methods, together with the optimization of computing infrastructure, will pave the way for more reliable subsurface modeling.[468] Next-generation AI frameworks are expected to advance our understanding of the lithosphere, deep resource exploration, and geological hazard assessment. This progress is driven by their enhanced capacity to tackle uncertainty, ensure physical consistency, and achieve system-level integration within coupled Earth system models.

Future perspective

Future AI development in geoscience is likely to coalesce around four interdependent directions: (1) construction of coupled Earth system foundation models that integrate atmospheric, hydrological, lithospheric, and biospheric processes; (2) rigorous embedding of physical constraints and interpretability mechanisms within AI architectures; (3) probabilistic and uncertainty-aware forecasting frameworks for risk-informed decision-making; and (4) scalable data infrastructures that enable continual model updating within emerging Earth digital twin initiatives.

A critical frontier is enhancing the interpretability and physical consistency of AI systems. The current evolution from black-box to "gray-box" models will accelerate, with AI increasingly constrained by learning explicit physical laws. This will lead to marked improvements in model reliability, transferability, and multi-scale inference capability while fostering greater trust in operational forecasts and enabling new scientific discoveries.

On the data side, multimodal integration of remote-sensing, reanalysis, *in situ* observations, and social sensing data will support continuously learning Earth system models. In parallel, atmospheric science is expected to achieve deeper operational assimilation, with AI enhancing initial conditions, parameterization schemes, and ensemble-based probabilistic forecasting.

Methodologically, causal discovery and physics-constrained generative models will enable closed-loop modeling of processes, feedbacks, and decisions, supporting high-resolution coupled simulations and dynamic early warning systems. In parallel, future developments need to explicitly address non-stationarity through continual learning, domain adaptation, and physics-informed and uncertainty-aware frameworks, including approaches that ensure reliable uncertainty quantification under distribution shift, to ensure robust generalization across evolving Earth system conditions, where feasible within the limits of available data and system stability, including shifts in climate regimes, extremes, and cross-scale process interactions, while acknowledging that robust

extrapolation beyond the support of available data remains a fundamental challenge.

Ultimately, by integrating physical constraints with uncertainty-aware inference, AI has the potential to transform geoscience research from a linear observation-simulation-decision pipeline into an adaptive, scenario-oriented Earth system framework that supports societal resilience and sustainable development.

DECODING, DESIGNING, AND DRIVING LIFE: THE CONVERGENCE OF AI AND BIOLOGICAL SCIENCES

The meeting of AI and the life sciences is accelerating a shift from observational description to predictive design. Biology, long limited by the inherent randomness of living systems and the throughput of manual experimentation, is moving into a stage defined by dense data, mechanistic engineering, and increasing automation while also raising concerns about black-box decision-making and the need for causal and mechanistic interpretability when models guide interventions. This synthesis surveys the frontier across five dimensions: computational decoding of biological big data, engineering of synthetic systems, optimization of agricultural paradigms, monitoring of macro-ecological complexity, and the emergence of autonomous biological agents that connect these workflows into self-driving research cycles.

Computational biology and big data: Reading the code of life

The explosive growth of biological data—genomes, single-cell omics, protein structures, and clinical records—has enabled a step change in analytical capacity. Classical methods struggle with data that are high-dimensional, heterogeneous, and noisy, making AI integration important for extracting latent structure and candidate mechanisms.[276,469] A prominent feature of the current moment is the rise of foundation models that treat biomolecular sequences as a language, moving from alignment-centric matching toward representation learning with functional semantics.

Sequence-structure-function analysis. AI-driven sequence analysis has progressed from statistical comparison to generative, generalizable modeling. Large-scale DNA models such as DNABERT-2[470] and frontier-scale systems such as Evo 2[471] use transformer-style learning to build reusable sequence representations. Trained on 9.3 trillion base pairs, Evo 2 adopts a Striped Hyena architecture to accommodate genomic context windows up to 1 million base pairs, supporting genome-scale sequence generation and prediction of epigenetic features.[472] This push toward multimodality is further illustrated by LucaOne, which places nucleic acid and protein sequences into a unified token set, capturing relationships implied by the central dogma more effectively than training separate models.[473] In protein science, the focus is expanding from structure prediction to integrated modeling of evolution and function. ESM-3 combines sequence, structure, and function and can simulate hundreds of millions of years of evolutionary change to generate novel proteins.[474] In parallel, platforms such as SaprotHub, NaRaDa,[475] and CRUSTADB[476] are broadening access by enabling community-driven fine-tuning of protein language models for targeted functional annotation tasks.[477]

Multi-omics and network modeling. Beyond sequences, a central challenge is integrating multi-omics tables (transcriptomics, proteomics, and metabolomics) into coherent biological explanations, as employed by HBIcloud.[478] DL approaches relax the linear assumptions underlying traditional methods such as principal component analysis (PCA) and CCA. The deep association model (DAM), for example, captures nonlinear disease associations and can substantially outperform conventional biomarker strategies.[479] To mitigate data scarcity and measurement noise, generative models such as scTranslator exploit central-dogma constraints to infer single-cell proteomes from transcriptomes.[480] The long-term aim is not only to summarize data but to help reconstruct the regulatory logic that produces phenotype. AI-based gene regulatory network (GRN) inference has advanced through methods such as RegDiffusion, which uses probabilistic diffusion modeling to infer networks from single-cell data with improved speed and accuracy.[481] Interpretability matters particularly in this setting, as association alone is often insufficient for intervention; frameworks such as CIMLA (counterfactual inference by ML and attribution) pair causal inference with DL to pinpoint disease-specific regulatory perturbations and identify candidate drivers in disorders such as Alzheimer's.[482] More broadly, causal AI methods (e.g., counterfactual evaluation, explicit intervention modeling, and invariant mechanism discovery across cohorts) can help test whether learned signals are stable under distribution shift and experimentally relevant.

Synthetic biology: Upgrading cognition in engineering and manufacturing. If computational biology is about reading, synthetic biology is about writing and building. Here, AI is not just an accelerator; it is increasingly serving as an enabling layer that broadens search, improves decision-making, and automates the design-build-test-learn (DBTL) loop, where the "learn" stage can incorporate causal validation and interpretability checks before designs are prioritized for wet-lab execution. The field is shifting from open-ended "discovery" toward explicit "specification," especially in non-clinical domains such as industrial bioproduction.

AI-driven protein and circuit design. A key enabler is *de novo* component design. Tools such as RFdiffusion and ProteinMPNN support inverse design of proteins with new folds and functions, reducing dependence on naturally occurring templates.[262,263] This capability is extending into enzyme "grammar": AI workflows can select catalytic motifs and sculpt active sites to meet practical industrial constraints.[483] At the systems level, designed parts must operate reliably inside complex cellular contexts. DL models that treat gene regulation as programmable logic are helping address the difficulty of predicting circuit behavior *in vivo*. By modeling allostery and dynamic conformational switching, AI supports the design of circuits that act as robust sensors or logic gates.[484] Related advances include synthetic enhancer and promoter generation with tunable activity, guided by models trained on large-scale reporter assay datasets.[485,486]

Metabolic engineering and biofoundries. In metabolic engineering, AI is increasingly used to manage the combinatorial complexity of multi-enzyme pathways, host physiology, and process constraints. Rather than optimizing enzymes in isolation, models integrate metabolic flux, pathway interactions, and fermentation realities to propose system-level edits for improved "cell factories." This is particularly important for sustainable manufacturing—such as converting lignocellulosic feedstocks into chemicals—where inhibitor tolerance and pathway bottlenecks must be modeled alongside yield.[487,488] Turning designs into reality in some organizations increasingly depends on biofoundries: automated facilities that physically execute the DBTL loop. In these settings, AI agents can propose DNA designs, robotics can assemble constructs, and experimental outputs can feed directly back into models. This form of closed-loop autonomy can reduce trial and error, but it also raises the bar for data and for model auditing (e.g., uncertainty calibration, counterfactual robustness, and constraint checks) so recommendations remain scientifically and operationally defensible across host strains and environmental conditions.[10,489]

Plants and smart breeding: Optimizing solutions in industry

Agriculture is moving from experience-driven breeding toward "Smart Breeding 4.0," powered by high-throughput multi-omics and AI.[490] The objective is to resolve complex agronomic traits—yield, stress tolerance, and quality—that often evade linear statistical modeling.

Foundation models for plant genomes. As in human biology, plant science is developing domain-specific foundation models. PlantCaduceus, a DNA language model trained on angiosperm genomes, uses the Mamba architecture to capture long-range evolutionary constraints and supports zero-shot prediction of variant effects.[491] At cellular resolution, scPlantLLM models genes as "words" and cells as "sentences," helping disentangle regulatory networks and cellular heterogeneity that shape developmental fate.[492]

Generative breeding and digital twins. GAI is starting to redefine the "build" phase of agricultural DBTL. Cas-SF01, an AI-designed gene-editing tool, shows how protein engineering can deliver high-fidelity nucleases optimized for crop genomes, reported to outperform natural variants in both efficiency and specificity.[493] To assess these designs without waiting for entire growing seasons, the field is increasingly turning to crop digital twins (CDT). By integrating real-time sensing with functional-structural plant models, CDTs simulate genotype-environment dynamics *in silico*, enabling breeders to probe how particular genetic combinations are likely to perform across diverse climate conditions.[494] This shift is further supported by XAI frameworks (e.g., EXGEP) that quantify the nonlinear contributions of genotype, soil, and weather, making yield prediction easier to interpret and more directly usable in decision-making.[495]

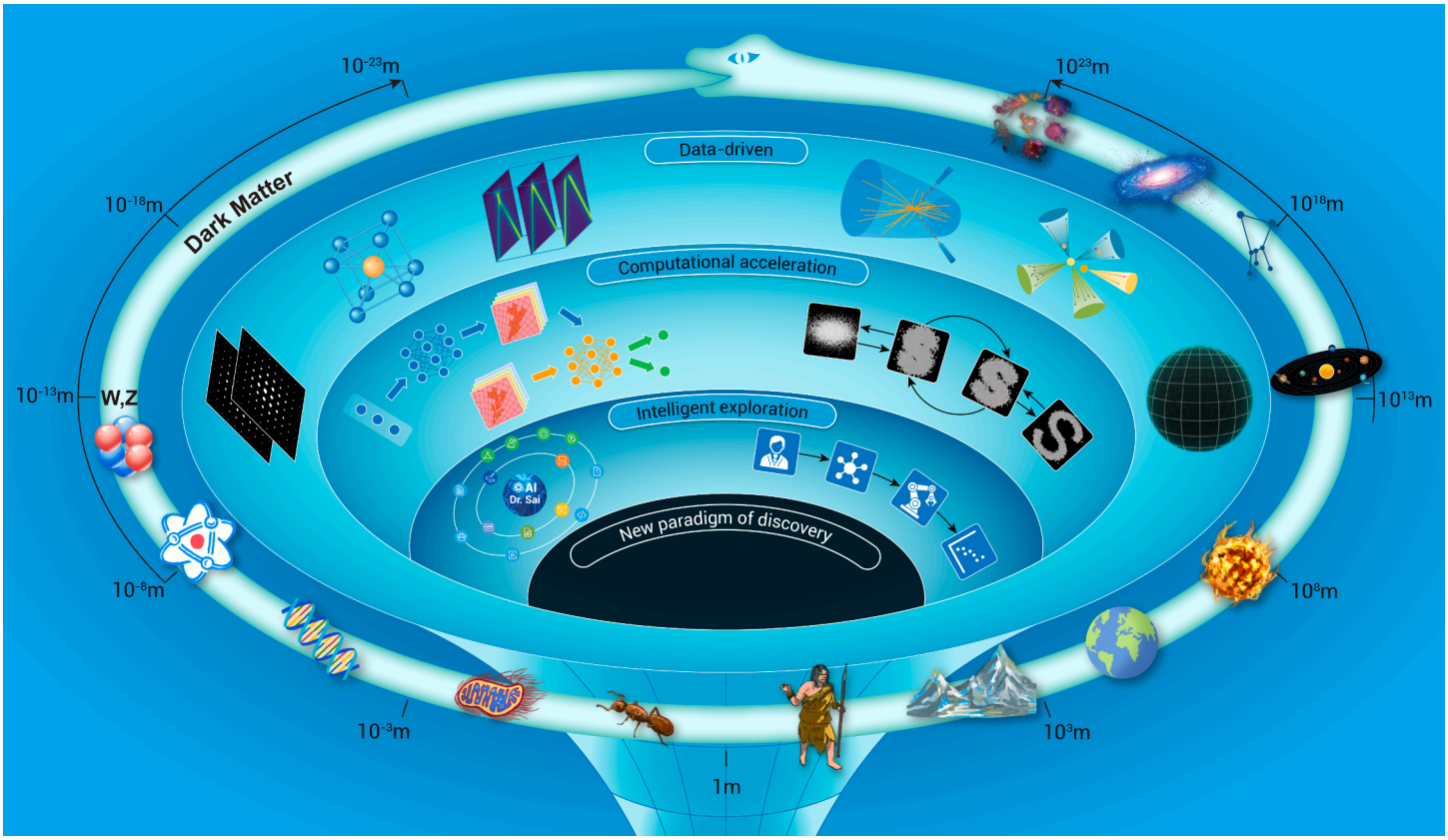

Figure 7. AI in physics

Biodiversity and ecology: Macro-scale constraints and exploration

While synthetic biology builds new systems, ecology measures and protects the existing biosphere. In this context, AI functions as a macroscope—converting heterogeneous environmental data into conservation intelligence while helping formalize the constraints that govern natural systems.

Multimodal data fusion and discovery. Ecological monitoring is shifting from field-only observation to AI-assisted discovery pipelines. Integrating environmental DNA (eDNA), acoustic sensors, drone imagery, and satellite remote sensing enables multimodal fusion at scale.[496] In high-altitude regions such as the Tanggula Mountains, AI can cross-validate infrared (IR) imagery with snow leopard vocalizations, improving species confirmation rates.[497] In addition, DL can act as a discovery engine by flagging anomalies that may indicate cryptic diversity or previously undescribed lineages.[498,499]

Dynamic profiling and "adaptive governance". The aim is to move from periodic reports to dynamic ecosystem profiling. AI workflows can estimate population density and connectivity and present these outputs through interactive dashboards to support near-real-time situational awareness.[500] This enables "precision conservation," where RL agents model long-term intervention outcomes—such as habitat corridor restoration—under constraints that include climate change trajectories and budget limits.[501] Ecological plausibility is critical. Models that explicitly encode phenology, habitat structure, or hard constraints (for example, elevation ranges for specific species) can improve both accuracy and interpretability.[497] In this way, adaptive governance can be framed as a feedback loop: management strategies are updated as new ecological observations arrive and system states change.[223,502]

Biological agents: Autonomous planning, experimentation, and iteration

Across these domains, a notable emerging development is the rise of AI agents—systems that plan task sequences, coordinate tools, and improve through feedback. Agents target a core bottleneck in modern biology: orchestrating multi-step reasoning across heterogeneous data types, software pipelines, and experimental procedures.[503]

Orchestrating the scientific loop. Mechanism-driven research typically requires chained operations: data curation, hypothesis formation, simulation, and experimental validation. AI agents are built to decompose high-level scientific goals into executable workflows. In drug discovery and molecular design, agents can explore vast chemical spaces, invoke property predictors and physics-based simulations, and iteratively optimize candidates through successive rounds of evaluation.[504–506]

Connecting the digital and physical. A particularly important potential of agents is their ability to connect "dry-lab" modeling with wet-lab experimentation. When integrated with biofoundry hardware and automated pipelines, agents can run partially automated and, in some prototype settings, end-to-end, closed-loop research cycles. They can translate biological objectives into robot-executable instructions while preserving provenance, traceability, and reproducibility.[507,508] In systems biology, for example, copilot agents can manage omics harmonization, convert raw outputs into testable mechanistic and causal hypotheses, and maintain a verifiable audit trail throughout the workflow.[509]

Constraints and future directions. Agent deployment still faces practical constraints in data quality and governance, highlighting operational and policy requirements for standardization, oversight, and accountability. Without standardized, high-quality inputs, agents can amplify noise and bias rather than reduce it.[510] In parallel, the opacity of some models reinforces the need for interpretability and for "scientifically meaningful" AI—systems whose decisions are linked to causal mechanisms, not only correlations.[511,512] The next stage will likely involve workflow-native agents embedded directly in research infrastructure, capable of autonomous verification and operating under strict safety and ethics protocols.[513–515]

AI IN PHYSICS: THE INTELLIGENT EVOLUTION OF RESEARCH PARADIGMS IN PHYSICS

Over the past 5 years (2021–2025), the role of AI in physics research has evolved shifting from a supportive data analysis tool toward a core computational paradigm, as shown in Figure 7. With the rapid growth of experimental data rates and the increasing complexity of physical systems, traditional methods have increasingly faced bottlenecks in scale, dimensionality, and response speed. Consequently, AI has transcended its early backend

applications to become deeply embedded in critical workflows—ranging from experimental operations to simulation—across high-energy, nuclear, and condensed matter physics. Crucially, AI does not replace existing frameworks; rather, it serves as a complementary modeling approach, expanding the boundaries of solvable physical problems.

This evolution is driven by synergistic advancements across three dimensions. Advances in computing architectures and hardware platforms have allowed AI methods to move beyond offline analysis and participate in real-time decision processes, particularly in applications subject to strict latency constraints. In scenarios such as trigger systems and accelerator control, this capability has demonstrated verifiable potential, effectively boosting operational efficiency. Simultaneously, DL and generative models have been systematically applied to a wide range of complex physical problems, demonstrating strong performance in high-dimensional and strongly coupled processes. These models offer efficient approximation pathways for complex simulations that were previously constrained by computational resources. Alongside these technical advances, research strategies have also begun to shift. Data-driven exploration approaches now conduct model-independent searches, thereby enhancing the potential for discovering new physics signals.

This section systematically outlines the evolution of intelligent methods across particle physics, nuclear physics, and condensed matter physics, focusing on three key stages: experimental operation and data reconstruction, accelerated calculation of complex physical systems, and the exploration of new physics coupled with the transformation of research paradigms.

Data driven: Efficient data processing and feature extraction

At the frontier of experimental observation, ML serves as a powerful engine for data processing and physical feature extraction, addressing the limitations of traditional analysis pipelines. The core objective is to enhance the operational efficiency of experimental facilities and accurately reconstruct physical events from massive, high-dimensional detector readings.

Event reconstruction based on geometric DL has emerged as a unifying framework across subfields. Methods based on GNNs and point-cloud networks (such as PointNet and ParticleNet)[516–518] treat detector signals as point clouds or graph structures with topological associations. This approach significantly improves the precision and robustness of track reconstruction and particle identification (PID) in complex environments with high background levels. Specifically, in high-energy collider environments such as the Large Hadron Collider (LHC),[519] ParticleNet represents jets as point clouds of final-state particles in momentum space. By capturing local geometric structures, it significantly outperforms traditional methods based on engineered features in tasks such as Higgs boson identification and top quark jet tagging, becoming a widely adopted benchmark model.[518–527] Addressing combined reconstruction tasks such as track finding, momentum and energy estimation, and PID, modern data-driven approaches—most prominently GNN-based models—learn the topological and feature correlations among detector hits and higher-level signals, enabling the simultaneous association of hits into tracks, regression of kinematic properties, and discrimination of particle species under severe pile-up conditions.[516,528–535] Similar high-dimensional reconstruction challenges also arise at synchrotron radiation facilities, where ML models are applied to invert diffraction patterns, scattering images, or spectroscopic maps into real-space structures and key physical parameters.[536,537] In condensed matter and materials physics, CNNs have become powerful tools for image recognition and analysis, particularly in processing data from scanning probe microscopy. They enable the handling of large datasets and complex tasks that are challenging for manual operation.[538,539]

AI leverages its immense data-processing capabilities to achieve deep data mining. DL algorithms were recently applied to mine nuclear emulsion data from the J-PARC E07 experiment, successfully rediscovering a new double-Λ hypernucleus 24 years after the initial data collection, highlighting the potential of AI for rare-event searches in nuclear physics.[540] Furthermore, data-driven nuclear models are breaking through the limitations of traditional semi-empirical models. By directly and accurately capturing complex nucleon interactions, these models significantly enhance the reliability of predictions for unknown nuclides, enabling precise research into parameters such as atomic mass, charge radius, decay half-life, and pairing gaps.[541–543] In neutrino detectors, DL models have been employed to analyze the spatial and temporal distributions of photons recorded by large arrays of photomultiplier tubes, enabling accurate reconstruction of neutrino direction and energy, as well as classification.[544–550] Meanwhile, in ultra-high-energy cosmic-ray experiments such as LHAASO,[551] ML-based approaches have been applied to the complex signals recorded by ground-based detector arrays and imaging Cherenkov telescopes to reconstruct the primary cosmic-ray energy, arrival direction, and mass composition, achieving robust event classification in the presence of strong backgrounds.[552–554] In condensed matter, AI-driven workflows can reconstruct electronic band structures by integrating large-scale simulation data with minimal experimental input. Recent advances employ generative models to perform deconvolution, extracting otherwise inaccessible intrinsic spectral functions from experimental data with enhanced accuracy.[555–557]

On the hardware front, lightweight neural networks deployed on front-end devices such as FPGAs now enable real-time trigger decisions within stringent low-latency constraints (tens of ns), greatly improving the efficiency of selecting rare physics processes.[558–560] Additionally, for accelerator beam control, DRL and Bayesian optimization are being used to automatically tune high-dimensional beam parameters, achieving rapid optimization and dynamic stabilization of collision luminosity.[561]

Computational acceleration: Fast simulation and efficient calculation of complex systems

In theoretical validation and mechanism-driven studies, ML, with the capacity to approximate high-dimensional physical functions and perform rapid inference, has achieved orders-of-magnitude acceleration in tasks such as lattice quantum chromodynamics (LQCD) calculations and detector simulations. The central challenge is to overcome substantial computational acceleration without compromising physical fidelity.

In LQCD, critical slowing down near phase transitions severely limits the efficiency of conventional Markov chain Monte Carlo (MCMC) sampling.[562,563] In particular, autocorrelation times grow rapidly close to critical points, making it prohibitively expensive to generate statistically independent gauge field configurations using traditional update schemes. Generative models based on normalizing flows have provided a promising alternative by learning invertible mappings from simple probability distributions to complex gauge field configurations, thereby mitigating autocorrelation issues and enabling more efficient exploration of configuration space. These approaches have been successfully demonstrated in 2D lattice gauge theories with active progress toward higher-dimensional generalizations.

Across the simulation chain, GAI has been introduced as a surrogate for traditional Monte Carlo techniques in specific components at multiple scales. At the detector-response level, fast simulation methods based on GANs and diffusion models can replace computationally intensive Monte Carlo simulations with high fidelity, effectively alleviating computing constraints in massive data analysis.[564] Representative examples include deep generative calorimeter simulators that accurately reproduce energy deposition patterns and spatial correlations in electromagnetic and hadronic showers while achieving orders-of-magnitude speedup compared to full Geant4 simulations, as validated in realistic LHC detector geometries.

At the level of fundamental interactions, neural-network-based event generators have been developed to model nuclear reaction cross-sections with high precision, facilitating efficient event production for experiments searching for neutrinoless double-beta decay or dark matter, where accurate recoil spectra are essential.[565] By learning multi-dimensional differential cross-sections directly from theory calculations or experimental data, these models enable rapid generation of physically consistent recoil spectra across broad parameter spaces, supporting large-scale sensitivity studies and systematic uncertainty evaluations under limited computational resources.

In condensed matter, AI offers promising pathways to overcome the "exponential wall" inherent in solving the many-body Schrödinger equation for complex quantum systems. One approach replaces traditional tensor networks with neural quantum states (NQSs) to represent the many-body wave function. This allows the simulation of large-scale quantum systems with parameters scaling polynomially rather than exponentially with system size.[164,566,567] Another significant development is the use of ML potentials (MLPs), such as those implemented in the DeepMD-kit. Trained on costly DFT calculations, these models provide surrogate potentials that achieve *ab initio* accuracy at a

fraction of the computational cost, enabling MD simulations of millions of atoms over nanosecond timescales[568,569]

Intelligent exploration: Discovering physical laws and automating research workflows

At the frontier of new-physics searches and upgrading research modes, AI is reshaping how discoveries are pursued by utilizing model-independent anomaly detection and full-process automation technologies. The underlying objective is to expand the search boundaries for unknown phenomena and to redefine the discovery paradigm in physics.

In the search for physics beyond the standard model (BSM), traditional hypothesis-driven strategies are often constrained by predefined signal models. Unsupervised anomaly detection methods based on autoencoders provide a complementary perspective.[570,571] This approach aims to directly learn the statistical distribution of standard model background data, enabling the keen capture of any statistically significant outliers. This model-independent framework has vastly broadened search sensitivity for unknown signals, including dark matter candidates and long-lived particles.

Regarding the automation of scientific workflows, AI-driven automation is transforming the organization and execution of scientific research. AI agents and robotic platforms are increasingly capable of handling complex experimental workflows,[572,573] from literature review and task planning to experimental design and execution. Examples range from the "Dr. Sai" system already implemented in the Beijing Spectrometer III (BES III) experiment, which assists with literature review and task planning,[573,574] to the OptoMate platform developed by MIT,[575,576] which integrates optical design, assembly, and measurement. Meanwhile, quantum ML (QML) represents a frontier exploration, leveraging quantum hardware to handle high-dimensional physical analysis tasks,[577,578] offering a potential pathway beyond classical computational limits.

Summary and outlook

AI has been deeply integrated in modern physics research, forming a multilayered support structure across data processing, calculation, and discovery. For data processing, it enables real-time triggers and precise reconstruction in high-throughput experiments; in the realm of simulation and calculation, it serves as a high-efficiency approximation tool that alleviates the prohibitive cost of simulating complex physical systems; and at the exploratory level, it introduces model-independent discovery strategies and automated research workflows. This evolution from tool-assisted analysis toward intelligent collaboration reflects a growing integration of data-driven inference, computational acceleration, and intelligent exploration.

Looking ahead, the central challenge is no longer whether ML techniques can improve performance in specific tasks, but whether they can be systematically adopted as verifiable, controllable, and sustainable computational components within physical research. Many current approaches emphasize empirical approximation and speedup, while the roles of physical consistency, uncertainty propagation, and systematic error control remain insufficiently defined. Addressing these issues will require the explicit integration of physical constraints, symmetries, and prior knowledge into model architectures and training strategies, together with uncertainty-aware methodologies compatible with high-precision measurements and hypothesis testing.

AI is evolving from an offline analysis tool into an integral part of the experimental operation and decision-making chain. This transition imposes unprecedented demands on model stability, interpretability, and the understanding of failure modes. The key issue is therefore whether physical inference can be safely delegated to data-driven models and which must remain anchored in explicit physical reasoning and human oversight. In this sense, the long-term role of AI in physics is best viewed as a physics-constrained research collaborator that extends the reach of human inquiry while preserving the reliability of scientific conclusions.

AI IN CHEMISTRY

AI transforms chemistry from a labor-intensive discipline into a generative and automated science. Chemistry occupies a central position in the natural sciences, serving as the bridge between fundamental physics and complex biological or material systems. Consequently, it is an inherently data-dense discipline, accumulating complexity from centuries of empirical observation and decades of computational simulation. This vast repository is exemplified by the Chemical Abstracts Service (CAS), which now indexes hundreds of millions of unique substances. Yet, despite this wealth of data, traditional chemical research remains a labor-intensive endeavor, constrained by the sheer magnitude of chemical space, which is estimated to contain overdrug-like molecules.[579]

The convergence of chemistry and AI has moved beyond the nascent exploratory phase of the early 2020s into a period of systemic integration and generative capability. The primary driver for this paradigm shift lies in the unique nature of chemical complexity. The combinatorial explosion of possible molecules, coupled with the intricate permutations of reaction conditions (solvents, catalysts, and temperature), creates a search space that is intractable for human intuition alone. While early applications of ML focused on navigating this space via predictive property screening, contemporary approaches utilize generative foundation models to inversely design molecular structures and synthetic pathways with targeted functionalities.

Furthermore, chemistry is characterized by emergent phenomena—concepts encapsulated by Anderson's axiom that "more is different."[580] Nontrivial complexities arise in macroscopic systems where symmetry breaking and environmental interactions render reductionist physical laws computationally prohibitive. Historically, this forced chemists to rely on heuristics and fragmented rules of thumb, advancing the field primarily through trial and error. Today, AI bridges the dichotomy between inductive data-driven approaches and deductive physical theories. PIML and neural operators now allow for the extrapolation of accurate quantum mechanical properties at a fraction of the computational cost, effectively democratizing high-fidelity simulation.[581] The impact of AI has become ubiquitous across all sub-disciplines. We are witnessing the rise of "self-driving" laboratories[582] that close the loop between computational hypothesis generation and robotic experimentation, drastically reducing the time to discovery. From the democratization of protein structure prediction via tools such as AlphaFold to the use of LLMs for extracting synthesis protocols from unstructured literature,[583] AI is no longer merely a tool for organizing data; it is an active participant in the scientific method. This review explores these transformative advances, detailing how the synergy of GAI, automated synthesis, and advanced theory is reshaping analytical chemistry, organic synthesis, catalysis, computational chemistry, and food chemistry (Figure 8).

AI in analytical chemistry

AI drives analytical precision through automated decision-making and intelligent spectral interpretation. The application of AI in the field of analytical chemistry has transcended the realm of mere data processing, advancing toward addressing complex structural elucidation, automated experimental decision-making, and green process design. It is steering the discipline from being experience driven to becoming an "intelligent and explainable" precise science. In spectroscopic analysis and structural validation, AI has notably enhanced the parsing capabilities and automation levels of multimodal data. Priessner et al. developed the multimodal spectral transformer model by integrating NMR (1H, 13C, HSQC, and COSY), IR, and MS data, overcoming the challenge of insufficient information from single spectra. After training on 4 million simulated compounds, the model achieved a top-1 structural prediction accuracy of 72% (top 3 reached 80%) and effectively bridged the distribution gap between simulated and experimental data through an "improvement cycle" strategy.[584] Moreover, GAI (ChatGPT-4) demonstrated potential in tackling specific metrology tasks. Wahab et al. utilized ChatGPT-4 to write MATLAB code that successfully converted free induction decay data of benzyl acetate into microwave rotational resonance spectra and achieved a 16-fold super-resolution reconstruction of biological images, indicating that LLMs can significantly lower the programming barrier for advanced data processing.[585]

AI has realized a transition from rules of thumb to a "quantitative science" in chromatographic separation and condition optimization. Addressing the issue of relying on chemists' intuition for column chromatography (CC) condition selection, Xu et al. constructed an automated platform and collected 5,984 data points from 192 compounds. Using symbolic regression, they discovered a clear mathematical equation linking thin-layer chromatography retention factor values with CC retention volumes, forming quantifiable "AI experiences."[586] DL has also accelerated the development of sustainable separation processes in green solvent design and process intensification. To tackle the high-energy-consumption problem associated with separating isopropanol (IPA)/water

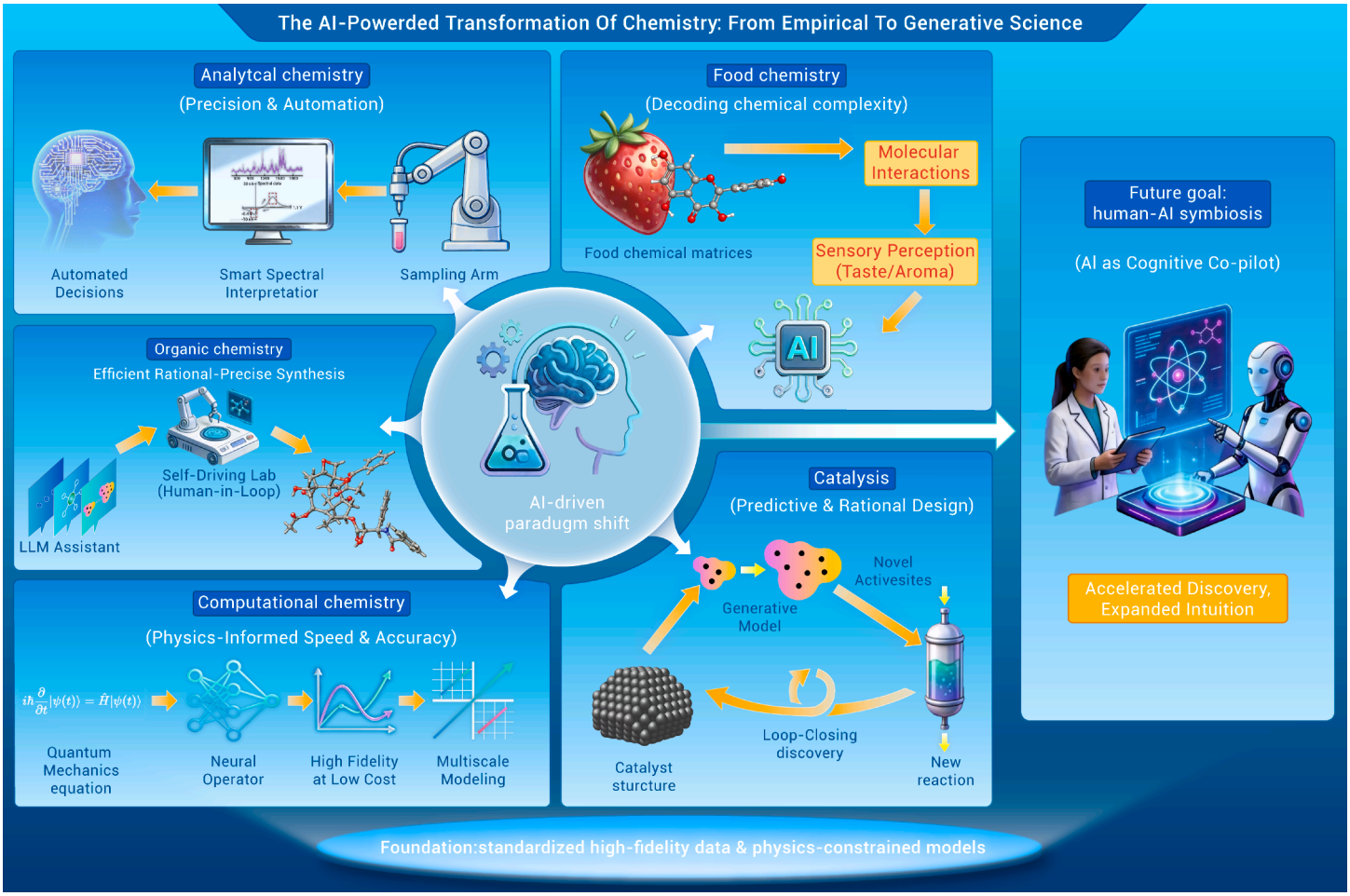

Figure 8. The AI-powered transformation of chemistry: From empirical to generative science

azeotropes, Wang et al. proposed a hybrid representation-based DNN (HrDNN) capable of accurately predicting physical properties such as melting and boiling points and toxicity (LD_{50}). Employing a DL-driven model, they screened five green solvents from 108 molecules for separating IPA/water azeotropes, achieving a 45.86% energy-saving potential when combined with heat pump distillation technology.[587] The application of AI in analytical chemistry is moving toward full-process automation and explainability, including the deep integration of AI decisions with robotic experimentation platforms, such as automating chromatographic data acquisition and converting AI's black-box predictions into explicit physicochemical rules understandable by chemists, thereby enhancing trust and practicality in precision analysis.

AI has become an essential tool for boosting sensor performance. An example is ion-sensitive field-effect transistors (ISFETs), where signals are notoriously affected by nonlinear drift, device variability, and environmental noise. In one study, an artificial neural network trained on data from a simple, non-functionalized graphene ISFET array showed remarkable versatility.[588] It achieved high accuracy in pH sensing, food adulteration detection, and pollutant identification without needing constant recalibration. This example underscores the power of a data-centric approach to compensate for hardware limitations, paving the way for intelligent sensing systems. Integrating AI closely into chemical sensing workflows goes beyond simply enhancing sensor functionality. This represents a rethinking of chemical sensor development. The focus is shifting from optimizing hardware specifications alone. Instead, the aim is to create robust, adaptable systems where sensing hardware and AI capabilities are tightly linked and mutually enhance each other.

The evolution of AI in bio-electrochemical sensing reveals a clear trajectory from *ex vivo* single-cell analysis to *in vivo* monitoring and from data-driven approaches to knowledge-integrated paradigms. In *ex vivo* single-cell analysis, Gao et al. pioneered the use of ML to precisely identify tumor cells from ion current fingerprints at nanopipettes, demonstrating AI's robust capability in processing discrete biophysical signals.[589] When the research focus shifted to the living brain, a far more complex and continuously dynamic system, earlier work by Xue et al. utilized a purely data-driven convolutional autoencoder.[590] This model, trained on extensive *in vitro* data and refined through transfer learning, enabled the resolution of dopamine, ascorbic acid, and ion signals. To address the challenges of limited data and complex interference *in vivo*, Li et al. then integrated electrochemical domain knowledge into AI models, proposing a "chemistry-informed" generative neural network (CIGNN).[591] By explicitly decoupling Faradaic and non-Faradaic currents, this approach significantly enhanced the accuracy and robustness of quantitative analysis. Looking ahead, this profound integration of AI with chemical principles is propelling sensing technology toward "edge intelligence." The polyelectrolyte-confined fluidic memristor developed by Xiong et al. further showcases the potential for sensors themselves to mimic synaptic plasticity and memory functions.[592] This implies that AI will no longer merely be an external analytical algorithm; instead, it is poised to be embedded within miniaturized, low-power hardware, performing brain-like computation and intelligent decision-making directly at the source of neurochemical signal acquisition. This trend of integrating AI into sensing hardware will reduce the latency between perception and response, providing the core hardware and intelligent foundation for next-generation high-performance, real-time neuro-modulation and diagnostic systems, such as implantable BCIs and closed-loop theranostic systems.

AI in organic chemistry

AI helps to usher in a new paradigm for the discovery of molecules, reactions, and mechanisms. AI is fundamentally reshaping organic chemistry, transitioning it from an experience-driven, trial-and-error discipline into a data-intensive and intelligent science. This paradigm shift is powered by synergistic advances across several core domains: LLMs as versatile chemical assistants, autonomous laboratories that execute closed-loop discoveries to produce high-quality

multimodal data, and predictive algorithms that overcome traditional data limitations.

LLMs have emerged as a prime engine for processing the "language" of organic chemistry. General-purpose AI agents such as ChemCrow[593] and Coscientist[594] integrated LLMs with tools and robotics to autonomously plan and execute complex tasks, from catalyst synthesis to experiment design. Simultaneously, domain-specific models such as Chemma,[595] which fine-tune for organic synthesis, have achieved remarkable accuracy in retrosynthesis planning. A critical trend in chemical LLMs is enhancing interpretability and data efficiency. Models such as LocalTransform[596] simplified reaction prediction by focusing on electron configuration changes, while approaches such as LocalMapper[597] demonstrated that minimal expert-labeled data can yield highly accurate atom mapping for reaction databases. The integration of GNNs with transformers in frameworks such as RXNGraphormer[598] enabled a unified approach to diverse tasks including reactivity prediction and synthesis planning.

The vision of autonomous discovery is physically realized in robotic laboratories. Systems such as Synbot[599] and "AI-Chemist"[600] adopted a fully integrated architecture where the AI software and specified method directed robotic hardware to perform synthesis with high efficiency and yield. This was complemented by ultra-high-throughput platforms, such as Fang's LCW-microfluidics system,[601] which can screen tens of thousands of photocatalytic conditions per day. These platforms addressed the reproducibility crisis through universal chemical programming languages such as X-DL, which allow for the precise, cross-platform execution of synthetic protocols.[602]

In predictive modeling, it moved beyond the bottleneck of data scarcity. Recent innovations included the creation of specialized benchmark datasets (e.g., NiCOlit[603]), transfer learning techniques that successfully apply knowledge across different reaction types, and multi-substrate screening strategies that evaluate catalyst generality from the outset. Retrosynthetic planning has also been revolutionized by GAI. Template-free and fragment-based models such as RetroTRAE,[604] along with edit-based and explainable frameworks such as Graph2Edits[605] and RetroExplainer,[606] achieved new levels of accuracy and diversity in route design. Furthermore, AI was unraveling complex reaction mechanisms. For instance, ML-based potential energy surfaces were providing atomic-molecular-level insights that can challenge long-held mechanistic assumptions, while DL models can now automatically classify reaction mechanisms directly from kinetic data.[607]

The integration of AI and related toolkits is forging a new paradigm of data- and knowledge-driven organic chemistry. The convergence of LLMs, predictive models, and autonomous laboratories is helping to overcome challenges in synthesis planning, mechanistic elucidation, optimization, etc. The foreseeable future of novel organic chemistry lies in developing lifelong learning systems that adapt to shifting data landscapes, embedding physical laws into AI frameworks more deeply, and building robust dry/wet experimental platforms that will unlock unprecedented opportunities for discovering novel organic molecules, reactions, and materials.

AI in computational chemistry

Physics-informed DL disrupts the traditional trade-off between computational efficiency and quantum accuracy. Computational chemistry has historically been defined by a rigid dichotomy: the high fidelity of ab *initio* quantum mechanics (QM) comes at a prohibitive computational cost, while the efficiency of classical force fields often sacrifices chemical accuracy. This accuracy-efficiency dilemma has long restricted high-throughput screening and long-timescale dynamics to simplified systems. In recent years, however, the integration of geometric DL and physics-informed architectures has fundamentally disrupted this paradigm, effectively bypassing the traditional scaling laws of computational theory.

ML is no longer merely an auxiliary tool for fitting parameters; it now serves as a central engine for multiscale modeling. The most significant advance has been the maturation of ML interatomic potentials (MLIPs) and equivariant neural networks.[608] These models, trained on vast datasets of DFT or coupled-cluster calculations, now routinely achieve *ab initio* accuracy at computational speeds orders of magnitude faster than traditional electronic structure methods. Crucially, the emergence of "foundational" universal potentials[609] has begun to replace system-specific force fields, allowing for zero-shot generalization across diverse chemical spaces, from drug-like organic molecules to complex inorganic lattices.

This transformative capability extends across all temporal and spatial scales. At the electronic level, neural wavefunction ansatzes are solving the Schrödinger equation for many-electron systems with unprecedented precision, challenging the "gold standard" of CCSD(T).[166] At the atomistic scale, ML techniques allow researchers to train on lower-level theory and correct to high-level accuracy with minimal data, drastically reducing the cost of high-fidelity thermodynamic predictions.[610] Furthermore, differentiable MD and generative coarse-graining methods are automating the discovery of reaction coordinates and collective variables, enabling the unbiased exploration of rare events and complex reaction kinetics that were previously intractable.[611–613] By encoding physical symmetries (rotation, translation, and permutation) directly into network architectures, modern AI does not just approximate physical laws—it enforces them, ensuring that the next generation of simulations is both data efficient and physically rigorous.[614]

AI in catalysis

AI converts catalysis research from an empirical art into a predictive and quantitative science. The development of highly efficient catalysts is a cornerstone of diverse chemical processes, as approximately 80% of industrial manufacturing relies on catalytic reactions at one or more stages of the production pathway.[615] Catalytic R&D has been referred to as "stir-frying" without clear experimental guidelines, and the catalytic system is referred to as a black box.[616] The integration of AI and ML is transforming catalysis from an empirical art into a quantitative, predictive science.[617–620] This paradigm shift confronts the field's inherent complexities: the vast composition-structure design space, the dynamic nature of active sites under working conditions, and the critical multiscale gap between atomic mechanisms and reactor-scale performance. AI's impact is manifested through four synergistic pillars. Predictive modeling (e.g., via GNNs) establishes structure-property relationships for rapid screening.[621,622] Generative design proposes novel catalysts tailored to specific performance targets.[616–624] AI-enhanced *operando* characterization deconvolutes complex data to track transient intermediates and surface dynamics.[625–627] These capabilities converge in autonomous laboratories, which close the loop between computational design, robotic synthesis, and testing to create accelerated discovery engines.[628,629] However, significant challenges persist: a scarcity of high-quality, standardized data limits model generalizability; the black-box nature of advanced models sacrifices interpretability; a disconnect remains between atomic-scale predictions and engineering reality; and a "realization gap" separates optimal computational designs from materials that are synthesizable, stable, and economical. Consequently, the field is pivoting from data-driven correlation to knowledge-informed causal inference. This requires developing physics-infused and XAI models, as well as community-driven efforts to build standardized data infrastructures. Ultimately, a human-AI symbiotic collaboration will elevate AI from a tool to a foundational component of catalytic science, systematically accelerating solutions for global energy and sustainability challenges.

AI in food chemistry

AI unveils real-world chemical complexity of food systems. Food chemistry represents one of the most chemically complex systems in nature, characterized by a high diversity of molecules, nonlinear interactions within multicomponent matrices, and diverse sensory and biological functions. Unlike conventional chemical systems that focus on isolated compounds, food embodies a hierarchical complexity spanning molecular, mixture, and perceptual-functional levels.[630]

AI is reshaping food chemistry from a theory-poor discipline into a discovery-driven science.[631,632] At the molecular level, AI enables the systematic mining of large-scale literature, spectral data, and chemical databases to uncover previously unrecognized key molecules and non-obvious structure-function relationships governing flavor, bioactivity, and stability.[633] At the mixture and matrix levels, where traditional reductionist approaches fail, AI reveals interaction patterns and behaviors arising from complex molecular combinations. ML models capture synergistic and antagonistic effects, uncovering hidden compositional rules that govern aroma release, texture formation, and chemical reactivity in food systems.[634,635] At the sensory and functional levels, AI bridges chemical composition with human perception and biological response, decoding how molecular ensembles are transformed into sensory experiences and

physiological effects.[636] By mapping chemical-sensory-functional data, AI provides new mechanistic insights into how food chemistry translates into taste, aroma, nutrition, and health outcomes. Together, AI-driven discovery across these three levels establishes a new paradigm for food chemistry that moves beyond prediction and optimization toward fundamental understanding of chemical complexity in real-world systems.

Perspective: Toward a human-AI symbiosis in chemical discovery

As AI matures from a novel tool into a foundational pillar of chemical research, the discipline stands at a critical inflection point. The transition described in this review—from empirical trial and error to predictive, generative science—suggests a future where the bottleneck of discovery is no longer human intuition or labor but the quality of data and the seamlessness of physical integration.

Looking forward, three key challenges must be addressed to fully realize this potential. First, the field must prioritize the construction of standardized, high-fidelity data infrastructures. The current reliance on fragmented, positive-result-biased literature limits the generalizability of models; a shift toward open, negative-data-inclusive repositories is essential. Second, we must bridge the interpretability gap. While DL offers unprecedented predictive power, trust in these systems requires the continued development of physics-informed and XAI that respects thermodynamic and quantum constraints, ensuring that predictions are not just accurate but chemically valid.

Finally, the realization gap between digital design and wet-lab synthesis remains the ultimate frontier. The future of chemistry lies in the complete closure of the loop, where GAI agents not only design molecules but directly drive robotic platforms to synthesize them. In this emerging era, AI will not replace the chemist; rather, it will evolve into a cognitive copilot, liberating researchers to focus on high-level conceptual creativity while intelligent systems navigate the vast complexities of chemical space.

AI IN ENERGY

Role of AI in paradigm change of energy research

Energy science is intrinsically characterized by the intersection of rigorous physical constraints and extensive engineering empiricism. It encompasses cross-scale analysis spanning working fluids, components, and systems, involving multi-objective thermodynamic, economic, and environmental performance, subject to strong coupling constraints among scenarios, parameters, and properties. Conventional research methods predominantly rely on empirical optimization algorithms and trial-and-error analyses for specific cases.[637] However, faced with the increasingly significant multi-objective requirements, strong nonlinear conditions, and complex dynamic scenarios of diversified renewable energy systems, traditional methods suffer from limitations such as weak generalization capabilities, slow computational inefficiency, rigid design modes, and suboptimal decision-making processes.[638]

These bottlenecks have necessitated a transition from rule-based modeling toward a new paradigm of computational intelligence, a shift deeply intertwined with the rapid evolution of AI. This technological convergence is driving the paradigm of energy research to transition from traditional empirical induction to AI-driven scientific discoveries. Recent progress in DL and RL has enabled data-driven approaches to learn complex nonlinear mappings and representations directly from high-dimensional variable spaces of data.[639] While these AI-based methods provide new pathways for nonlinear regulation, uncertainty analysis, and decision optimization in future energy systems, the current wave of paradigm shift is not merely the adoption of "more AI tools" but fundamentally the emergence of LLMs as intelligent collaborators for researchers and workflow orchestrators in energy science.[640]

Driven by the transformer architecture, LLMs have been increasingly explored as a unifying framework for various tasks, including natural-language processing, and are being extended to domains such as time-series analysis, knowledge mining, inference, and decision-making. In recent years, LLMs such as ChatGPT and DeepSeek, endowed with powerful cross-modal semantic understanding and reasoning capabilities, have demonstrated the capacity to translate complex constraints into readable, traceable decision rationales, thereby enhancing the efficiency of human-machine collaboration and the consistency of governance.[641] Furthermore, the development of domain-specific large models for energy and power systems to support more reliable and interpretable intelligent decision-making has emerged as a prominent research frontier.[642] China has developed several trillion-parameter energy and power industry large models, such as the "Guangming Large Model," "Watt Large Model,"[643] and "Qingyuan."[644] Internationally, the Electric Power Research Institute has established the Open Power AI Consortium to promote the development and deployment of open AI models tailored for the power sector,[645] while Westinghouse Electric has also developed "Bertha," an LLM for the nuclear field.[646]

As LLMs demonstrate distinctive capabilities in knowledge transfer, semantic understanding, and few-shot reasoning, they are increasingly evolving into intelligent collaborators for researchers and automation engines for scientific workflows, thereby not only changing energy research but also influencing energy education by catalyzing the emergence of new disciplinary knowledge within the field. This LLM-driven shift is increasingly influencing methodological transformations in energy research, although its effectiveness remains contingent on domain adaptation and validation.[647] Conventional energy systems research relies heavily on researchers' domain expertise and manual operations. From model formulation and code implementation to parameter tuning and result validation, each stage requires substantial human intervention, often extending research cycles to months or even years. By embedding physical models into LLM-driven workflows, this human-AI collaborative paradigm enables intelligent collaboration across the full life cycle from problem formulation to simulation and experimental validation.[648] Such a hybrid architecture not only mitigates the scientific hallucinations of GAI but also substantially lowers the expertise barrier for complex physical simulations through natural-language interaction, thereby accelerating the closed-loop process from theoretical modeling to experimental validation.

Furthermore, multi-agent collaborative frameworks based on LLMs can assist users in rapidly translating problem descriptions into optimization models. This human-AI collaborative mode liberates researchers from labor-intensive programming and hyperparameter tuning, allowing them to focus on problem formulation, solution evaluation, and innovation, thus shifting their role from "model builders" to "research decision-makers."[649] Building on human-AI collaboration, the energy field is observed to be gradually advancing toward an automated research pipeline, which can be defined as "energy research automation (ERA)" based on LLMs.

However, as probabilistic language models, LLMs lack inherent mathematical reasoning and physical constraint mechanisms, making them susceptible to logical errors in precise numerical computations and optimization tasks. A fundamental limitation lies in their black-box nature. Most models rely on high-dimensional, non-transparent representations that lack explicit physical interpretability, making it difficult to trace how inputs are transformed into outputs. This limitation becomes particularly critical in energy systems, where safety, reliability, and regulatory compliance are essential. In applications such as power system operation, nuclear energy management, and integrated energy system optimization, the inability to provide transparent and physically consistent explanations may hinder trust, validation, and large-scale deployment.[650] Therefore, a more rational strategy is to position LLMs as a language-driven orchestration layer, focusing on high-level cognitive functions such as natural-language interaction, knowledge retrieval, and task decomposition, while delegating numerical computation, physical constraint validation, and real-time control to specialized algorithms based on clear mechanistic foundations. Through tool integration interfaces, LLMs can flexibly coordinate numerical solvers, professional simulation software, and physics engines.[651] Coupled with RAG technology, the system can dynamically link authoritative external knowledge bases, providing auditable evidence chains with complete citations and logical foundations.[650] For example, drawing inspiration from the field of organic chemistry, LLM-based research pipelines have already proven effective in fine-grained building energy simulation.[652] This automated pipeline ERA, while maintaining engineering controllability, holds the potential to significantly reduce the cycle from problem formulation to conclusion verification, offering a new paradigm for next-generation energy research.[653]

This methodological revolution is firmly underpinned by the modernization of energy infrastructure and computational capacity. The widespread application of large-scale monitoring and data acquisition systems (SCADA), advanced metrology architectures (AMIs), and multimodal Internet of Things (IoT) sensors has accumulated massive amounts of observation data.[654] Concurrently, high-fidelity simulation technology based on GPU acceleration is becoming

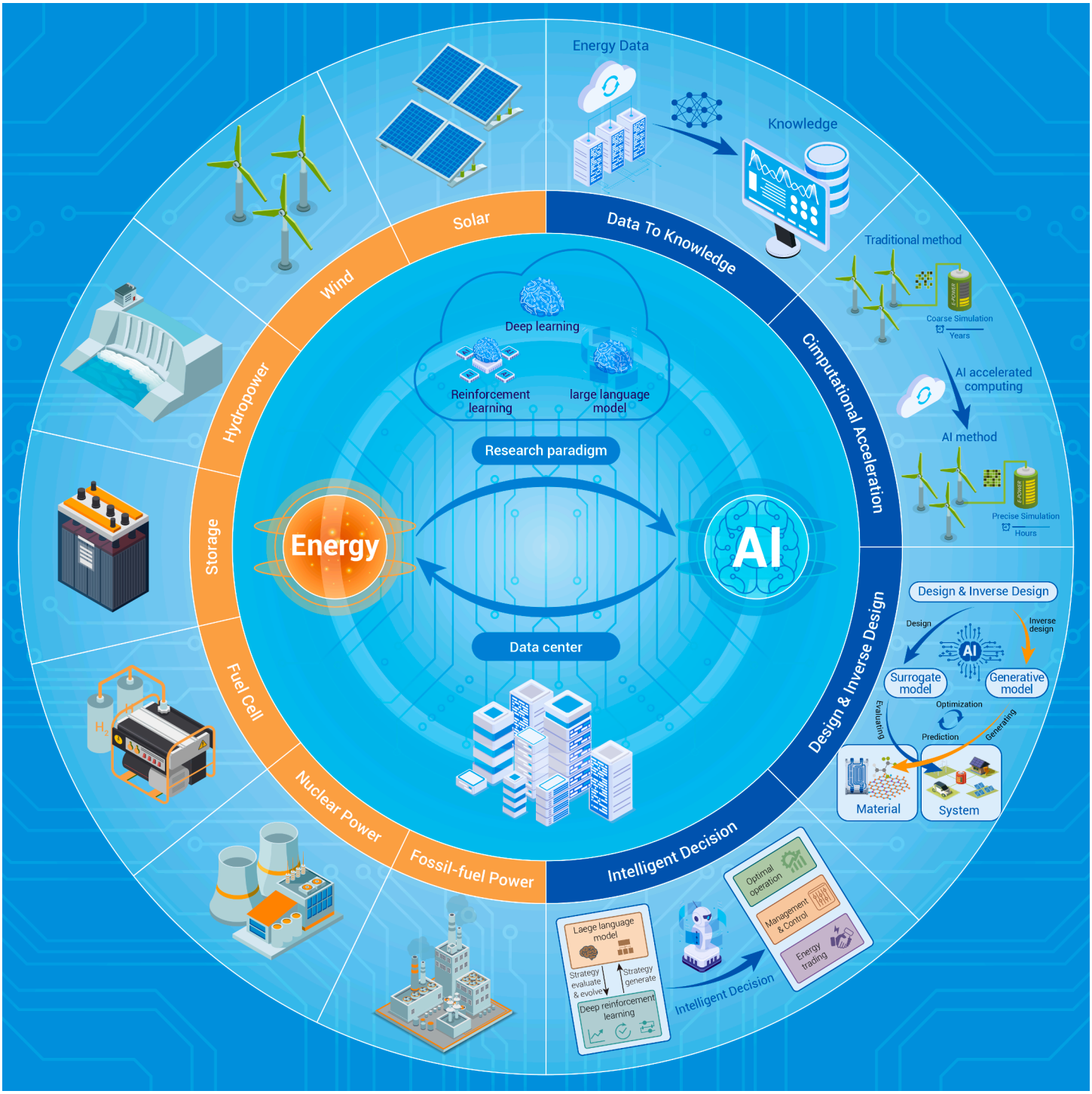

Figure 9. AI in energy

mature, generating large-scale simulation data for energy systems.[655] Further support is derived from the development of HPC clusters and cloud-native architectures, which offer efficient and stable computing resources for data processing and model training.[656]

This section focuses on the functional roles of AI in the energy domain, analyzing in detail the research progress of AI and LLMs in knowledge mining, computational acceleration, inverse design, and intelligent decision-making optimization in the energy domain, as shown in Figure 9.

Functional roles of AI in the energy domain

Energy data to domain-specific knowledge. The energy domain is undergoing a transition from model scarcity to data overload. From renewable energy generation, load behavior, and equipment operation and degradation to experimental measurements and high-fidelity simulation outputs, energy-related data are characterized by large-scale, heterogeneous sources, strong correlations, and deeply embedded physical mechanisms. How to transform massive and heterogeneous data into stable patterns and reusable domain knowledge has therefore emerged as one of the key bottlenecks constraining further progress in energy research and engineering.[657]

In recent years, AI-based approaches have demonstrated distinct advantages in enabling this transition from data to knowledge. AI provides a systematic means to uncover latent patterns and governing structures that are difficult to formalize through human intuition or predefined analytical assumptions. A large body of work has initially focused on prediction and state awareness. For example, deep temporal learning models have been widely applied to photovoltaic (PV)/wind power prediction,[658] energy load prediction,[659] and lithium-ion

battery diagnosis.[660] Compared with conventional statistical, ML, or shallow learning methods, these approaches typically achieve error reductions exceeding 20% and exhibit improved robustness under highly volatile and uncertain conditions.[659] By learning high-dimensional feature representations, such models map environmental variables, operational states, and historical behaviors into a unified latent space. This not only enhances predictive accuracy but also provides a data-driven basis for analyzing the coupled effects of multiple influencing factors. Similarly, in critical assets such as wind turbines, batteries, and deepwater energy equipment, DL-based fault diagnosis and health assessment methods enable much earlier anomaly detection, extending warning lead times from hours to days or even longer, thereby substantially reducing the risk of unplanned outages.[661,662]

Beyond predictive performance, AI is increasingly being used for explicit knowledge discovery, which is of greater methodological significance. Through explainable learning and sensitivity analysis, models can quantify the relative contributions of individual parameters to efficiency, reliability, or degradation behavior, allowing factors that were traditionally identified through expert intuition to be systematically validated or revised based on data.[663] In complex energy conversion and transport processes, such approaches often compress high-dimensional input spaces into a small number of dominant variables, achieving effective parameter reductions of more than 50% while maintaining comparable accuracy.[664] Meanwhile, learning frameworks incorporating sparsity constraints, symbolic regression, and physical consistency assumptions have enabled the recovery of approximate governing equations or equivalent evolutionary relationships directly from experimental or simulation data. For representative energy processes involving flow, heat transfer, and reactions, reconstruction errors can be limited to within 10%.[665,666] These results indicate that AI-based methods are not confined to black-box mappings but are increasingly capable of extracting conservation laws, scaling relationships, and dominant physical mechanisms.

AI has demonstrated the potential to reshape the "data-to-knowledge" pipeline in energy research, marking a broader paradigm shift from experience- and hypothesis-driven approaches toward a framework in which data and domain knowledge co-evolve in a mutually reinforcing manner.

Computational acceleration in energy science. As energy systems increasingly demonstrate characteristics of multiscale behavior, multi-physics coupling, and high-dimensional parameter spaces, traditional modeling approaches frequently fail to satisfy the requirements for high-frequency strategy optimization and large-scale scenario analysis due to limitations in computational resources and response times. AI-accelerated computing denotes the application of ML, DNNs, and physics-informed data-driven modeling to approximate or structurally reconstruct computationally demanding numerical simulations and optimization tasks in typical energy sectors, including fuel cells,[667] wind farms,[668] and country-level energy systems.[669] The objective is to significantly improve computational efficiency and broaden modeling capacity, while maintaining physical consistency and prediction accuracy, to facilitate swift simulation, optimization, and decision-making across various parameters and scenarios.

Despite AI's powerful capacity to model complex nonlinear relationships, several significant challenges remain in energy applications. A central issue is the incorporation of physical constraints such as thermodynamic laws and conservation principles into data-driven frameworks in a way that ensures both physical consistency and reliable generalization beyond well-observed operating regimes rather than producing purely correlational models that fail under distribution shifts. In addition, the inherently multiscale and multi-physics nature of energy processes poses further difficulty, as capturing fine-grained microscopic behaviors while preserving coherent macroscopic system trends remains a nontrivial task.[670]

In recent years, numerous physics-informed AI modeling methodologies have developed, including neural differential equations, PINNs, and mixed-fidelity surrogate models.[176,464] Incorporating control equations and structural priors into the learning framework has markedly enhanced model robustness in extrapolation tasks and state estimation. However, critical challenges remain, including guaranteeing closed-loop stability in learning-based controllers and attaining real-time multi-objective trade-off optimization. Importantly, AI-accelerated computing goes beyond numerical acceleration; it seeks to convert complex domain knowledge into computable and constrained model frameworks. It should be noted that as AI-accelerated computing increasingly merges with HPC platforms, parallel simulation engines, and diverse physical data sources, the design, scheduling, and dynamic simulation of energy systems will progressively achieve fine-grained resolution and extensive scenario generation.[671]

Though neural-network-based AI algorithms have been extensively used in these AI-accelerating frameworks, LLMs, as sophisticated model generators and coordinators, are also increasingly utilized to analyze unstructured physical knowledge, formulate design rules, and govern logic within the energy domain. These can subsequently be converted into explicit priors or constraints for subsequent models, thereby improving physical consistency and model reliability. The development of domain-specific LLMs that integrate causal reasoning with mechanistic knowledge is expected to further propel AI-accelerated computing from a paradigm of black-box approximation toward causality-driven physical modeling.

Energy system design and inverse design. Design and inverse problems in energy technologies focus on evaluating existing designs or generating new ones that meet specific objectives and constraints.[672,673] AI plays a crucial role in these challenges, using surrogate models to evaluate designs and generative models to propose new solutions. Surrogate models rely on data-driven methods to compute performance metrics such as thermodynamic properties, cycle efficiency, cost, environmental impact, and safety regulations,[674] offering greater exploration breadth and prediction accuracy at lower computation costs. Generative models use AI-driven optimization and algorithms, including RL[675] and GANs,[676] to generate solutions that meet predefined goals and constraints. While conventional design and inverse problems are solved via optimization or equation solving, the AI-based approach focuses more on the direct generation or end-to-end solution by enabling the systematic expansion of design spaces and the automation of design exploration beyond conventional trial-and-error approaches.

For the molecular design of working fluids, AI enables parameterized molecular design driven by application scenarios and performance objectives. The combination of supervised learning and quantitative structure-property relationship (QSPR) can directly correlate thermodynamic properties with molecular characteristics, simplifying thousands of candidate molecules into dozens of features, thus avoiding the time-consuming process of individually evaluating each molecule.[677]

For material design, AI can predict the key properties and identify novel materials by high-throughput molecular design (e.g., electrolyte materials for electricity storage,[678] phase-change materials for thermal energy storage,[679] perovskite materials for solar cells,[680] thermoelectric materials for thermoelectric power generation,[681] adsorbents for carbon capture and utilization,[682] radiative cooling materials for passive cooling,[683] and thermally conductive materials for thermal management)[684] and then identify the optimum candidate to avoid the time-consuming screening.

At the equipment level, data-driven surrogate models learn implicit mappings between design parameters, operating conditions, and performance metrics, enabling rapid evaluation and efficient exploration of large design spaces. Learning-based inverse design frameworks further allow direct inference of optimal configurations and operating strategies that meet prescribed performance targets. Representative applications include topological optimization of heat exchangers,[685] modular reactor design,[686] and thermal storage design.[687] Such approaches have reduced design iterations by an order of magnitude and computational cost by over 80% while achieving comparable or improved performance relative to conventional workflows.[686]

AI applications in design and inverse problems across the energy domain enable professionals to personalize energy systems by optimizing working fluids and materials based on surrogate and generative models. This AI-based inverse design can lead to significant improvements in efficiency, compactness, cost-effectiveness, and sustainability, ultimately tackling global energy challenges more effectively.

Intelligent decision-making and optimization. Modern energy systems are rapidly evolving into highly complex and tightly coupled infrastructures. At the same time, the potential influences of environmental science, economics, and sociology on system operation are being increasingly uncovered. Traditional approaches struggle under real-world constraints such as massive system scale, enormous volumes of multimodal data, intertwined multi-spatiotemporal scales, and deep multidisciplinary integration. These limitations have exposed

critical tensions, for example, between model fidelity and computational speed and between economic efficiency and operational reliability.

To address these systemic challenges, AI offers a new organizational framework for perception and decision-making in energy systems. AI methods—including DL, RL, and LLMs—can uncover complex relationships and latent structures from multimodal data. Meanwhile, AI4Science[10] enhances capabilities in data processing, hypothesis generation, and experimental simulation, accelerating AI adoption in energy system decision-making, scheduling, and control.[688] According to our Web of Science (WOS) database search,[689] over 20% of recent research in energy system operation and control now employs AI techniques.

A major advantage of AI technologies lies in their capacity to autonomously distill knowledge from massive, non-standardized, multimodal data without constructing precise mechanistic models,[690] thereby accurately identifying latent operational risks in energy systems and underpinning system-level operations, maintenance, and intelligent decision-making.[691] Among these approaches, DRL, which fuses neural networks with RL, has delivered substantial advances in efficient control,[692] optimal dispatch,[693] load management,[694] and the design and simulation of energy market mechanisms.[695] Especially when dealing with problems that involve high levels of uncertainty, DRL demonstrates unique advantages due to its characteristic of not relying on precise environmental models,[696] which is crucial for ensuring the resilience, coordination, and operational optimization of the energy system.

Now, AI technologies such as DRL address the challenges of massive data processing and uncertainty in energy systems, enabling risk identification and optimal control without precise models. However, they still face shortcomings in areas such as interpretability and reliability. One possible solution is to introduce LLMs to empower human-machine collaboration through cross-modal reasoning and traceable decision-making, partially resolving the limitations of existing AI technologies.

Watt-limiting AI

Energy has become the physical foundation of AI development. AI is transforming not only computational capability but also energy demand at scale, with profound implications for modern energy systems. The rapid expansion of AI workloads, especially in large-scale models and cloud services, is driving exponential growth in data center electricity consumption, straining grid capacity and threatening stability in regions with limited transmission and generation infrastructure. Globally, the data center market is projected to expand from USD 4.4 billion in 2022 to USD 6.6 billion by 2027, corresponding to a compound annual growth rate of 8.7%,[697] while the number of hyperscale data centers worldwide has been nearly doubling every 5 years. This infrastructure expansion is accompanied by a sharp rise in energy consumption. In 2024, global data center electricity use was estimated at approximately 415 terawatt-hours (TWh) and is projected to increase by approximately 100% by 2030, roughly equivalent to the current annual electricity consumption of Japan.[698] For scientific research, this paradigm shift implies that the cost of discovery is increasingly coupled with electricity prices and hardware availability. The training of a single large-scale foundational model (e.g., GPT-4 or specialized scientific transformers) can consume megawatt-hours of power, with costs escalating from thousands to hundreds of millions of dollars,[699] creating a "computational divide" that may restrict frontier research to only the most well-funded institutions.

More fundamentally, the high energy demand of AI and data center operations raises sustainability concerns if powered by fossil-fuel-dominant grids. In the United States (US), approximately 56% of electricity consumed by data centers is generated from fossil fuels, resulting in more than 105 million tons of CO_2 emissions annually.[700] In tandem with the rapid expansion of data centers, infrastructure historically relying on gas turbines or other fossil sources for reliability may exacerbate carbon emissions unless clean energy integration is prioritized. In contrast, policy and infrastructure innovations are emerging to address this challenge. China has introduced direct green electricity supply mechanisms,[701] such as requiring the proportion of green electricity used in newly built large data centers to exceed 80%,[702] under which each unit of electricity consumed by data centers is physically traceable to specific renewable energy projects. In practice, several large-scale data center projects in Inner Mongolia have already achieved power usage effectiveness (PUE) values as low as 1.15, with renewable electricity shares exceeding 90%.[703] These metrics suggest that the feasibility of the AI paradigm depends not only on algorithmic efficiency but also on the geographical and policy-driven alignment with renewable energy grids. Ultimately, ensuring a stable, low-carbon, and economically accessible energy supply is a critical prerequisite for the sustainable development of AI infrastructure in science.

Future trends and development paths

The future evolution of AI for energy is expected to move beyond isolated applications toward integrated, trustworthy, and autonomous intelligence across the energy domain. A central trend is the development of LLMs trained on large-scale, multimodal datasets spanning experiments, simulations, operations, and literature. Such models have the potential to provide transferable representations and serve as shared cognitive infrastructures for diverse energy tasks, from materials discovery to system-level analysis.

Another key direction lies in deep integration of physical principles and learning architectures. Physics-informed, constraint-aware, and hybrid modeling frameworks will play an essential role in improving generalization, data efficiency, and scientific credibility and interpretability, enabling AI systems to extrapolate beyond observed conditions and support decision-making in safety-critical energy applications. Closely related is the growing emphasis on uncertainty quantification and trustworthiness, which is crucial for deploying AI in long-term planning, infrastructure operation, and policy assessment.

AI is also poised to accelerate the transition toward autonomous energy research and engineering, where learning agents coordinate experiments, simulations, and optimization loops with minimal human intervention. Finally, advances in human-AI collaboration, including explainable interfaces and natural-language-based interactions, will lower disciplinary barriers and enhance knowledge sharing across the energy community. Together, these developments suggest a shift from AI as a computational tool to AI as a foundational enabler of next-generation energy science and technology.

The core non-technical bottleneck in applying AI technology to energy science is insufficient energy data openness and accessibility. Many energy-relevant datasets, including cost information, design specifications, operational records, and environmental measurements, remain fragmented, proprietary, or inaccessible, which restricts large-scale model training, cross-scenario validation, and the development of generalizable and reproducible AI solutions. The sensitivity of the energy and power industries makes the openness and sharing of such data extremely challenging, particularly in cross-border scenarios.

Finally, it lies in the development of supporting energy infrastructure. Data centers are electricity consumers while also being waste heat producers. By co-optimizing computing, power supply, waste heat recovery, and thermal networks, integrated architectures can harness renewable energy locally and recover otherwise wasted heat for district heating. In such systems, waste heat can be effectively elevated in value through heat pumps, reducing overall energy consumption and enabling low-carbon heating, with good economic feasibility.[704] From an operational perspective, data centers can be important power flexible regulators on the user side, as they can modulate computing loads in response to real-time grid signals, provide ancillary services, and participate in demand response programs.[705] AI can facilitate this interaction by forecasting demand, optimizing workload schedules, and aligning computational tasks with periods of high renewable availability, thereby lowering power usage by 25% for 3 h during peak demand while maintaining AI quality of service guarantees.[706] This grid-interactive paradigm offers potential benefits for both energy networks and digital infrastructure: improved stability, reduced reliance on fossil-based peaking resources, and more efficient use of renewable generation. Addressing these challenges requires reimagining data centers not as static loads but as active, flexible participants in integrated energy systems.

CONCLUSION

AI is rapidly evolving from a supporting computational technique into a potentially important enabler of energy domain. Across knowledge mining, computational acceleration, inverse design, and intelligent decision-making optimization, AI provides scalable tools to manage complexity, reduce computational cost, and extract actionable knowledge from high-dimensional systems. Importantly, the integration of physical principles with data-driven learning ensures robustness, interpretability, and generalizability. Together,

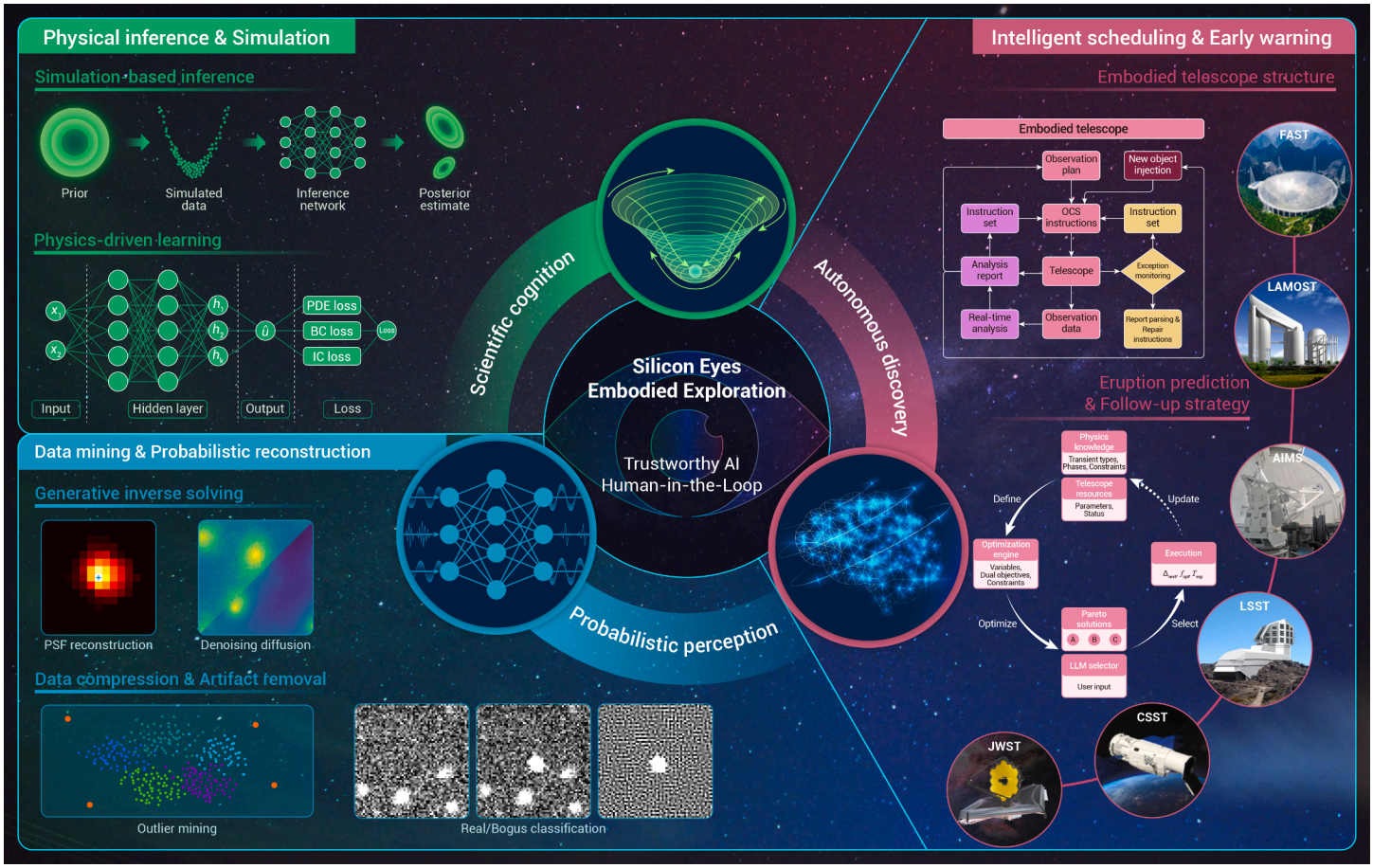

Figure 10. AI in astronomy

these advances signal a paradigm shift in the energy field, where data, models, and intelligence are increasingly co-developed to accelerate innovation and support the transition toward more efficient, reliable, and sustainable energy technologies.

FROM SILICON EYES TO EMBODIED EXPLORATION: THE AI-DRIVEN REVOLUTION IN ASTRONOMICAL EPISTEMOLOGY

Role of AI in the epistemic shift of astronomy

The history of astronomy is, in essence, a chronicle of the minimal compression of cosmic information and the expansion of human cognitive dimensions. Driven by next-generation survey projects, astronomy is undergoing a cognitive phase transition from data scarcity to a regime of extreme abundance governed by probabilistic uncertainty. AI has transcended its role as an auxiliary instrument to become the foundational medium restructuring the epistemology of the discipline. This section synthesizes a dynamic scientific closed loop composed of perception, cognition, and embodied discovery: at the perception layer, data analysis with probabilistic programming is employed to resolve uncertainty, transforming observational noise into probabilistic representations of physical parameters; at the cognition layer, foundation models and symbolic regression facilitate hypothesis formulating by distillation of analytical physical laws from cross-scale correlations; and at the discovery layer, embodied intelligence—forged through digital twins and RL—drives telescopes to execute the observation autonomously and report the explorations rather than passively recording. While addressing deep-seated challenges such as physical embedding, interpretability, and the redefinition of the scientist's role, we posit that just as Kepler distilled laws from chaotic data and Newton revealed the gravitational order of the cosmos, the AI of today is poised to guide us in transmuting vast oceans of survey data into physical truths that strike at the very heart of the universe (Figure 10).

The cognitive phase transition: From photon harvesting to bit mining. For millennia, astronomers have stood as both sentinels and interpreters of cosmic signals. As Stephen Hawking observed, the exploration of the universe represents the ultimate release of the human spirit—a spirit that has driven astronomy to transcend cognitive boundaries since Galileo initiated the era of empirical science. However, with the advent of astronomical mega-science facilities, the discipline is experiencing a paradigm shift from photon harvesting to bit mining.[707–714] Facing the constructed Square Kilometre Array (SKA), projected to archive 700 PB of data annually, and the initial operating Legacy Survey of Space and Time (LSST), which will detect millions of transient events per night, the traditional mode of manual verification has become epistemologically obsolete. This data deluge precipitates a profound cognitive paradox: while possessing the richest observational samples in history, our insight into the universe risks stagnation due to information overload, necessitating the emergence of a highly efficient parsing paradigm.

Paradigm fusion: Constructing the "perception-cognition-embodied" closed loop. The intervention of AI has catalyzed a data-driven analysis paradigm capable of identifying laws directly from high-dimensional data, thereby orchestrating a deep fusion with traditional theory-driven approaches. By hard coding physical priors (such as conservation laws) into neural networks, or utilizing symbolic regression for mathematical reverse engineering, astronomy will soon construct a hybrid research paradigm that is both flexible and rigorous. This paradigm converges into a full-life-cycle scientific closed loop covering perception, cognition, and embodied discovery, elevating AI from a supporting instrument to the underlying medium that reconstructs the epistemological framework of the field.

Probabilistic perception: The probabilistic reconstruction from observational noise to physical truth

As a non-interventional observational science, astronomy faces the challenge of verifying irreproducible phenomena. To address this, the discipline has already built a verification system centered on statistical consistency and multi-band cross-validation. The deep integration of AI has given rise to

a probabilistic analytical paradigm, the core logic of which lies in constructing physical forward models via probabilistic programming. This approach deconstructs observational noise into probabilistic representations of physical parameters, achieving a fundamental physical reduction of the observational process.

In practice, this paradigm drives an evolution from discriminative tasks to generative inverse problem solving. For discriminative analysis, AI substantially improves the statistical efficiency of classifying galaxy morphologies, gravitational lenses, and transient sources in large-scale surveys, tasks that have traditionally relied heavily on manual inspection.[715–723] Conversely, generative reconstruction abstracts imaging as an inverse problem; diffusion models are employed to enhance the recovery of faint signals while simultaneously employing uncertainty calibration mechanisms to circumvent "algorithmic hallucinations," preventing system errors from being misread as physical structures.[724–726] Furthermore, this paradigm captures genuine physical outliers within statistical fluctuations through a rigorous uncertainty quantification system and extracts robust, cross-survey universal representations using self-supervised pre-training, thereby laying the groundwork for a global collaborative detection network[727–731] and allowing an automatic data mining process.

Scientific cognition: A paradigm revolution in physical inference and simulation

Relying on the deep representational capabilities of astronomical foundation models for cross-scale, multimodal data, researchers are now able to capture implicit physical correlations within multi-dimensional latent spaces.[732–737] Even more revolutionary is the developing technique of symbolic regression, which distills high-dimensional correlations captured by neural networks into concise analytical equations, realizing a sublimation from regression prediction to the extraction of physical laws.[738–741]

Two frontier pathways drive the current inference paradigm. Likelihood-free probabilistic inference utilizes generative models to bypass explicit likelihood calculations, coordinating with differentiable simulation techniques to efficiently search high-dimensional spaces under gradient guidance while ensuring statistical rigor through simulation-based calibration (SBC).[742–749] Simultaneously, physics-constrained learning hard codes physical equations into the algorithmic foundation, enhancing the model's extrapolation stability and analytical interpretability.[750–756] By deeply decoupling physical priors from complex observational effect models, astronomy relies on multi-physics simulation benchmarks such as CAMELS to establish a scientific argumentation framework that aligns with physical intuition while withstanding the scrutiny of massive datasets.[757,758]

Autonomous discovery: The embodied loop of "observe-infer-reobserve"

Once perception and cognition possess physical fidelity, the research paradigm leaps toward intent-driven closed-loop exploration. This architecture, powered by embodied intelligence, will soon construct a complete "perception-cognition-action" loop, similar to that shown in Figure 10. At the input, large model agents proactively retrieve and synthesize alerts, literature, and archive data, identifying "unknown unknowns" of high scientific value.[759–761] Meanwhile, the decision layer employs RL and Bayesian optimization to solve high-dimensional sequential decision problems, achieving optimal configurations under scarce observational resources and translating strategies into control scripts via tool invocation to realize automated data acquisition.[762–766] Furthermore, distinct from traditional automation, embodied intelligence systems will possess introspective perception capabilities that can monitor telemetry data to sense faint anomalies, facilitating a shift from passive maintenance to active self-healing.[767–769]

Under this paradigm, human scientists will soon sublimate into architects of intent and adjudicators of logic, while AI handles most noise and foundational inference, allowing humans to focus on the ultimate philosophical scrutiny of latent laws.

Outlook and summary

AI has evolved from an auxiliary tool into the core infrastructure for exploring the cosmic frontier. Future transformative breakthroughs will concentrate on three dimensions.

First, from "low explainability" to "physical transparency": establishing XAI frameworks to realize the semantic mapping between model logic and physical parameter spaces, rendering inference logic transparent. Second, ensuring the self-consistency of data-driven approaches through physical information embedding. Finally, the human role in the AI era requires redefinition; scientists will be liberated from tedious operations to dedicate themselves to the top-level logical construction of cosmic essence and the adjudication of physical hypotheses.

In summary, only by establishing a traceable, reproducible chain of physical evidence can AI4Astronomy truly evolve from a toolbox into an engine of scientific discovery, helping us transmute the vast oceans of survey data into physical truths that strike at the very heart of the universe.

INTERDISCIPLINARY FRONTIERS: AI-DRIVEN SCIENTIFIC CONVERGENCE

AI in BCIs

BCIs function as vital channels that convert neural activity into executable commands for machines.[276,770] Despite their transformative potential, traditional BCI systems face three core limitations: (1) the inherently low signal-to-noise ratio (SNR) characteristic of non-invasive recording methods such as EEG, often falling below 0.5 dB; (2) significant variability across individuals, which necessitates extensive personalized calibration; and (3) latency periods exceeding 300 ms, impairing real-time responsiveness. The integration of AI has fundamentally reshaped this domain by directly addressing these constraints. DL methodologies facilitate end-to-end feature extraction to boost the SNR. Transfer learning techniques have reduced calibration durations, while neuromorphic computing platforms enable closed-loop latencies, thereby supporting real-time control capabilities (Figure 11).

AI-enhanced BCI workflows. *Signal acquisition and enhancement.* A persistent obstacle in BCI signal acquisition is the contamination of neural recordings by non-stationary noise, which can affect 30%–60% of the data. AI-driven denoising strategies have proven highly effective in mitigating this challenge. Cutting-edge hybrid architectures that combine CNNs with GANs have emerged as leading solutions. For example, Yin et al. introduced GCTNet,[771] a GAN-guided parallel CNN-and-transformer network, which achieved an 11.15% reduction in relative root-mean-square error (RRMSE) and a 9.81% improvement in the SNR compared to existing methods, underscoring substantial gains in signal clarity and decoding accuracy.

Neural decoding architectures. In the context of motor intent decoding, motor BCIs have historically focused on controlling single effectors, such as cursors or robotic arms, moving fingers collectively. Wodlinger et al. advanced this field by decoding four distinct grasp postures.[772] However, true individuated finger control—enabling complex tasks such as typing or playing musical instruments—has remained limited, with continuous decoding demonstrated only for two finger groups in non-human primates. Willsey et al. overcame this barrier by employing a shallow feedforward neural network decoder to continuously decode three independent finger groups, including 2D thumb control, totaling four degrees of freedom—effectively doubling previous capabilities.[773] This system enabled precise manipulation of a virtual quadcopter, exemplifying a high-performance, intuitive intracortical BCI that fosters natural brain-to-digital interfacing, thereby enhancing user engagement in recreation, socialization, and empowerment. For linguistic decoding, Chen et al.[774] achieved a significant breakthrough by developing a DL framework capable of decoding human speech directly from neural signals, a critical advancement for restoring communication in individuals with neurological impairments. Their approach integrates an electrocorticographic (ECoG) decoder that translates cortical activity into meaningful speech parameters, coupled with a novel differentiable speech synthesizer that converts these parameters into spectrograms. To improve training efficacy, they introduced a speech-to-speech autoencoder generating reference parameters. Tested on 48 participants, the system produced natural-sounding speech with high reproducibility and strong correlation, even under real-time constraints. Remarkably, it successfully decoded speech from both hemispheres, offering promising applications for speech prostheses in patients with unilateral brain damage.

High-impact application domains. *Medical neurorehabilitation.* AI-driven closed-loop BCIs have demonstrated remarkable success in restoring motor functions. Herring and colleagues pioneered a method to restore upper limb

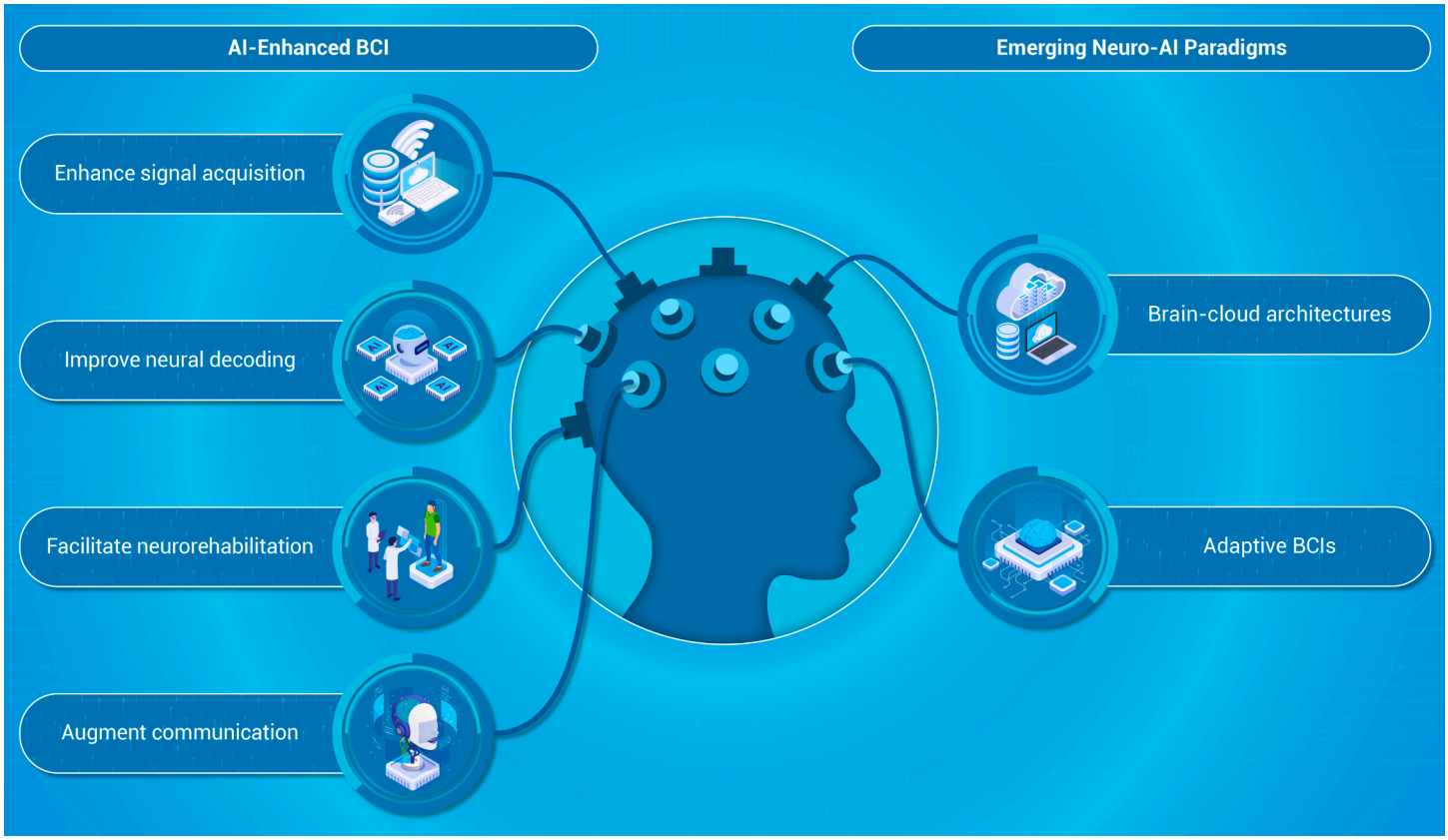

Figure 11. AI in brain-computer interfaces Integrating AI into the BCI workflow enhances signal acquisition, improves neural decoding, facilitates neurorehabilitation, and augments communication. Emerging paradigms, including brain-cloud architectures and adaptive BCIs, further advance the integration and development of AI and BCI.

function in a 27-year-old man with high cervical tetraplegia caused by spinal cord injury, which disrupts brain-to-limb nerve communication.[775] Their approach combined selective peripheral nerve stimulation with intracortical microelectrode recordings from sensorimotor regions, alongside intracortical microstimulation to reinstate tactile sensation. Implanted arrays captured neural signals associated with intended movements, while targeted nerve stimulation elicited precise muscle contractions, enabling both virtual and physical control of arm and hand movements. This bidirectional brain-machine interface demonstrated safety and feasibility, offering promising avenues for restoring motor and sensory functions following neurological injury.

Augmented communication. Addressing speech delays in individuals with paralysis, Littlejohn and colleagues utilized high-density surface recordings from the speech sensorimotor cortex of a severely paralyzed participant to control a real-time, naturalistic speech synthesizer.[776] Employing DL RNN transducer models, they achieved fluent, large-vocabulary speech synthesis personalized to the participant's original voice with rapid 80 ms neural decoding latency. The system demonstrated continuous, implicit speech detection and generalized effectively to other silent-speech interfaces, presenting a promising neuroprosthetic solution to restore natural spoken communication for people with paralysis.

Emerging neuro-AI paradigms. *Brain-cloud architectures.* As BCI applications transition from controlled laboratory environments to real-world settings, hybrid edge-cloud frameworks are gaining prominence.[777] While multisensory devices and hybrid methods enhance accuracy, they also generate large volumes of data and increase computational complexity, straining both field and on-site resources. This often necessitates costly hardware upgrades and leads to inefficient resource utilization. To mitigate these challenges, offloading computation to cloud platforms has become increasingly attractive due to their scalable resources, parallel processing capabilities, and ubiquitous accessibility. Advances in internet technology facilitate seamless integration of portable BCI headsets with cloud infrastructure. However, indiscriminate cloud usage incurs high costs for both users and providers, underscoring the need for research into efficient cloud-BCI integration strategies. Future efforts must focus on optimizing this balance to ensure cost-effective, reliable cloud adoption in BCI systems.

Adaptive BCIs. BCIs enable direct brain-to-device communication, supporting human-machine integration and neurological therapies. Yet the brain's dynamic and fluctuating signals pose challenges to consistent BCI performance, necessitating adaptive decoders that evolve alongside neural activity. Conventional decoders depend on pre-labeled datasets and are limited by the energy inefficiency and latency inherent in von Neumann architectures. Neuromorphic computing, particularly memristor-based chips, offers a compelling alternative by enabling in-memory computing with high energy efficiency and real-time adaptability. Liu et al. developed a memristor-based adaptive decoder tested in real-time brain-controlled drone flight, achieving accuracy comparable to software-based methods but with drastically reduced energy consumption and faster processing speeds.[778] Their interactive update scheme fosters co-evolution between brain signals and the decoder, improving accuracy by approximately 20% over 6 h sessions across multiple participants, marking a significant advance toward practical, efficient BCIs.

Critical challenges and AI solutions. *Technical barriers.* The development of DL models for BCIs is hindered by data scarcity and heterogeneity. These models require extensive and diverse neural datasets to perform reliably, yet such comprehensive data are often difficult to acquire due to variability in neural signals and limited availability. To address these issues, advanced techniques such as data augmentation, transfer learning, and synthetic data generation are actively pursued. These strategies enhance model robustness and generalization, enabling effective interpretation of neural signals across diverse sources and conditions. Furthermore, real-time processing of large-scale neural data imposes stringent demands on computational efficiency and energy consumption. Traditional hardware frequently struggles to meet these requirements, resulting in latency and excessive power usage. Neuromorphic hardware platforms have emerged as promising solutions by emulating the

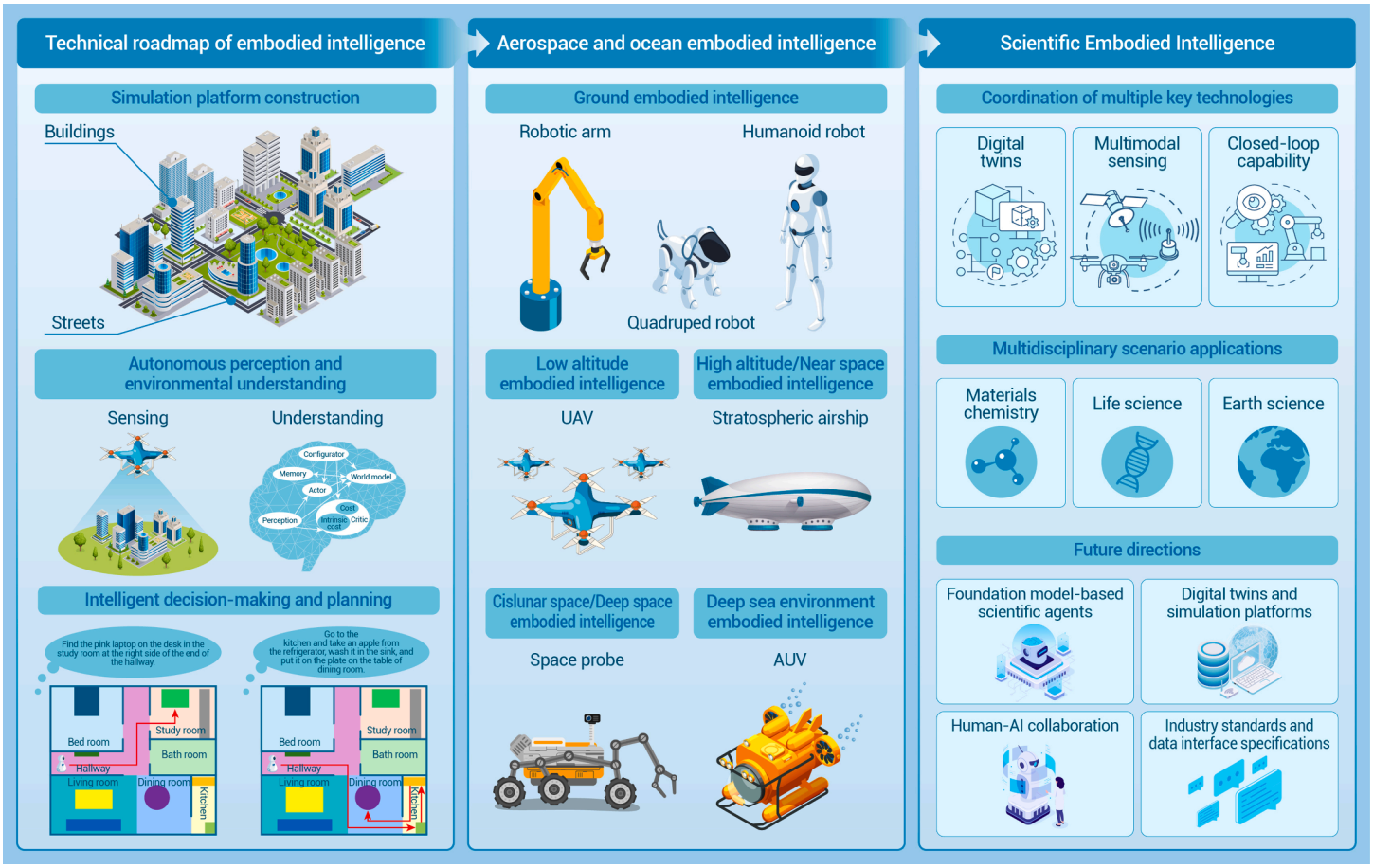

Figure 12. Embodied intelligence

brain's structural and functional principles and employing event-driven computation. These platforms enable rapid, energy-efficient processing, which is critical for integrating AI algorithms into real-time BCI applications, ensuring timely and accurate neural signal interpretation while maintaining sustainable power consumption.

Ethical and security risks. Adversarial attacks pose a significant threat to BCI reliability,[779] as carefully crafted perturbations can consistently deceive EEG classifiers with alarming success rates. To counter these vulnerabilities, advanced defense mechanisms incorporating adversarial training and robust optimization have demonstrated substantial effectiveness in enhancing system resilience. Additionally, neuroprivacy concerns are being proactively addressed through privacy-preserving federated learning frameworks.[780] Techniques such as differential privacy and secure multi-party computation facilitate decentralized model training, ensuring that sensitive user data remain protected and uncompromised throughout the learning process.

Future directions and conclusion. Causal neuro-AI approaches leveraging GNNs to capture effective connectivity hold great promise for elucidating complex biomarkers, particularly those associated with depression within limbic system circuits. The integration of LLMs with BCIs is expected to enhance thought-completion capabilities and improve communication efficacy. Moreover, quantum neural interfaces (QNIs) represent a future direction by combining the intricate complexity of neural activity with the unparalleled computational power of QM. By exploiting quantum phenomena such as superposition, entanglement, and interference, QNIs offer the potential to mitigate signal degradation, reduce energy consumption, and vastly expand bandwidth—addressing current limitations of existing BCI systems. AI has evolved beyond a mere signal processing tool to become the cognitive engine propelling modern BCIs, enabling revolutionary progress in precision neurotherapeutics, naturalistic communication, and adaptive embodiment.[276] Neuroprosthetics optimized through RL provide proprioceptive feedback that significantly enhances users' sense of embodiment. Looking forward, it is imperative to establish standardized validation protocols for AI-BCI systems, develop ethical frameworks tailored to neurotechnology, and advance integrated hardware-software co-design for implantable AI chips. These initiatives are essential to ensure seamless integration and facilitate the clinical translation of next-generation neurotechnologies.

Embodied intelligence for science

Research landscape of embodied intelligence. *Conceptual introduction.* Embodied intelligence refers to an AI paradigm wherein intelligent agents learn and accomplish tasks through real-time perception, interaction, and action with their environment. Its core lies in the physical embodiment of the agent and the closed-loop feedback generated through interaction with the environment. Unlike traditional AI agents that process information solely in virtual spaces, embodied agents must understand instructions, perceive their surroundings, plan actions, and execute tasks within complex physical worlds. Their intelligence develops through the "interaction-learning-adaptation" cycle with the environment, making it a vital path toward achieving AGI.[781] As illustrated in the Figure 12, owing to its powerful capabilities in high-fidelity scene modeling and simulation, which provides safe, parallelizable training environments and enables simulation-to-real (sim-to-real) transfer, multimodal environmental perception and understanding (integrating vision, LiDAR, force, and chemical sensors to form a comprehensive, context-aware representation of the physical world), and intelligent, closed-loop decision-making and planning (decomposing high-level goals into executable action sequences and adapting in real time based on perceptual feedback), embodied intelligence excels in a diverse spectrum of task scenarios—ranging from ground and low-altitude operations (e.g., flexible manufacturing and autonomous inspection) to extreme environments (e.g., planetary surface exploration and deep-sea resource assessment). More importantly, by transforming the scientific process from passive data analysis into an active, physical closed-loop of "perception-decision-action," it holds significant promise for driving novel scientific discoveries. This is evidenced by its emerging role as an "autonomous experimental operator" in fields such as materials science (e.g., A-Lab for autonomous materials synthesis), chemistry (e.g.,

Table 3. Paradigm shift from traditional AI4S to embodied scientific discovery

Dimension	Traditional AI for science	Embodied intelligence for science discovery
Core role of AI	data analysis, predictive model	autonomous operator
Interaction mode	offline, passive, relying on historical/simulated datasets	online, active, and physically embodied closed-loop interaction
Objective	discovering patterns in data and optimizing known variables	actively intervening in experimental processes, exploring unknown parameter spaces, and generating novel hypotheses
Representative works	AlphaFold,[782] computational chemistry simulations[594]	A-Lab,[7] Coscientist[783]

LLM-driven systems for complex reaction execution), and space exploration (e.g., cognitive intelligence-guided sample collection), fundamentally enabling a more autonomous, dynamic, and parallelizable research paradigm. In the following sections, we will systematically introduce the key enabling technologies, exemplary applications, and contributions to scientific discovery of embodied intelligence.

First, embodied intelligence represents a fundamental paradigm shift in the field of AI. Its core thesis posits that intelligence formation does not rely solely on abstract symbol processing or the analysis of offline data but also originates from the closed-loop feedback generated by continuous, real-time perception, interaction, and action between an intelligent agent (possessing a physical or virtual body) and its environment. This concept emphasizes that physical embodiment and environmental interaction are necessary conditions for the emergence of intelligence. At the technical level, embodied agents must overcome significant challenges in understanding high-level instructions, accurately perceiving surrounding environmental states, planning sequential actions, and executing tasks safely and reliably within complex and uncertain real-world physical environments. Consequently, embodied intelligence is regarded as a crucial pathway toward overcoming the bottlenecks of generalization, adaptability, and commonsense reasoning in AI, ultimately contributing to the achievement of AGI.[781]

Second, the introduction of the embodied intelligence paradigm into scientific research has catalyzed a second leap in the scientific paradigm. The first leap involved AI4Science utilizing powerful computational and algorithmic capabilities to identify patterns in vast datasets, predict molecular properties, and accelerate simulations, yet it remained essentially an "offline" analytical tool based on historical data. The integration of embodied intelligence, however, enables agents to actively operate instruments, intervene in experimental processes, observe immediate phenomena, and adjust research strategies, much like human scientists. This mode, which combines AI's rapid reasoning capabilities with the embodied operational abilities in the physical world, is transforming scientific research from the traditional linear, time-consuming cycle of "hypothesis-design experiment-manual operation-data analysis" into a highly autonomous, dynamically closed-loop, and parallelizable "active exploration" mode. This not only significantly enhances research efficiency but also holds the potential to catalyze disruptive scientific discoveries by executing complex experimental strategies that are either inaccessible to humans or previously unimagined.

The shift from traditional AI4Science to the paradigm of embodied intelligence represents a fundamental transformation in how AI interacts with and contributes to the scientific process. This evolution is not merely incremental but constitutes a radical redefinition of the AI agent's role—from a passive analytical tool to an active, embodied participant within the experimental loop. To systematically elucidate the core distinctions between these two generations, we delineate their contrasting characteristics across several key dimensions in Table 3.

Technical roadmap. Due to the high cost and risks associated with training robots in the real world, constructing high-fidelity, physically accurate simulation environments (e.g., NVIDIA Isaac Sim and Microsoft AirSim) is a primary foundation. These platforms simulate not only visuals and sounds but also accurately model rigid-body dynamics, fluid dynamics, and various sensor data. They provide a safe, unlimited, parallelizable, and efficient training ground for algorithms and enable the transfer of trained policies to physical robots through sim-to-real techniques.[784]

Building on this foundation, agents also need to integrate multimodal sensor information (e.g., RGB-D cameras, LiDAR, force/torque sensors, and chemical sensors) to form a comprehensive, context-aware understanding of their environment. This includes not only object and scene recognition but also the comprehension of spatial relationships between objects, physical properties (e.g., hardness and fluidity), and the current state of tasks (e.g., progress of a chemical reaction), thereby providing rich and accurate data for decision-making.

Furthermore, intelligent decision-making and planning constitute the "brain" of the system. Utilizing methods based on RL, imitation learning, and planning that incorporate symbolic knowledge, agents must be able to decompose high-level goals into sequences of executable actions and replan in real time based on perceptual feedback to handle unexpected situations, ensuring task completion.

For many large-scale scientific tasks, the capabilities of a single agent are limited, necessitating collaboration among multiple agents through communication and coordination mechanisms. This involves complex issues such as distributed perception, task allocation, and cooperative path planning, aiming to achieve a "1 + 1 > 2" collective intelligence effect.

Typical applications of embodied intelligence. *Ground embodied intelligence (vehicles, quadrupeds, and humanoids).* Ground-based embodied intelligent systems primarily include wheeled, legged, and humanoid robots, which play significant roles in smart manufacturing, national defense security, and high-risk operations. In smart manufacturing, mobile manipulators with autonomous navigation and fine manipulation capabilities enable flexible production line transformation, performing tasks such as material handling and component assembly. In national defense, quadruped robots adapt to complex terrains for reconnaissance, explosive ordnance disposal (EOD), and other hazardous missions, effectively ensuring personnel safety. These systems are typically equipped with various sensors such as LiDAR and IR cameras, possessing capabilities for environmental 3D reconstruction and dynamic obstacle avoidance. For high-risk operations, humanoid robots can replace humans in hazardous environments such as nuclear plants or chemical facilities for equipment inspection and emergency response. These robots feature bipedal locomotion and bimanual coordination, enabling them to use standard tools in complex tasks such as valve operation and equipment disassembly.

Low-altitude embodied intelligence. Low-altitude embodied intelligent systems, primarily using UAVs as platforms, show great potential in urban logistics, emergency response, infrastructure inspection, and precision agriculture. In urban logistics, multi-rotor UAVs enable rapid "last-mile" delivery, achieving precise building localization and obstacle avoidance through visual SLAM technology. In emergency response, UAV swarms can quickly reach disaster sites, constructing situational awareness panoramas through collaborative perception to provide real-time data support for rescue decisions. For infrastructure inspection, fixed-wing UAVs equipped with high-resolution cameras and multispectral sensors can automatically identify defects such as power line damage and pipeline leaks. In precision agriculture, UAVs utilize multispectral imaging to accurately monitor crop growth, enabling variable-rate fertilization and precise pesticide application, significantly enhancing agricultural productivity.

High-altitude and near-space embodied intelligence. High-altitude and near-space embodied intelligent systems mainly include stratospheric airships, solar-powered UAVs, and other platforms, offering unique advantages in atmospheric monitoring, wide-area communication relay, and Earth observation. For atmospheric monitoring, near-space vehicles equipped with high-precision sensors enable long-term, continuous observation of atmospheric composition, providing crucial data for climate change research. These platforms often employ lightweight structures and efficient energy management systems, capable of sustained flight for months or even years. In wide-area communication relay, high-altitude pseudo-satellites (HAPSs) can establish airborne communication networks, providing emergency communication services for remote areas. In Earth observation, high-altitude balloons carrying high-resolution imaging systems enable large-scale land surface monitoring, playing important roles in land resource survey and environmental monitoring.

Cislunar space and deep space embodied intelligence. Cislunar and deep space embodied intelligent systems represent key technologies for future space exploration, holding strategic significance for lunar base construction and operational support, autonomous planetary surface exploration, asteroid sampling, and deep space celestial body science. For lunar base construction, robots with autonomous operational capabilities can perform tasks such as site preparation and structure assembly, significantly reducing astronaut workload. These systems typically feature radiation-hardened designs and reliable operation capabilities in extreme temperatures and vacuum. In planetary surface exploration, rovers such as the "Perseverance" Mars rover achieve autonomous navigation and scientific operations through multi-sensor fusion technology; its intelligent sampling system autonomously collects and caches rock samples. Asteroid sampling missions require a probe-autonomous approach and rapid decision-making capabilities, as demonstrated by "Hayabusa2's" successful surface sampling via its visual navigation system. In deep space, scientific intelligent probes can autonomously select observation targets to optimize scientific output.

Deep-sea environment embodied intelligence. Deep-sea embodied intelligent systems primarily include autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), and other platforms, having made significant progress in seabed topographic mapping and resource assessment, submarine cable and pipeline inspection, and marine biological observation. In seabed mapping, AUVs equipped with multibeam bathymetry systems can achieve centimeter-resolution seabed terrain detection. For submarine pipeline inspection, ROVs utilize machine vision technology to automatically identify defects such as corrosion and free spans, enabling predictive maintenance of infrastructure. In marine biological observation, biomimetic robots allow for natural companion observation of marine life, providing novel technological means for marine ecological research.

Embodied intelligence for scientific discovery. Embodied intelligence represents a new research paradigm for scientific discovery. The core is to view an agent as an autonomous experimental operator that actively interacts with the environment. Through a perception-decision-action closed loop, embodied intelligence enables a shift from passive observation to active exploration in scientific discovery.[785]

Coordination of multiple key technologies. Embodied intelligence in research environments relies on coordinated advances in high-fidelity simulation, multimodal sensing, and autonomous operation. First, high-fidelity simulation and digital twins construct virtual replicas of research environments, providing safe and controlled platforms for algorithm training and risk assessment. For example, Lo et al.[786] proposed a "frugal twin" that applied a low-cost proxy system to simulate high-cost research processes, which enabled rapid iteration of data infrastructure and automation policies before they were transferred to real SDLs. Second, multimodal sensing enables an accurate understanding of experimental states and environmental changes by integrating vision, touch, and force information. For example, Mao et al.[787] developed a flexible vision-touch glove that allowed robots to perform visual localization while acquiring tactile cues such as pressure, temperature, and surface interaction, thereby improving the robustness of grasping and recognition. Finally, autonomous operation requires a closed-loop capability (including hypothesis formulation, testing, and replanning) to support workflow organization, multi-source result interpretation, and iterative decision refinement in physical environments. For example, Zhang et al.[380] applied an embodied cognitive intelligence framework with a dynamic cognitive map that fused real-time vision, 3D geometry, and expert knowledge, which enabled efficient autonomous operation and precise sample collection for Chang'e-6 in a lunar environment.

Multidisciplinary scenario applications. By empowering agents with the capabilities of continuous interaction and adaptive learning in physical environments, embodied intelligence is fundamentally driving the evolution of disciplines such as Earth science, materials chemistry, and life sciences. These fields are gradually transitioning from human-dominated paradigms toward autonomous, intelligent, and systematic directions. In Earth science, Liu et al.[788] utilized large-scale swarm collaborative planning algorithms to enable UAV swarms to perform dynamic task allocation autonomously, achieving continuous monitoring of wide-area environmental changes without human intervention. Zhang et al.[380] applied embodied cognitive intelligence to the Chang'e-6 lunar mission, constructing a system that combined cognitive mapping with autonomous environmental interaction for precise sample collection. In materials science, Szymanski et al.[7] developed A-Lab as an embodied system tightly integrating robotic synthesis with AI-driven planning, autonomously closing the physical-digital feedback loop to successfully synthesize 41 previously unknown inorganic materials. Building on the capacity of LLMs to significantly enhance cognitive understanding and decision-making capabilities of agents,[789] Boiko et al.[594] developed an LLM-driven system (i.e., Coscientist) with autonomous experimental design capabilities, which successfully executed complex palladium-catalyzed cross-coupling reactions. In life sciences, Rapp et al.[790] constructed an embodied intelligence system for protein engineering, achieving directed enzyme evolution by autonomously driving the closed loop of wet experiments. Gottweis et al.[783] introduced Coscientist as a multi-agent AI system powered by LLMs to generate and validate novel research hypotheses, demonstrating human-level discovery capabilities in drug repurposing.

Key challenges and current limitations. Despite the significant promise demonstrated by embodied intelligence in scientific discovery, its widespread adoption and scaling face several fundamental challenges that highlight where progress remains limited. (1) The reality gap in sim-to-real transfer: while high-fidelity simulation platforms are foundational, the extreme complexity and stochasticity of physical, chemical, and biological processes create a residual reality gap. Current strategies such as domain randomization and system identification are effective in structured lab settings but face significant hurdles in unstructured, real-world environments such as the deep sea, wilderness, or planetary surfaces, where exhaustive modeling of all physical contingencies is intractable. (2) The gap in integrated multimodal understanding: although agents integrate various sensors, achieving a unified, physics-aware understanding of experimental states—beyond simple feature concatenation—remains a core challenge. The system must reason from fused visual, tactile, force, and chemical sensory data to infer states. Technologies such as visuotactile gloves represent a step forward, but a generalized machine scientist perception capable of commonsense physical reasoning is still distant. (3) Long-horizon planning and commonsense reasoning for complex tasks: current systems excel at predefined or short-horizon tasks. However, they struggle with open-ended scientific exploration. Formulating appropriate subgoals, learning efficiently from abundant negative results, and adjusting strategies based on scientific commonsense and causal reasoning still heavily rely on human-designed frameworks or extensive domain-specific knowledge injection. While LLMs enhance task comprehension, their integration with grounded, causal world models is an ongoing research frontier. (4) Evaluation metrics and reproducibility: unlike games with clear reward functions, quantifying the performance of an autonomous discovery system is nontrivial. Furthermore, ensuring reproducibility across different labs and agents requires standardized orchestration interfaces, data formats, and procedural descriptions, which are still in nascent stages of development. (5) Paradigms for human-AI collaboration and governance: full autonomy is neither the sole nor always the most responsible goal. A more viable and immediate path is human-in-the-loop collaboration. However, designing intuitive interfaces that allow scientists to set goals, interpret results, and guide the process at key junctures is a significant barrier to adoption. Concurrently, emerging ethical and governance questions regarding responsibility attribution and intellectual property for discoveries made by autonomous systems require urgent attention.

Future directions. To develop a more scalable and general framework for embodied intelligence in scientific discovery, future research can be summarized into four directions. First, embodied intelligence is moving toward foundation model-based scientific agents that translate high-level objectives into executable experimental plans and iteratively refine them based on feedback, enabling closed-loop discovery rather than single-pass automation.[785] Second, digital twins and simulation platforms will become increasingly important because they allow models and workflows to be trained and evaluated in virtual environments, reducing experimental cost and risk while improving robustness under practical laboratory constraints. Third, practical systems still need human oversight and governance. Autonomy may increase, but people should still supervise key checkpoints to ensure safety, reliability, and reproducibility.[791] Finally, broad impact will require standardized orchestration and data interfaces that enable modular integration, traceable logging, and interoperability across instruments and laboratories, so that workflows and models can be

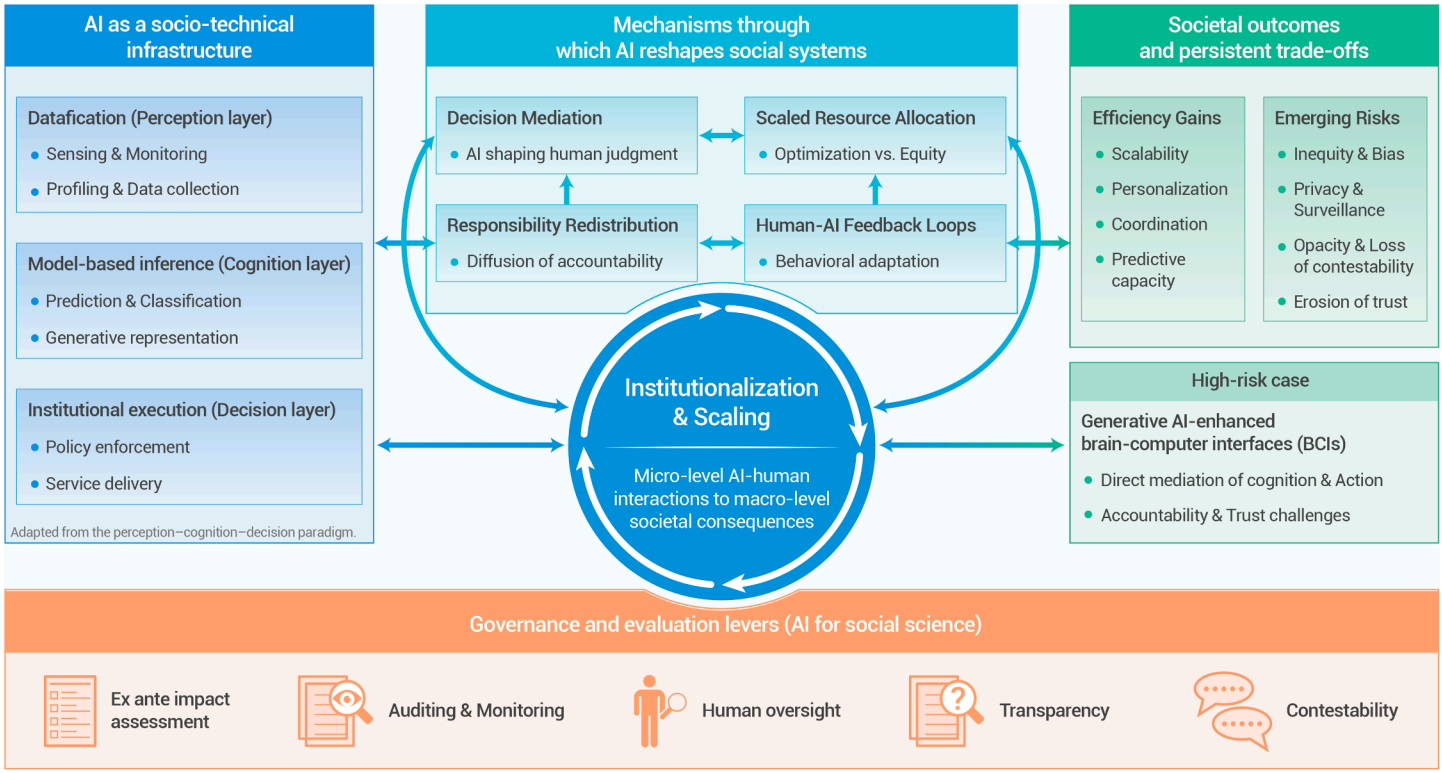

Figure 13. A mechanism-based framework for understanding the structural trade-offs of AI in social systems

reused and compared consistently.[792] In summary, embodied scientific discovery is shifting from isolated automation tools to integrated closed-loop systems; the scalability of embodied scientific discovery will depend on unifying planning, validation, oversight, and standardization into a coherent cross-lab infrastructure.

AI IN SOCIAL SCIENCE

From AI as a scientific paradigm to AI as a social infrastructure

Over the past decade, AI, together with ML techniques, has moved from being a specialized computational paradigm to a principal component of contemporary scientific and technological practice.[793,794] Early developments primarily served laboratory research and industrial optimization, supporting pattern recognition, control systems, and large-scale data analysis in closed technical environments. More recently, AI systems have become embedded in the routine operation of social institutions, including education, employment, healthcare, media environments, and public service delivery.[795,796] In these contexts, AI no longer merely processes information; it increasingly mediates how information is produced, how opportunities are distributed, and how decisions are enacted at an institutional scale.[797,798] This transition has become even more visible with the recent rise of GAI, which extends AI's role from recognition and prediction to the large-scale production of text, images, code, and multimodal content, thereby further expanding its influence over communication, knowledge production, and institutional interaction.

Large-scale reviews of AI applications addressing sustainability and societal challenges consistently document rapid expansion across domains closely tied to human well-being, such as health, education, energy, and environmental governance.[799–801] At the same time, these reviews reveal a persistent imbalance. Only a limited share of existing studies meaningfully integrates advanced AI techniques with domain-specific social or sustainability expertise.[802] Most applications remain oriented toward technical performance—prediction accuracy, optimization efficiency, or scalability—while paying comparatively little attention to institutional context, normative goals, or governance implications.[803,804] This pattern suggests that AI's societal impact is shaped less by technical capability itself than by how AI systems are embedded within social decision structures.[805]

Evidence from multiple empirical domains further indicates that AI rarely produces uniform benefits. Instead, efficiency gains, scalability, and personalization often emerge alongside heightened risks related to fairness, privacy, accountability, and public trust.[62,802,806,807] These tensions become more pronounced as AI systems are institutionalized and deployed at scale.[808,809] Algorithmic rankings, recommendations, and risk scores increasingly influence life-relevant outcomes, yet the processes through which these outputs are generated often remain difficult to observe, interpret, or contest.[810,811] Over time, such dynamics can subtly reshape power relations between institutions and individuals subject to algorithmic decision-making.[812] These issues may be further intensified by GAI, which not only filters or ranks existing information but can also actively produce synthetic content and interactive outputs, thereby complicating questions of authorship, authenticity, persuasion, and responsibility.

A useful conceptual bridge can be found in the commonly referenced AI capability layers of perception, cognition, and decision-making. When translated into social systems, these layers correspond to datafication through continuous observation and profiling; model-based inference via prediction, classification, and ranking; and institutional execution through policy enforcement, service allocation, and organizational action.[813] Once coupled, these layers extend AI's influence beyond technical performance to encompass distributional outcomes, institutional legitimacy, and trust.[814,815] Emerging developments in embodied intelligence further extend this trajectory by linking AI not only to informational inference but also to situated perception-action loops in physical and social environments. Although embodied intelligence is not yet as widely institutionalized across social domains as GAI, it signals an important direction in which AI may increasingly mediate action, coordination, and human-machine interaction in real-world settings.

This section advances an AI-for-social-science perspective that treats AI as a socio-technical infrastructure rather than a standalone analytical tool.[816,817] In contrast to AI4Science, where evaluation typically prioritizes accuracy, efficiency, and discovery speed, AI deployed in social systems must be assessed in terms of its effects on distribution, accountability, contestability, and long-term trust.[818] From this perspective, GAI and embodied intelligence should be understood not merely as technical advances but as developments that amplify AI's capacity to reshape social communication, institutional processes, and human agency. Building on this perspective, the section develops

a mechanism-based framework, synthesizes evidence across key application domains, and outlines governance levers necessary for responsible integration of AI into social systems. Figure 13 summarizes this perspective by illustrating how AI functions as a socio-technical infrastructure that reshapes social systems through recurring mechanisms, generates persistent trade-offs, and requires governance levers to align technical performance with social values.

Mechanisms of AI-induced structural change in social systems

Although AI applications differ across domains, their societal consequences tend to follow a limited set of recurring patterns.[819,820] These patterns help explain why AI's effects are often structural rather than confined to individual sectors. Four mechanisms are particularly consequential: decision mediation, scaled resource allocation, responsibility redistribution, and human-AI feedback loops.[821,822] These mechanisms remain broadly relevant across different generations of AI systems, but recent advances in GAI and emerging forms of embodied intelligence may intensify them by expanding AI's role from prediction and classification to content generation, interactive communication, and situated action.

Decision mediation occurs when AI outputs are inserted into judgment processes that were previously guided by professional expertise, deliberation, or rule-based procedures.[823] Algorithmic recommendations can improve consistency and reduce cognitive or administrative burdens, especially in high-volume or complex decision environments.[824,825] At the same time, mediation can narrow the space of legitimate reasoning by privileging information that is model legible, potentially sidelining contextual knowledge and ethical judgment that resist formalization.[826] This issue becomes especially salient as GAI systems increasingly shape not only evaluative outputs but also the framing, wording, and communicative presentation of recommendations, thereby extending mediation from decision support to the construction of decision environments themselves.

Scaled resource allocation reflects AI's capacity to distribute opportunities and services across large populations.[827,828] Educational support, employment screening, healthcare access, and public services can be dynamically allocated based on predicted performance or risk.[829] While such approaches enhance efficiency and targeting, they also raise the risk of allowing latent inequalities in data and institutional practice to be reproduced or amplified at scale.[830,831] As AI systems become more deeply embedded in communication platforms, service infrastructures, and interactive interfaces, allocation may increasingly be shaped not only by predictive classification but also by how AIGC guides attention, interpretation, and behavioral response.

The responsibility of redistribution emerges as algorithmic systems participate more directly in decision-making.[832,833] Accountability becomes dispersed across developers, deploying institutions, frontline professionals, and regulators, complicating redress and weakening institutional responsibility.[815] These difficulties may become even more pronounced in systems that rely on generative outputs or semi-autonomous interaction because responsibility then extends beyond model prediction to include the interpretation, communication, and downstream effects of AI-produced content or action.

Human-AI feedback loops arise through repeated interaction with AI systems. Behavioral research suggests that individuals adapt their judgment strategies in response to algorithmic feedback, gradually internalizing system outputs.[834] When institutionalized and scaled, such adaptations can reshape organizational norms, reduce diversity of judgment, and amplify initial biases over time.[835] In environments increasingly shaped by conversational AI and GAI, such feedback loops may become more pervasive because users are not only responding to algorithmic scores or rankings but also interacting with systems that simulate explanation, dialogue, and adaptive assistance, potentially strengthening reliance and reducing critical distance.

Taken together, these mechanisms generate persistent trade-offs between efficiency and equity, automation and discretion, and scalability and legitimacy. They also clarify why governance challenges associated with AI cannot be addressed through technical optimization alone. At their core, these challenges are institutional and social. For this reason, emerging technological trajectories should be evaluated not only in terms of capability gains but also in terms of how they reinforce or transform these recurring structural mechanisms within social systems.

Domain evidence: Benefits, risks, and trade-offs

This section briefly synthesizes domain-level evidence to illustrate how recurring benefits, risks, and trade-offs appear across major social sectors, without repeating the mechanism-based discussion developed above.

Education and skills. In education, AI-driven learning analytics, automated assessment, and adaptive tutoring systems are increasingly deployed to support personalized instruction and scalable feedback. These systems can be particularly valuable in large or resource-constrained educational settings. More recently, GAI has further expanded these functions by enabling conversational tutoring, automated content support, and individualized feedback at scale. However, data-driven profiling and automated evaluation also introduce risks of over-standardization and path dependence, whereby early predictions constrain future opportunities. These developments suggest that evaluation in education should attend not only to learning outcomes but also to interpretability, fairness, and students' ability to question or contest algorithmic assessments.

Employment and organizations. AI applications in labor markets commonly support recruitment screening, performance evaluation, and algorithmic management. While such systems can improve matching efficiency and reduce administrative costs, they also intensify monitoring and erode professional discretion. The growing use of GAI in organizational communication and workflow support may further broaden AI's influence over everyday work practices and managerial processes. Empirical and theoretical research on automation and labor indicates that productivity gains may coexist with increased inequality and weakened worker agency when institutional safeguards are insufficient. Governance efforts in this domain should therefore emphasize auditability, transparency, and fair allocation of error costs.

Healthcare and public trust. In healthcare, AI systems offer substantial potential to support diagnosis and operational decision-making.[305] These benefits, however, depend on extensive data integration and raise concerns about uneven performance across population groups.[836] Opaque decision processes further complicate accountability.[837] Importantly, healthcare AI directly affects institutional trust: even isolated failures or perceived unfairness can undermine legitimacy, regardless of average technical performance.[305,836] Trust should thus be treated as an outcome requiring active governance rather than as an assumed by-product of accuracy gains.[837]

Media and information ecosystems. AI-driven recommendations, moderation, and content generation systems increasingly shape information exposure and public attention.[838] This issue has become especially salient with the rise of GAI, which expands AI's role from filtering information to actively producing text, images, and other communicative content at scale. Although these systems enhance efficiency and personalization, feedback mechanisms can amplify polarization and misinformation risks.[839,840] Evaluation frameworks in this domain should prioritize the quality of information, diversity of exposure, and effectiveness of corrective mechanisms rather than relying solely on engagement-based metrics.[841]

Public service delivery and urban systems. AI is also used to support traffic management, emergency response, environmental monitoring, and public service allocation.[842] These applications illustrate AI's role as a coordination infrastructure.[843] Nevertheless, uneven model performance across contexts can exacerbate service disparities, and responsibility diffusion complicates accountability when failures occur.[844] As AI becomes more deeply embedded in public-facing systems, maintaining legitimacy requires not only technical reliability but also transparent criteria, clearly defined accountability chains, and meaningful contestability.[845]

Governance and evaluation: Why AI for social science is necessary

Across domains, a consistent pattern emerges: AI enhances efficiency and coordination by mediating decisions and scaling allocation yet simultaneously introduces structural risks that cannot be addressed through technical fixes alone. Large-scale reviews of AI for sustainability further indicate that most applications prioritize prediction and optimization, while relatively few embed social, institutional, or normative knowledge.[1] This imbalance reinforces the need for an AI-for-social-science perspective. The point becomes even more salient as recent developments such as GAI and embodied intelligence expand AI's role from analytic support to communicative production, interactive assistance, and, increasingly, situated forms of action. As AI systems become more deeply

embedded in how institutions communicate, coordinate, and intervene, the limits of purely technical evaluation become more apparent.

Experience from multiple policy and institutional settings suggests that effective governance relies on a combination of complementary levers. *Ex ante* impact assessments can help identify affected populations and potential distributional effects prior to deployment. Continuous auditing and monitoring are necessary to detect performance drift and feedback-induced harms over time. Meaningful human oversight must preserve genuine decision authority rather than serving as a symbolic safeguard.[14] Finally, transparency and contestability are essential to institutional legitimacy, enabling individuals and organizations to challenge and correct algorithmic decisions.[846] These requirements are especially important in settings where AI systems not only classify or recommend but also generate content, shape interaction, or influence action in ways that may be less visible yet more deeply consequential for institutional trust and social legitimacy.

Crucially, evaluation frameworks must extend beyond accuracy and efficiency to incorporate equity, accountability, and trust. Aligning AI deployment with broader societal objectives, including sustainability agendas, requires integrating technical performance with institutional design and social values.[847] AI for social science offers the conceptual and analytical tools needed to support this integration. In this sense, the value of AI for social science lies not in opposing technological progress but in making visible the institutional conditions, distributional consequences, and governance choices through which AI becomes socially sustainable and publicly legitimate.

Summary

AI is reshaping social systems through mechanisms that reconfigure decision-making, resource allocation, and responsibility. Across education, employment, healthcare, media, and public services, efficiency gains coexist with persistent trade-offs involving fairness, privacy, accountability, and trust. Recent developments in GAI, as well as the broader emergence of embodied intelligence, further reinforce this transformation by extending AI's role beyond prediction and classification into content generation, interactive communication, and potentially action in real-world settings. The central challenge is therefore not whether AI should be adopted but how it is embedded within institutional processes and governed over time. By emphasizing mechanisms, distributional outcomes, and governance levers, an AI-for-social-science perspective complements AI4Science and broadens the evaluation of AI beyond technical performance. This perspective is essential for realizing AI's potential to support socially legitimate and sustainable transformation.

CROSS-DOMAIN LESSONS AND METHODOLOGICAL PRIORITIES FOR AI4Science

The preceding sections show that, despite major differences in disciplinary objects, data modalities, experimental settings, and evaluation cultures, AI4Science is beginning to display several recurring cross-domain patterns. These patterns clarify what is genuinely changing in scientific practice, which limitations remain shared across fields, and which methodological priorities will determine whether current progress can be translated into reliable scientific outcomes. Rather than revisiting domain-specific examples, this section distills the common lessons from the reviewed literature and highlights key priorities for rigorous and scalable AI4Science.

Cross-domain lessons from current AI4Science progress

A first cross-domain lesson is that the most important shift in AI4Science is not simply better task-specific prediction but the gradual move from isolated analytical tools to workflow-level integration. Across domains, AI is increasingly used not only for data analysis and pattern recognition but also for literature retrieval, hypothesis generation, experimental planning, simulation, and report production. The significance of recent foundation models, multimodal learning, and agent-based systems therefore lies less in single benchmark gains than in their ability to connect previously separate stages of scientific inquiry into more continuous research pipelines.

A second lesson is that the maturity of AI4Science remains highly uneven across disciplines, while the major bottlenecks are remarkably similar. Across medicine, materials, physics, astronomy, geoscience, and embodied scientific discovery, progress is repeatedly constrained by limited standardized high-quality data, insufficient interpretability, incomplete incorporation of physical or domain-specific constraints, and weak evaluation protocols. The central question is therefore no longer only whether AI can improve performance on narrow tasks but whether it can function as a reliable and scientifically meaningful component of heterogeneous research environments.

A third lesson is that benchmark success, promising prototypes, and robust scientific reliability should not be treated as equivalent. Many reviewed studies show impressive capabilities in screening, prediction, optimization, and closed-loop automation, but scientific value depends on more than retrospective performance gains. In many fields, the real bottlenecks remain prospective validation, experimental confirmation, reproducible workflow design, and clear characterization of uncertainty, failure modes, and applicability boundaries. Thus, the most credible path for AI4Science is not unrestricted autonomy but progressively stronger human-AI collaboration under explicit validation and governance constraints.

Methodological priorities for AI4Science

The first priority is to strengthen benchmarking and evidence standards. Cross-domain evaluation should move beyond isolated accuracy or score reporting toward clearer descriptions of data provenance, task boundaries, baseline selection, distribution shift, and intended use conditions. Benchmark-level success should also be distinguished from domain-grounded and real-world validation. A practical framework is to separate three levels of evidence: offline benchmark validation, domain-grounded validation against accepted scientific constraints or expert interpretation, and prospective or experimental validation in realistic research settings. This distinction can reduce overinterpretation and improve transparency across domains.

The second priority is to place uncertainty quantification, interpretability, and domain constraints closer to the center of AI4Science methodology. Scientific applications often operate in data-sparse, safety-critical, or experimentally costly settings, where point predictions alone are insufficient. Models should therefore provide calibrated confidence estimates, reveal important failure modes, and support risk-aware decision-making. At the same time, explainability should not be treated only as a post hoc requirement. Across many domains, scientific utility improves when models are developed together with physical, chemical, biological, or mechanistic priors that enhance extrapolation, align inference with established knowledge, and narrow the gap between statistical fitting and scientific understanding.

The third priority is to improve reproducibility, traceability, and human oversight in increasingly integrated AI4Science workflows. Reproducibility depends not only on releasing models but also on maintaining versioned datasets, transparent configurations, traceable execution logs, and interoperable interfaces across tools, instruments, and laboratories. These requirements become even more important as AI systems move from passive analysis to active participation in planning and experimentation. Even where autonomy increases, key checkpoints involving high-stakes inference, experimental intervention, clinical deployment, or socially consequential decision-making should remain under human supervision. In this sense, scalable AI4Science will depend less on model capability alone than on trustworthy infrastructures that unify planning, validation, traceability, and oversight.

Overall, the cross-domain evidence reviewed here suggests that the next phase of AI4Science will be determined not only by larger models or broader automation but by whether the field can establish rigorous methodological foundations for evaluation, uncertainty handling, validation, reproducibility, and accountable human-AI collaboration. These priorities also provide a practical bridge to the broader discussion of governance and strategic development in the following section.

THE STRATEGIC LANDSCAPE OF AI4Science

The integration of AI into scientific research represents a fundamental shift in how nations approach innovation. Major technological powers have elevated AI4Science from an academic agenda to a core national strategy, competing to construct comprehensive ecosystems that underpin AI-driven discovery, as shown in Figure 14. The US deploys a whole-of-government approach to mobilize public-private resources, the European Union (EU) relies on federalized mechanisms to foster transnational collaboration, the United Kingdom (UK) pursues focused breakthroughs in areas of comparative advantage, and China

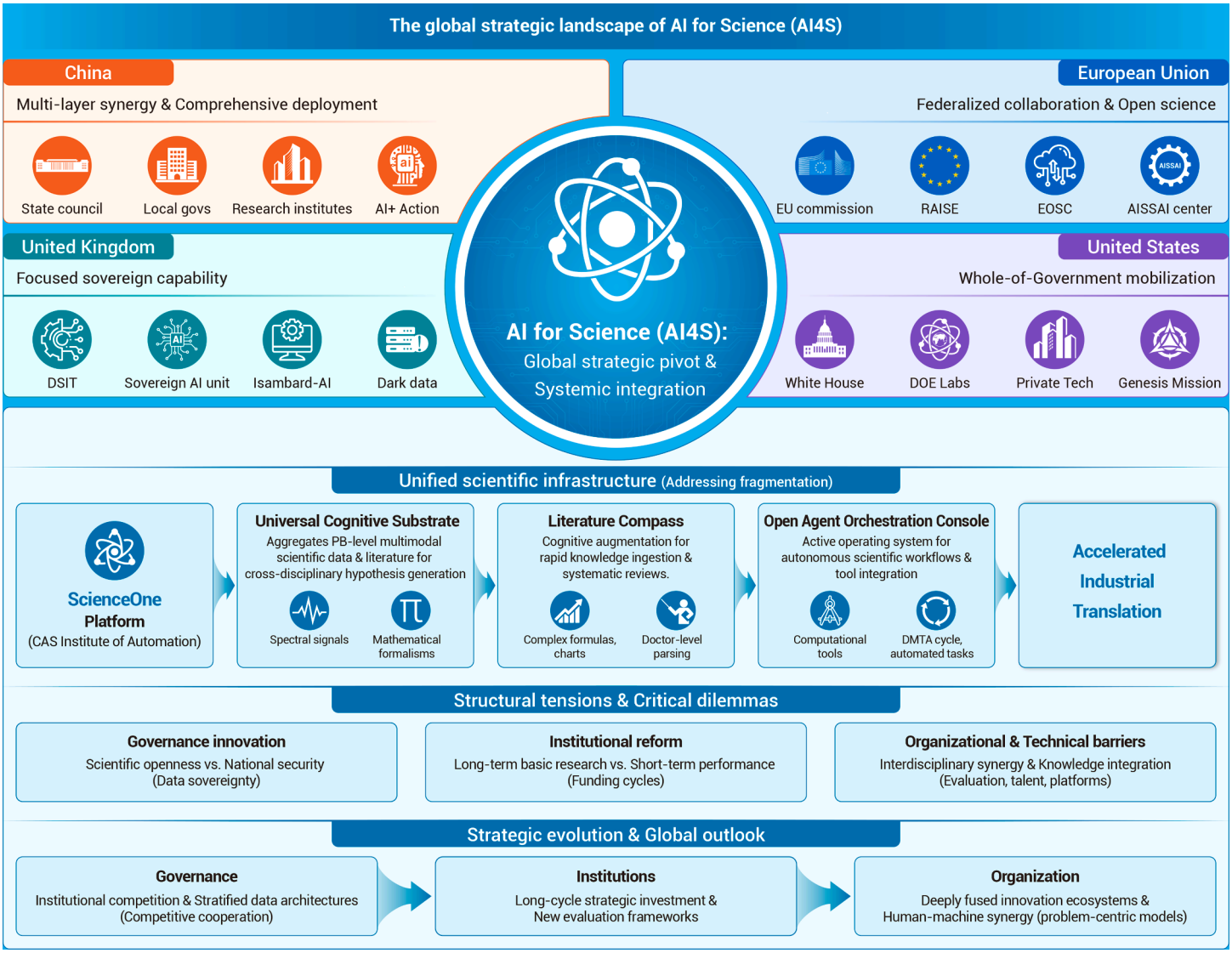

Figure 14. The global strategic landscape of AI for science

leverages its institutional strengths to drive comprehensive deployment. The locus of strategic competition has shifted from isolated technical innovations to the systemic integration of data resources, computational infrastructure, and talent pipelines. Nations are navigating complex trade-offs between open science and data sovereignty, long-term investment and short-term performance metrics, and disciplinary specialization versus convergent innovation. How effectively these strategies are executed in the coming years will likely shape the global competitive positioning of nations in this new research paradigm. This strategic layer is directly relevant to the scientific advances reviewed above because it shapes whether domain-specific breakthroughs can be sustained by durable data infrastructures, compute resources, shared tools, and reproducible workflows. Because this section draws on strategy documents, institutional reports, and platform announcements in addition to peer-reviewed literature, the discussion distinguishes stated priorities from reported implementation outcomes where possible.

From research tool to national strategy

AI-driven scientific research has evolved from a frontier topic of academic exploration into a core strategic priority for global technological competition. In recent years, major powers, including the US, the EU, the UK, and China, have released specific strategy documents or policy frameworks related to AI4Science, marking systematic deployment at the state level.[848–851] This strategic pivot reflects recognition that AI4Science serves as a critical lever for reshaping national innovation systems and building long-term competitive advantages.[852]

The core logic of these national strategies revolves around five key elements: computing power, data, models, talent, and institutions. Regarding resource allocation, there has been a distinct shift from project-based funding to the systematic construction of national-level research infrastructure, creating platforms that integrate data, compute, and models.[853] In terms of organizational architecture, nations are universally establishing cross-departmental coordination mechanisms to integrate federal agencies, national laboratories, universities, and industrial forces.[854] Regarding strategic objectives, states are no longer satisfied with mere academic output but seek to align AI4Science directly with the technical breakthrough requirements of strategic industries such as energy, materials, and biomedicine.[849,851] This signifies a transition in research strategy from supporting dispersed innovation to cultivating holistic ecosystems.

National pathways and strategic approaches

While major technological powers share consensus on the direction of this paradigm shift, their strategic paths exhibit significant divergence rooted in differences in resource endowments, political institutions, and strategic priorities.

The US is characterized by whole-of-government mobilization through presidential executive orders, deeply integrating private sector capabilities through public-private partnerships. The White House released the special report "Supercharging Research: Harnessing Artificial Intelligence to Meet Global

Challenges"[855] and the "AI Action Plan,"[856] which identify increased investment in AI-driven research and AI infrastructure as pillar strategies. The "Genesis Mission," launched in November 2025, explicitly aims to expand the productivity and impact of federal R&D within a decade through AI, positioning this as critical to restoring American technological leadership and addressing geopolitical competition.[848,853] At the execution level, the Department of Energy leverages the specialized talent, large-scale scientific facilities, and supercomputing resources of its 17 national laboratories, while the Office of Critical and Emerging Technologies supports department-wide coordination relevant to ecosystem development.[857] The National Science Foundation continues to fund the National AI Research Institutes program, establishing multiple institutes covering foundational sciences such as chemistry, astronomy, and materials.[858] Tech giants such as NVIDIA, OpenAI, and Google DeepMind have also announced collaborations with the national laboratory system.[859–861] This model combines national strategic deployment with market innovation vitality, aiming to leverage scale advantages in computing infrastructure and data accumulation to establish long-term leadership. Its practical relevance is especially visible in materials, chemistry, and energy research, where access to national laboratory facilities, specialized datasets, and large-scale computing resources strongly influences AI4Science outcomes. At the same time, this model may also concentrate access to key resources and increase dependence on proprietary infrastructure.

The EU focuses on federalized collaboration mechanisms to achieve transnational integration of member state resources, balancing demands for strategic autonomy with the philosophy of open science. The "EU Strategy for AI in Science," released by the European Commission in October 2025, acknowledges that single member states cannot bear the prohibitive costs of training frontier scientific models. Consequently, it established RAISE (Resource for AI Science in Europe) as a virtual institute to pool and coordinate AI research forces across Europe.[850,862] The strategy further provides support for scientific access to AI Gigafactories through Horizon Europe funding of up to 600 million euros.[863] In terms of disciplinary integration, the AISSAI center established by the French National Centre for Scientific Research organizes thematic trimesters, enabling mathematicians and computer scientists to collaborate deeply with physicists and biologists over extended periods.[864–866] Europe is also promoting a data governance system compliant with FAIR principles through the European Open Science Cloud, laying the data foundation for training European scientific AI models.[867] The EU strategy attempts to compensate for limitations of individual national capabilities through institutional innovation and resource aggregation while upholding core values of data sovereignty and scientific collaboration. This pathway is particularly relevant to cross-domain data reuse, benchmark construction, and reproducibility across the scientific fields reviewed above, although its effectiveness still depends on coordination speed and resource alignment across member states.

The UK adopts a focused strategy, aiming to establish global leadership in specific areas of advantage through sovereign capability building and mission-oriented resource allocation. The "AI for Science Strategy," published in November 2025, builds on the Sovereign AI Unit and related Department for Science, Innovation and Technology initiatives as part of the UK's broader effort to strengthen sovereign AI capabilities and research infrastructure. Rather than constituting a standalone program, the strategy draws on broader UK investment in AI research resources, including the AI Research Resource and related national compute infrastructure such as the Isambard-AI supercomputer in Bristol.[868] The strategy outlines five priority areas: engineering biology, fusion energy, materials science, medical research, and quantum technology. The inaugural mission focuses on accelerating drug discovery and therapeutics, building on the UK's strengths in AI-enabled structural biology and its biopharmaceutical industry.[869] This gives the strategy a direct connection to the life-science applications discussed earlier in the review. The strategy also emphasizes the use of dark data, unpublished negative experimental results, to train more robust AI models[849] and highlights the Materials Innovation Factory model from the University of Liverpool, where AI-controlled robotic chemists conduct autonomous experimentation.[849] Together, these initiatives link national strategy to concrete workflows in therapeutics and materials discovery, although the overall scale of the UK approach remains more limited than that of larger coordinated programs.

China leverages its institutional strengths to pursue multi-layer coordination across government, industry, academia, and research, driving comprehensive deployment. The State Council's 2025 opinion on deeply implementing the "AI+" initiative includes a dedicated "AI+ science and technology" component and explicitly proposes deep integration of AI into the entire scientific research process.[851,870] Organizationally, local governments are actively responding to national strategic deployment. Beijing and Shanghai, for example, have introduced related policy measures and institutional initiatives to support interdisciplinary AI4Science activity.[871,872] Technologically, China is building a comprehensive domestic technology stack from underlying chips and computing frameworks to upper-layer models and applications. This has cultivated prominent domain-specific models and systems, including Sigma Geography, a multimodal large model for geographic sciences[873]; the Sai Boshi agent by the Institute of High Energy Physics; and Huawei Cloud's Pangu Weather, which surpasses traditional NWP in both accuracy and speed.[32] At the same time, cross-platform comparative evaluation remains underdeveloped.

Complementing these domain-specific advances, research institutions are also exploring unified scientific infrastructure to address fragmentation challenges. The ScienceOne platform, developed by the Institute of Automation at the Chinese Academy of Sciences, represents one such effort to construct comprehensive technical substrates for AI-driven research.[874] As one representative platform case within China's broader AI4Science landscape, the platform targets three structural bottlenecks that constrain current AI4Science development: the reasoning gap where general-purpose LLMs lack deep domain knowledge for complex scientific tasks, data silos created by incompatible formats across disciplines, and inefficient workshop-style development where researchers independently fine-tune models without shared foundations.

ScienceOne's strategic architecture is presented in public descriptions through three integrated capabilities relevant to the scientific workflow. First, it brings together large-scale scientific literature and domain resources into a shared knowledge base intended to support cross-disciplinary reasoning. Second, the "literature compass" component is described as supporting literature retrieval, formula and figure interpretation, and structured review generation based on a large scientific corpus. Third, the Open Agent Orchestration Console integrates hundreds of specialized computational tools and professional models so that researchers can assemble multi-step workflows across different research scenarios without deep programming expertise.[874] Reported deployments suggest potential utility, although broader independent benchmarking is still needed to assess generalizability and research impact.

The Chinese strategy emphasizes serving the needs of the real economy, striving for tight coupling between technological breakthroughs and industrial upgrading.[875] ScienceOne thus illustrates how coordinated strategy may be translated into routine scientific workflows and application-oriented research.

Implementation tensions and structural challenges

Despite massive strategic resources dedicated to AI4Science, implementation faces deep-seated structural challenges that transcend technical issues and are rooted in the intersection between scientific research and national strategy. These tensions directly influence whether the autonomous experimentation pipelines, cross-domain foundation models, and data-sharing protocols discussed in preceding sections can be benchmarked, shared across institutions, and scaled in practice.

The first challenge is the strategic tension between scientific openness and national security. The nature of scientific research demands data sharing and international cooperation, yet AI-driven research requires massive amounts of high-quality data, much of which involves sensitive national security domains. The US Genesis Mission explicitly links AI-enabled science to cybersecurity, biosecurity, and dual-use governance concerns, reflecting concerns over dual-use applications in sensitive areas.[848] China's AI governance framework emphasizes national sovereignty in data management, maintaining regulatory oversight on data flows while promoting domestic infrastructure.[851] The EU promotes data federation through the European Open Science Cloud, achieving cross-border scientific data sharing within constraints of the "General Data Protection Regulation."[867] This tension is not merely a technical implementation challenge but a fundamental governance dilemma requiring institutional innovation. The international community has yet to establish a consensus mechanism that effectively reconciles principles of open scientific exploration with legitimate national security concerns and data protection requirements. As computational demand for AI model development grows exponentially, this

structural contradiction is likely to intensify, necessitating new governance frameworks that protect data sovereignty while enabling collaborative infrastructure essential for scientific progress.

The second challenge is the cyclical dislocation between long-term basic research and short-term performance demands. Major scientific breakthroughs driven by AI often require long-cycle basic research accumulation and high-risk exploratory investment, yet strategic resource allocation inevitably faces pressure for short-term performance assessment. Misalignment between political cycles and technological cycles makes it difficult to sustain long-term strategic resolve. Nations are exploring institutional arrangements to address this. The US National Science Foundation's National AI Institutes program attempts to mitigate this through 5-year long-term funding mechanisms.[858] The UK locks in multi-year resource allocation through mission-oriented funding,[849] and China ensures continuity of budgetary input through its Five-Year Plan system.[876] However, these mechanisms still face challenges in ensuring strategic investment is not curtailed during technological adjustment periods or government transitions and in establishing institutional arrangements that tolerate failure and encourage long-term exploration. This involves fundamental reform of research evaluation systems, shifting from pursuing short-term output metrics to encouraging long-term paradigmatic breakthroughs.

The third challenge lies in organizational and technical barriers to interdisciplinary synergy and knowledge integration. AI4Science implementation inherently requires deep fusion of CS, mathematics, and various basic disciplines, yet existing disciplinary evaluation systems, talent training models, and research organizational architectures remain dominated by single-discipline paradigms. Diverging evaluation standards mean interdisciplinary results often struggle for full recognition. CS prioritizes algorithmic innovation while domain sciences prioritize discovery, creating tension where interdisciplinary work often falls between disciplinary evaluation criteria. Talent training systems lack composite talents proficient in both computational methods and domain knowledge, and compartmentalized resource allocation further exacerbates collaboration difficulties.[852] Beyond organizational inertia, the field also faces technical fragmentation that reinforces these structural barriers. General-purpose models lack deep domain knowledge, incompatible data formats create persistent silos, and workshop-style development prevents systematic knowledge accumulation across research groups. Countries are actively exploring solutions at multiple levels. At the organizational level, France's AISSAI center enables sustained cross-disciplinary collaboration through thematic trimesters that bring together mathematicians, computer scientists, physicists, and biologists for extended periods.[864–866] The UK strategy explicitly invests in cultivating truly interdisciplinary teams and launches comprehensive researcher upskilling programs.[849] At the strategic coordination level, the US Genesis Mission exemplifies cross-institutional resource orchestration to integrate capabilities across government laboratories and industry. At the technical infrastructure level, platforms such as China's ScienceOne demonstrate efforts to construct unified substrates that provide shared foundations across domains, addressing fragmentation through integrated data resources and computational tools. However, fundamentally resolving these dual challenges requires systemic reform that begins with the educational foundation itself, establishing new interdisciplinary degree programs and career pathways while developing technical platforms that enable rather than constrain collaborative innovation.

Strategic trends and outlook

Looking ahead, national AI4Science strategies will evolve along three main dimensions: governance innovation, institutional reform, and organizational restructuring. This evolution represents both a systemic response to structural challenges and a determinant of whether nations can truly unlock the transformative potential of AI-driven research.

In governance, strategies will shift from simple resource competition to institutional competition over data governance systems. The tension between research openness and national security requires governance innovation rather than technical solutions. The core direction of future evolution is establishing stratified and classified data governance architectures that maintain open sharing principles for basic scientific data to guarantee collaborative foundations required for progress while implementing sovereign control over data in sensitive strategic domains to safeguard national interests. The degree to which this balance is achieved will be a key indicator of AI4Science strategy maturity. Despite intensifying geopolitical competition, the global-public-good attribute of scientific research dictates that total decoupling is neither realistic nor desirable. A more probable path is the formation of a competitive cooperation landscape, maintaining limited data interoperability and result sharing in basic science while engaging in fierce competition in strategic technologies, with nations seeking dynamic equilibrium between openness and closure.

In institutions, strategies will transition from project-based funding to long-cycle strategic investment mechanisms. The dislocation between long-term research and short-term performance reflects a structural contradiction between political and scientific cycles. Key to future evolution is establishing institutional arrangements that can span political cycles by cementing long-term investment commitments through legislation, establishing strategic execution agencies independent of administrative cycles, and constructing research evaluation systems that tolerate failure and encourage exploration. This requires shifting from quantitative short-term output metrics to new evaluation frameworks that assess potential for paradigm shifts and long-term strategic value. The ability to complete this institutional transition will determine whether strategic investment translates into sustained innovation capacity or remains short-term behavior fluctuating with political winds.

In organizations, strategies will evolve from parallel disciplinary development to deeply fused innovation ecosystems. Barriers to interdisciplinary synergy are rooted in modern traditions of disciplinary institutions, and their dissolution requires systemic change in talent training, evaluation incentives, and organizational architecture. Future evolution points toward building problem-centric rather than discipline-centric research models by cultivating composite talents, establishing evaluation standards that recognize interdisciplinary contributions, and creating organizational structures that facilitate continuous deep collaboration. These organizational transformations will be enabled and accelerated by the emergence of unified technical platforms that lower barriers to cross-disciplinary tool access and data integration. A deeper transformation lies in the shift of the research paradigm itself, from human-led linear division of labor to a new paradigm of human-machine synergy, where AI is upgraded from an auxiliary tool to a research agent capable of autonomously proposing hypotheses and designing experiments. This will fundamentally reshape the organizational logic and knowledge production methods of science.

From a global perspective, the next decade will be a critical test period for the efficacy of AI4Science strategic execution. Current national strategies are increasingly taking shape, and the focus of competition will shift from strategic planning to execution capability. The key to success lies not in the absolute size of the investment but in the effective integration of data resources, computing facilities, and talent pools and the establishment of governance institutions and organizational cultures adapted to the new research paradigm. Nations that can more effectively complete the transformation from supporting dispersed innovation to building complete ecosystems will be better positioned to secure early coordination advantages in the new era of AI-driven science. If sustained, such advantages can become self-reinforcing through talent attraction, data accumulation, and model iteration, further widening the gap with follower nations. As revealed by national strategy documents, AI is no longer merely a tool for research but is becoming a strategic force reshaping national innovation systems and determining long-term competitiveness.

CONCLUSION

In summary, AI has evolved into a central cognitive infrastructure that is reshaping the methodology of scientific discovery. The progress achieved by 2026 demonstrates that by integrating data-driven inference with reasoning-oriented intelligence, AI can effectively address systemic challenges and drive a paradigm shift from empirical trial and error toward predictive and generative science. However, fully realizing this potential requires overcoming several critical challenges, including the realization gap between digital design and automated synthesis, the lack of explicit physical constraints in DL architectures, and the structural tensions between scientific openness and data sovereignty. Furthermore, the maturity of closed-loop automation remains highly heterogeneous across disciplines, and overcoming the reality gap between benchmark success and real-world deployment (e.g., non-stationarity in geosciences, synthesizability in materials, and hallucinations in medicine) remains a top priority. Finally, future roadmaps must address the severe resource barriers and environmental energy footprints of scaling laws, necessitating a shift toward

energy-efficient algorithms and sustainable computing infrastructures. Looking forward, future perspectives should emphasize the intrinsic integration of physical priors with neural operators and the widespread deployment of closed-loop SDLs. Ultimately, rather than replacing scientists, AI will serve as an indispensable cognitive copilot, facilitating a profound human-AI symbiosis in the exploration of natural laws.

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AUTHOR CONTRIBUTIONS

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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